Quasi-Bayesian sequential deconvolution

Stefano Favaro*[1] and Sandra Fortini†[2]

[1]Department of Economics and Statistics, University of Torino and Collegio Carlo Alberto, Italy
[2]Department of Decision Sciences, Bocconi University, Italy

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Abstract

Density deconvolution is the inverse problem of estimating a probability density from observations contaminated by additive noise. Traditionally studied in static or batch settings, it increasingly arises with streaming data, where existing frequentist and Bayesian procedures face substantial computational bottlenecks. We develop a quasi-Bayesian nonparametric method for sequential density deconvolution based on Newton's recursive algorithm. The resulting estimate is straightforward to evaluate and scalable to massive datasets, as its per-observation computational cost remains constant as new data arrive. The quasi-Bayesian interpretation enables uncertainty quantification: local and uniform central limit theorems yield asymptotic credible intervals and bands, respectively. Under a frequentist data-generating model, we establish L^1-consistency for the proposed estimate and show that it asymptotically agrees with the estimate that would be obtained if the uncontaminated variables were directly observed. Further, under additional regularity conditions, we derive an L^1-Wasserstein convergence rate and establish merging, at an explicit rate, with the Bayesian nonparametric posterior mean estimate under a Dirichlet process mixture model. Synthetic-data experiments, together with an acquisition-ordered flow-cytometry application, demonstrate accuracy comparable to Bayesian nonparametric and Fourier deconvolution methods, while offering a substantial computational advantage.

Keywords: Credible bands; Newton's algorithm; quasi-Bayesian learning; sequential density deconvolution; Wasserstein distance

1 Introduction

1.1 Background and motivations

Density deconvolution, or simply deconvolution, is the inverse problem of estimating a probability density from observations contaminated by additive noise. More precisely, one observes $n \geq 1$

*stefano.favaro@unito.it
†sandra.fortini@unibocconi.it

real-valued random variables $Y_1, \ldots, Y_n$ generated according to the convolution model

$$Y_i = X_i + Z_i, \qquad i = 1, \ldots, n,$$

where the X_i's are i.i.d. unobserved signal variables with unknown density f_X, and the Z_i's are i.i.d. noise variables with known density f_Z, independent of the X_i's. The objective is to estimate f_X from $Y_1, \ldots, Y_n$, whose density satisfies the convolution identity $f_Y = f_X * f_Z$. The frequentist literature is extensive, encompassing parametric and nonparametric methods with strong theoretical guarantees (Carroll and Hall, 1988; Stefanski and Carroll, 1990; Diggle and Hall, 1993; Fan, 1991, 1993; Pensky and Vidakovic, 1999; Delaigle and Gijbels, 2002; Hesse and Meister, 2004; Butucea and Tsybakov, 2008a,b); see Yi et al. (2021, Chapters 10–14) for an overview. By comparison, the Bayesian literature is more recent and limited, focusing primarily on theoretical guarantees under Dirichlet process mixture models for the signal, with methodological development remaining less mature (Gao and Van der Vaart, 2016; Scricciolo, 2018; Su et al., 2020; Rousseau and Scricciolo, 2024).

Density deconvolution arises when observations are blurred or distorted, often by measurement error in the underlying signal (Buonaccorsi, 2010; Yi et al., 2021). Traditionally studied in static or batch settings, it has applications across the natural and social sciences, including medical imaging, bioinformatics, genomics, chemistry, spectroscopy, astronomy, and econometrics. The increasing availability of streaming data in these areas, together with emerging applications in social media analytics, remote sensing, e-commerce, and privacy-preserving data analysis, creates a growing need for online or sequential methods that enable timely inference and rapid adaptation as new observations arrive. Although existing frequentist and Bayesian procedures can, in principle, be adapted to streaming data, they typically incur substantial computational costs due to repeated optimization or posterior simulation. Moreover, no principled sequential method for density deconvolution currently combines theoretical guarantees with demonstrated empirical effectiveness. We fill this gap with a scalable and theoretically grounded framework specifically designed for streaming data.

1.2 Preview of our contributions

We propose a nonparametric method for sequential density deconvolution, assuming that the unknown density f_X of the unobserved X_i's admits a mixture representation of the form

$$f_G^{(X)}(x) = \int_\Theta k(x \mid \theta) G(\mathrm{d}\theta), \qquad x \in \mathbb{R}, \tag{1}$$

where $k(\cdot \mid \theta)$ is a known positive kernel, $\theta \in \Theta \subset \mathbb{R}$, and G is an unknown mixing distribution on Θ. Since $f_Y = f_X * f_Z$, the density of the observed Y_i's has a mixture representation over Θ with kernel $k * f_Z$ and the same mixing distribution G. Starting from an initial guess $\widetilde{G}_0$, our method updates $\widetilde{G}_{n-1}$ recursively upon receiving each new observation Y_n, according to the procedure of Smith and Makov (1978), known as Newton's algorithm (Newton et al., 1998; Martin and Ghosh, 2008). A sequential estimate of $f_G^{(X)}$ is then obtained from (1) by replacing G with $\widetilde{G}_n$. This recursive procedure admits a quasi-Bayesian interpretation, providing a predictive construction of a Bayesian model that is asymptotically equivalent (Fortini and Petrone, 2020). That is, for large n, it implicitly induces a prior on G under which $f_{\widetilde{G}_n}^{(X)}$ is the posterior mean of $f_G^{(X)}$. Moreover, $f_{\widetilde{G}_n}^{(X)}$ is straightforward to evaluate and scalable to massive datasets, as its per-observation computational cost remains constant as new data arrive.

Interpreting Newton's algorithm as a quasi-Bayesian learning model makes it possible to quantify uncertainty in the estimate $f_{\widetilde{G}_n}^{(X)}$ of $f_G^{(X)}$. For density deconvolution at a fixed point $x \in \mathbb{R}$,

we establish a local central limit theorem that enables the construction of a sequential asymptotic credible interval for $f_G^{(X)}(x)$. We then extend this result to density deconvolution over a bounded interval $I \subset \mathbb{R}$ by proving a uniform central limit theorem, which yields a sequential asymptotic credible band for $f_G^{(X)}$ on I. These new central limit theorems also apply to the direct density estimation problem, in which the X_i's are observed, thus extending the local uncertainty quantification results of Fortini and Petrone (2020) for the mixing distribution G to the induced density, both locally and uniformly.

We establish asymptotic guarantees for the estimate $f_{\widetilde{G}_n}^{(X)}$ under the frequentist modeling assumption that the X_i's are i.i.d. from a mixture model with kernel k and a "true" mixing distribution G^*. This frequentist framework was first considered by Tokdar et al. (2009) for the direct density estimation problem; see also Martin and Tokdar (2009); Martin (2012); Dixit and Martin (2023). First, under G^*, we prove that $f_{\widetilde{G}_n}^{(X)}$ is a consistent estimate of $f_{G^*}^{(X)}$ in L^1 and asymptotically agrees with the estimate that would be obtained if the X_i's were directly observed. Second, still under G^* and under additional regularity conditions, we derive a rate of convergence, in the L^1-Wasserstein distance, of the probability measure induced by $f_{\widetilde{G}_n}^{(X)}$ to that induced by $f_{G^*}^{(X)}$. We also establish that the associated distribution function $F_{\widetilde{G}_n}^{(X)}$ asymptotically merges, at an explicit rate, with the Bayesian nonparametric posterior mean estimate under a Dirichlet process mixture model (Ferguson, 1973; Lo, 1984).

We illustrate the proposed methodology through synthetic- and real-data analyses. In the synthetic-data experiments, we consider Gaussian location-scale mixtures under Laplace (ordinary-smooth) and Gaussian (super-smooth) noise distributions. We compare the resulting estimates with those obtained by a Bayesian nonparametric procedure based on the Dirichlet process mixture model, implemented using sequential Monte Carlo and batch algorithms, finding comparable empirical performance but a substantial computational advantage over both. We then analyze flow-cytometry data on reporter fluorescence in differentiating mouse embryonic stem cells. By retaining the acquisition order and considering nested prefixes of the data stream, this application illustrates sequential learning as observations become available. The resulting estimates are compared with the aforementioned sequential Bayesian nonparametric estimates and batch ridge-regularized Fourier deconvolution estimates.

1.3 Related work

Our work is connected to a growing literature on predictive or recursive inference, referred to as the "post-Bayes" framework. This literature includes recent developments on predictive distributions and martingale posteriors (Cappello, 2018; Hahn et al., 2018; Fong et al., 2023; Fong and Yiu, 2024; Holmes and Walker, 2023; Agnoletto et al., 2025; Battiston and Cappello, 2025; Fortini and Petrone, 2025; Yung et al., 2025), as well as on generalized Bayesian updates (Bissiri et al., 2016; Knoblauch et al., 2022). Although these contributions provide important insights and bring a broad range of procedures under a common quasi-Bayesian perspective, they have so far focused on direct problems, in which the X_i's are observed. By contrast, our work addresses the inverse setting, in which only the contaminated Y_i's are available, thereby extending the scope of predictive inference to a new class of problems.

1.4 Organization of the paper

Section 2 introduces the quasi-Bayesian learning model for density deconvolution, Section 3 develops sequential estimation and uncertainty quantification, and Section 4 establishes asymptotic frequentist guarantees for the resulting estimates. Sections 5 and 6 present, respectively, the synthetic- and real-data analyses. Section 7 concludes with a discussion of directions for future research. The Supplementary Material contains proofs and additional numerical illustrations.

2 Quasi-Bayesian density deconvolution

2.1 A learning process for density deconvolution

In the Bayesian framework, a likelihood and a prior specify a model, with the resulting predictive distributions determining the learning process, that is, how inference is updated as data become available. The predictive framework reverses this perspective by specifying the learning process directly through a sequence of predictive distributions; see Fortini and Petrone (2025) and the references therein. Under suitable conditions, this sequence implicitly determines both a prior and the corresponding (random) parameter of interest. The predictive distributions describe a learning mechanism rather than a data-generating process, so the observations need not be independent. We adopt this framework for density deconvolution.

Consider a stream of real-valued random variables $(Y_n)_{n\geq 1}$ that are generated according to

$$Y_n = X_n + Z_n \qquad n \geq 1, \tag{2}$$

where the unobserved variables $(X_n)_{n\geq 1}$ are independent of the noise variables $(Z_n)_{n\geq 1}$, with the Z_n's being i.i.d. with known density f_Z with respect to a σ-finite measure λ. Denoting by $(\mathcal{G}_n)_{n\geq 0}$ the natural filtration generated by $(Y_n)_{n\geq 1}$, for each $n \geq 0$: i) X_{n+1} is conditionally independent of $(X_{1:n}, Z_{1:n})$, given $\mathcal{G}_n$; ii) the conditional distribution of X_{n+1} given $\mathcal{G}_n$ is a mixture of a known positive kernel $k(\cdot \mid \theta)$, with $\theta \in \Theta \subset \mathbb{R}$, whose density with respect to λ is

$$f^{(X)}_{\widetilde{G}_n}(x) = \int_\Theta k(x \mid \theta)\widetilde{G}_n(\mathrm{d}\theta) \qquad x \in \mathbb{R}, \tag{3}$$

where the mixing distribution $\widetilde{G}_n$ on Θ is defined recursively by means of Newton's algorithm (Newton et al., 1998), i.e.,

$$\widetilde{G}_{n+1}(\mathrm{d}\theta) = (1-\widetilde{\alpha}_{n+1})\widetilde{G}_n(\mathrm{d}\theta) + \widetilde{\alpha}_{n+1}\frac{(f_Z * k(\cdot \mid \theta))(Y_{n+1})\widetilde{G}_n(\mathrm{d}\theta)}{\int_\Theta (f_Z * k(\cdot \mid \theta'))(Y_{n+1})\widetilde{G}_n(\mathrm{d}\theta')}, \tag{4}$$

given an initial guess $\widetilde{G}_0$ and a sequence $(\widetilde{\alpha}_n)_{n\geq 1}$ of real numbers in $(0,1)$ with $\sum_{n\geq 1}\widetilde{\alpha}_n = +\infty$ and $\sum_{n\geq 1}\widetilde{\alpha}_n^2 < +\infty$. The sequence $(\widetilde{\alpha}_n)_{n\geq 1}$ is referred to as the learning rate, for which a standard choice is $\widetilde{\alpha}_n = (\alpha+n)^{-\gamma}$ for $\alpha > 0$ and $\gamma \in (1/2, 1]$. See Fortini and Petrone (2020) for details.

From (2)–(4), for each $n \geq 0$ the density of the conditional distribution of Y_{n+1} given $\mathcal{G}_n$, with respect to λ, is

$$f^{(Y)}_{\widetilde{G}_n}(y) = \int_\Theta \widetilde{k}(y \mid \theta)\widetilde{G}_n(\mathrm{d}\theta) \qquad y \in \mathbb{R}, \tag{5}$$

where

$$\widetilde{k}(y \mid \theta) = (f_Z * k(\cdot \mid \theta))(y).$$

Since f_Z is known, $f^{(Y)}_{\widetilde{G}_n}$ is itself a mixture over Θ, with known convolution kernel $\widetilde{k}(y \mid \theta)$ and the same mixing distribution $\widetilde{G}_n$ defined in (4). In particular, $\widetilde{G}_n$ is updated as the new observation Y_{n+1} becomes available, by taking a weighted average, with respect to $\widetilde{\alpha}_{n+1}$, of $\widetilde{G}_n$ itself and of its posterior distribution, based on Y_{n+1}. The sequence of predictive densities $p_n(y) := f_{\tilde{G}_n}(y) = \int_\Theta \tilde{k}(y \mid \theta)\tilde{G}_n(\mathrm{d}\theta)$ characterizes the predictive learning mechanism for the observable Y_n's. The next proposition shows that this predictive specification induces a well-defined learning process for the (X_n, Z_n, Y_n)'s, and hence for the (X_n)'s; see B.2.1 for the proof.

Proposition 2.1. *There exists a probability space $(\Omega, \mathcal{F}, P)$ and three sequences of random variables $(X_n)_{n\geq 1}$, $(Y_n)_{n\geq 1}$ and $(Z_n)_{n\geq 1}$, which are defined on $(\Omega, \mathcal{F}, P)$, such that the following statements hold: i) (2), (4) and (5) hold true; ii) the Z_n's are independent of the X_n's and i.i.d. with known density f_Z with respect to λ; iii) for every $n \geq 1$, X_{n+1} is conditionally independent of $(X_{1:n}, Z_{1:n})$, given $\mathcal{G}_n$, and the conditional distribution of X_{n+1}, given $\mathcal{G}_n$ has density (3) with respect to λ; iv) the conditional distribution of X_{n+1} given $\{X_i, Y_i, Z_i : i \leq n\}$ is uniquely defined by (2), (4) and (5), as well as by the assumptions on $(Z_n)_{n\geq 1}$.*

2.2 Quasi-Bayesian properties of the learning process (2)–(4)

The next theorem, building on Fortini and Petrone (2020), shows that the learning process (2)–(4) is quasi-Bayesian, providing a predictive construction of a Bayesian model that is asymptotically equivalent as $n \to +\infty$. Whereas Fortini and Petrone (2020) considered direct density estimation with the X_n's observed, deconvolution requires an additional argument because only the contaminated Y_n's are observed. The predictive structure of $(Y_n)_{n\geq 1}$ fits the framework of Fortini and Petrone (2020), since the conditional distribution of Y_{n+1} given $\mathcal{G}_n$ is a mixture with mixing distribution $\widetilde{G}_n$. It remains to characterize the induced probabilistic structure of $(X_n)_{n\geq 1}$ and establish the required properties relative to the filtration $(\mathcal{G}_n)_{n\geq 0}$.

Theorem 2.2. *Consider the learning process (2)–(4), and assume that $k(x \mid \theta)$ is bounded and continuous in $\theta \in \Theta$ for every $x \in \mathbb{R}$ and that $\int \sup_{\theta\in\Theta} k(x \mid \theta)\lambda(dx) < +\infty$. As $n \to +\infty$, $\widetilde{G}_n$ converges weakly P-a.s. to a random probability distribution $\widetilde{G}$ on Θ, such that, by defining*

$$f^{(X)}_{\widetilde{G}}(x) = \int_\Theta k(x \mid \theta)\widetilde{G}(\mathrm{d}\theta) \qquad x \in \mathbb{R}, \tag{6}$$

the following statements hold: i) P-a.s. $f^{(X)}_{\widetilde{G}}$ is the limit pointwise and in $L^1(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ of $f^{(X)}_{\widetilde{G}_n}$; ii) $\int_A f^{(X)}_{\widetilde{G}}(x)\lambda(dx) = \lim_{n\to+\infty} \frac{1}{n}\sum_{k=1}^n I(X_k \in A)$; iii) for every $n \geq 0$, $k \geq 1$ and $A \in \mathcal{B}(\mathbb{R})$, $P(X_{n+k} \in A \mid \mathcal{G}_n) = \int_A f^{(X)}_{\widetilde{G}_n}(x)\lambda(dx) = E\left(\int_A f^{(X)}_{\widetilde{G}}(x)\lambda(dx)\middle| \mathcal{G}_n\right)$ with $f^{(X)}_{\widetilde{G}_n}$ as in (3); iv) for any $n \geq 1$, $P(X_n \in A) = \iint_A f^{(X)}_G(x)\lambda(dx)\pi(dG)$, where π is the distribution of $\widetilde{G}$.

See B.2.2 for the proof of Theorem 2.2. It follows from (2)–(4) that, for every $n \geq 0$, the random variables $(X_{n+k})_{k\geq 1}$ are conditionally identically distributed given $\mathcal{G}_n$, the information available after the n-th step. Since $(X_{n+k})_{k\geq 1}$ and $(X_i, Z_i)_{i\leq n}$ are conditionally independent given $\mathcal{G}_n$, the sequence $(X_n)_{n\geq 1}$ is conditionally identically distributed in the sense of Berti et al. (2004), with directing random probability measure $f^{(X)}_{\widetilde{G}}(x)\lambda(dx)$. Consequently, by Berti et al. (2004, Theorem 2.2), conditionally on the random limit $\widetilde{G}$, the sequence $(X_n)_{n\geq 1}$ is asymptotically i.i.d. with density $f^{(X)}_{\widetilde{G}}$ relative to λ, while its empirical distribution converges weakly, P-almost surely,

to the random probability measure with density $f^{(X)}_{\widetilde{G}}(x) = \int_\Theta k(x \mid \theta)\widetilde{G}(\mathrm{d}\theta)$. This is precisely the quasi-Bayesian framework of Fortini and Petrone (2020), in which $f^{(X)}_{\widetilde{G}}$ is the random mixture density assigned in the infinite-sample limit, making $\widetilde{G}$ the natural parameter of interest in the density deconvolution problem.

3 Sequential estimation and uncertainty quantification

3.1 Preliminaries

By Theorem 2.2, the density $f^{(X)}_{\widetilde{G}}$ in (6) is the inferential target of the quasi-Bayesian learning process (2)–(4). Theorem 2.2 also shows that $f^{(X)}_{\widetilde{G}_n}$ in (3), is a natural sequential estimate of $f^{(X)}_{\widetilde{G}}$. Unlike in a Bayesian model, however, uncertainty about $f^{(X)}_{\widetilde{G}}$ cannot be quantified analytically, since $f^{(X)}_{\widetilde{G}}$ is defined implicitly through the limiting mixing distribution $\widetilde{G}$, rather than through a prior specification. Within the quasi-Bayesian learning process (2)–(4), the conditional distribution of $f^{(X)}_{\widetilde{G}}$, given $Y_{1:n}$, plays the role of the posterior distribution, with $f^{(X)}_{\widetilde{G}_n}$ as its center. We establish local and uniform approximations of this conditional distribution, leading to asymptotic credible intervals and bands for $f^{(X)}_{\widetilde{G}}$. Approximations arise from central limit theorems under the following assumptions: A1) $\mathrm{Supp}(\widetilde{G}_0) = \Theta$, with Supp denoting closed support; A2) λ is the Lebesgue measure on $\mathbb{R}$; A3) $k(x \mid \cdot) > 0$, bounded and continuous for every $x \in \mathbb{R}$, and $\int \sup_{\theta\in\Theta} k(x \mid \theta)\lambda(dx) < +\infty$; A4) $\int_\Theta k(\cdot \mid \theta)G(\mathrm{d}\theta)$ and $\int_\Theta \widetilde{k}(y \mid \theta)G(\mathrm{d}\theta)$ identify G; A5) f_Z is strictly positive on $\mathbb{R}$ and bounded.

The assumptions A1)–A5) imply that $\widetilde{k}(y \mid \theta) > 0$ for every $y \in \mathbb{R}$ and $\theta \in \Theta$. For the kernel $\widetilde{k}(y \mid \theta) = (f_Z * k(\cdot \mid \theta))(y)$, A3)–A5) hold, for instance, for the Gaussian kernel k with bounded expectation and variance bounded away from zero, and with f_Z the density of either a Laplace (ordinary-smooth) or Gaussian (super-smooth) distribution. A closed form for $\widetilde{k}$ is not required but improves computational efficiency; otherwise, it can be evaluated numerically once, before the sequential updates, so that no additional integration is required during the recursion.

Remark 3.1. *The assumption $k(x \mid \theta) > 0$ for all x and θ can be relaxed to allow θ-dependent supports. In this case, (3) and its local and uniform asymptotic properties remain valid on regions where $f^{(X)}_{\widetilde{G}}$ is strictly positive. We retain the positivity assumption to simplify the statements.*

The central limit theorems below exploit the martingale structure associated with the X_n's being conditionally identically distributed, as established in Theorem 2.2, together with a suitable almost-sure conditional convergence result for martingales; we refer to B.1.1 for details.

3.2 Asymptotic credible intervals

We establish an asymptotic Gaussian approximation of the conditional distribution of $f^{(X)}_{\widetilde{G}}$, given $Y_{1:n}$, at a fixed point $x \in \mathbb{R}$, i.e. local approximation. According to Theorem 2.2, under A1)–A5), $f^{(X)}_{\widetilde{G}_n}(x)$ converges P-a.s. to $f^{(X)}_{\widetilde{G}}(x)$, and, for every $y \in \mathbb{R}$, $f^{(Y)}_{\widetilde{G}_n}(y)$ converges P-a.s. to

$$f^{(Y)}_{\widetilde{G}}(y) = \int_\Theta \widetilde{k}(y \mid \theta)\widetilde{G}(\mathrm{d}\theta). \tag{7}$$

Define the limiting conditional density of $(X_n)_{n\geq 1}$, given $(Y_n)_{n\geq 1}$, as

$$f^{(X)}_{\widetilde{G}}(x \mid y) = \int_{\Theta} k(x \mid \theta)\widetilde{G}(\mathrm{d}\theta \mid y), \tag{8}$$

with

$$\widetilde{G}(\mathrm{d}\theta \mid y) = \frac{\widetilde{k}(y \mid \theta)\widetilde{G}(\mathrm{d}\theta)}{f^{(Y)}_{\widetilde{G}}(y)}. \tag{9}$$

By A3) and A5), $f^{(X)}_{\widetilde{G}}(x)$, $f^{(Y)}_{\widetilde{G}}(y)$ and $f^{(X)}_{\widetilde{G}}(x \mid y)$ are strictly positive for every x and y. The next theorem provides a local central limit theorem, at a fixed point $x \in \mathbb{R}$, for $f^{(X)}_{\widetilde{G}}$ centered at $f^{(X)}_{\widetilde{G}_n}$.

Theorem 3.2. *For $x \in \mathbb{R}$, let $f^{(X)}_{\widetilde{G}_n}(x)$ and $f^{(X)}_{\widetilde{G}}(x)$ be defined as in (3) and (6), respectively. Under A1)–A5), if $(b_n)_{n\geq 1}$ is a sequence of strictly positive numbers such that $b_n \uparrow +\infty$, $\sum_{n\geq 1} \widetilde{\alpha}_n^4 b_n^2 < +\infty$, $b_n \sup_{k\geq n} \widetilde{\alpha}_k^2 \to 0$ and $b_n \sum_{k\geq n} \widetilde{\alpha}_k^2 \to 1$, then, for λ-almost every $x \in \mathbb{R}$, as $n \to +\infty$ the conditional distribution of $b_n^{1/2}(f^{(X)}_{\widetilde{G}}(x) - f^{(X)}_{\widetilde{G}_n}(x))$, given $\mathcal{G}_n$, converges P-a.s. to a centered Gaussian kernel with random variance function $v(x)$. In particular, there holds*

$$v(x) = \int \left(f^{(X)}_{\widetilde{G}}(x \mid y) - f^{(X)}_{\widetilde{G}}(x)\right)^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(\mathrm{d}y), \tag{10}$$

with $f^{(X)}_{\widetilde{G}}$, $f^{(Y)}_{\widetilde{G}}$ and $f^{(X)}_{\widetilde{G}}(\cdot \mid y)$ defined in (6), (7) and (8), respectively. In particular, if the learning rate is $\widetilde{\alpha}_n = (\alpha + n)^{-\gamma}$, with $\alpha > 0$ and $\gamma \in (1/2, 1]$, then $b_n = (2\gamma - 1)n^{2\gamma - 1}$, for $n \geq 1$.

See B.3.1 for the proof of Theorem 3.2. If $(\widetilde{\alpha}_n)_{n\geq 1}$ is non-increasing, then $b_n \sup_{k\geq n} \widetilde{\alpha}_k^2 \to 0$ follows from $\sum_{n\geq 1} \widetilde{\alpha}_n^4 b_n^2 < +\infty$. A typical choice for $(\widetilde{\alpha}_n)_{n\geq 1}$ is given by $\widetilde{\alpha}_n = 1/(1+n)$, for which $b_n = n$.

Let Φ_{0,σ^2} denote the cumulative distribution function of a Gaussian random variable with mean 0 and variance σ^2, with the proviso that $\Phi_{0,0}$ corresponds to the case of a distribution degenerate at zero. Under the assumptions of Theorem 3.2, for every $z \neq 0$, as $n \to +\infty$

$$P\left(b_n^{1/2}(f^{(X)}_{\widetilde{G}}(x) - f^{(X)}_{\widetilde{G}_n}(x)) \leq z \,\middle|\, \mathcal{G}_n\right) \to \Phi_{0,v(x)}(z), \qquad P\text{-a.s.} \tag{11}$$

Since the variance $v(x)$ depends on the unknown mixing distribution $\widetilde{G}$, (11) cannot be applied to constructing credible intervals for $f^{(X)}_{\widetilde{G}}(x)$. Then, we introduce the empirical variance $v_n(\cdot)$, which is defined as $v(\cdot)$ in (10) with $\widetilde{G}_n$ in place of $\widetilde{G}$; accordingly we set $f^{(X)}_{\widetilde{G}_n}(\cdot \mid y)$ and $\widetilde{G}_n(\cdot \mid y)$ as $f^{(X)}_{\widetilde{G}}(\cdot \mid y)$ in (8) and $\widetilde{G}(\cdot \mid y)$ in (9), respectively, with $\widetilde{G}_n$ in place of $\widetilde{G}$. In particular, as $n \to +\infty$, $v_n(x)$ converges to $v(x)$, P-a.s.; we refer to Lemma B.7 in B.3 for details.

Let $(b_n)_{n\geq 1}$ be a sequence of strictly positive numbers as in Theorem 3.2. Then, for every $t_1, t_2 \in \mathbb{R}$, as $n \to +\infty$

$$E\left(e^{it_1 b_n^{1/2}\left(f^{(X)}_{\widetilde{G}}(x) - f^{(X)}_{\widetilde{G}_n}(x)\right) + it_2 v_n(x)} \,\middle|\, \mathcal{G}_n\right) \to e^{it_1^2 v(x) + it_2 v(x)}, \qquad P\text{-a.s.}$$

Therefore, by an application of the Portmanteau theorem, for every $z > 0$ and every $\epsilon > 0$, it holds that

$$\liminf_{n\to+\infty} P\left(b_n^{1/2}\left|f^{(X)}_{\widetilde{G}}(x) - f^{(X)}_{\widetilde{G}_n}(x)\right| < z(\max(v_n(x), \epsilon))^{1/2} \,\middle|\, \mathcal{G}_n\right) \geq 2\Phi_{0,1}(z) - 1.$$

If $z_{1-\beta/2}$ is the $(1-\beta/2)$-quantile of the standard Gaussian distribution, with $\beta \in (0,1)$, then, as $n \to +\infty$

$$\liminf_{n\to+\infty} P\left(\left|f^{(X)}_{\widetilde{G}}(x) - f^{(X)}_{\widetilde{G}_n}(x)\right| \le b_n^{-1/2} z_{1-\beta/2}\big(\max\{v_n(x),\epsilon\}\big)^{1/2} \,\middle|\, \mathcal{G}_n\right) \ge 1-\beta,$$

i.e., for every $\epsilon > 0$, a sequential asymptotic credible interval for $f^{(X)}_{\widetilde{G}}(x)$, with level at least $(1-\beta)$ is

$$f^{(X)}_{\widetilde{G}_n}(x) \pm b_n^{-1/2} z_{1-\beta/2}\big(\max\{v_n(x),\epsilon\}\big)^{1/2}. \tag{12}$$

3.3 Asymptotic credible bands

We establish a Gaussian approximation of the conditional distribution of $f^{(X)}_{\widetilde{G}}$, given $Y_{1:n}$, on a bounded interval $I \subset \mathbb{R}$, i.e. uniform approximation. In addition to A1)–A5), we assume: A6) $k(x \mid \theta)$ is continuously differentiable with respect to x, with $\sup_{\theta\in\Theta,\, x\in\mathbb{R}} |k(x \mid \theta)| < +\infty$ and $\sup_{\theta\in\Theta,\, x\in\mathbb{R}} |\frac{\mathrm{d}}{\mathrm{d}x} k(x \mid \theta)| < +\infty$. Let $C(\mathbb{R})$ be the space of continuous functions on $\mathbb{R}$, endowed with the topology of uniform convergence on compact sets. Under A1)–A6), $f^{(X)}_{\widetilde{G}_n}(\omega)$ is continuous for every $\omega \in \Omega$. Then, as $n \to +\infty$, $f^{(X)}_{\widetilde{G}_n} \to f^{(X)}_{\widetilde{G}}$ P-a.s., i.e. as $n \to +\infty$

$$\sup_{x\in I} \left|f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}}(x)\right| \to 0, \qquad P\text{-a.s.}$$

The next theorem provides a uniform central limit theorem, on $I \subset \mathbb{R}$, for $f^{(X)}_{\widetilde{G}}$ centered at $f^{(X)}_{\widetilde{G}_n}$. The limiting Gaussian process is characterized by a centered Gaussian process kernel.

Theorem 3.3. *Let $f^{(X)}_{\widetilde{G}_n}$ and $f^{(X)}_{\widetilde{G}}$ be defined as in (3) and (6), respectively. Under A1)–A6), if $(b_n)_{n\ge1}$ is a sequence of strictly positive numbers such that $b_n \uparrow +\infty$, $\sum_{n\ge1} \widetilde{\alpha}_n^4 b_n^2 < +\infty$, $b_n \sup_{k\ge n} \widetilde{\alpha}_k^2 \to 0$ and $b_n \sum_{k\ge n} \widetilde{\alpha}_k^2 \to 1$, then, as $n \to +\infty$ the conditional distribution of $b_n^{1/2}(f^{(X)}_{\widetilde{G}} - f^{(X)}_{\widetilde{G}_n})$, given $\mathcal{G}_n$, converges P-a.s. in $C(\mathbb{R})$ to a centered Gaussian process kernel with random covariance function $R(x_1,x_2)$, for every $(x_1,x_2) \in \mathbb{R}^2$. In particular, there holds*

$$R(x_1,x_2) = \int (f^{(X)}_{\widetilde{G}}(x_1 \mid y) - f^{(X)}_{\widetilde{G}}(x_1))(f^{(X)}_{\widetilde{G}}(x_2 \mid y) - f^{(X)}_{\widetilde{G}}(x_2)) f^{(Y)}_{\widetilde{G}}(y)\lambda(\mathrm{d}y), \tag{13}$$

with $f^{(X)}_{\widetilde{G}}$, $f^{(Y)}_{\widetilde{G}}$ and $f^{(X)}_{\widetilde{G}}(\cdot \mid y)$ defined in (6), (7) and (8), respectively. In particular, if the learning rate is $\widetilde{\alpha}_n = (\alpha+n)^{-\gamma}$, with $\alpha > 0$ and $\gamma \in (1/2, 1]$, then $b_n = (2\gamma-1)n^{2\gamma-1}$, for $n \ge 1$.

See B.3.2 for the proof of Theorem 3.3. The random covariance function R in (13) is measurable with respect to the σ-algebra generated by $\widetilde{G}$, and it is positive semi-definite for every $\widetilde{G}$. Indeed, for every $J \ge 1$ and for every $x_1, \ldots, x_J \in \mathbb{R}$ and $u_1, \ldots, u_J \in \mathbb{R}$, there holds

$$\begin{aligned}
&\sum_{i,j=1}^{J} u_i u_j \int_{\mathbb{R}} \left(f^{(X)}_{\widetilde{G}}(x_i \mid y) - f^{(X)}_{\widetilde{G}}(x_i)\right)\left(f^{(X)}_{\widetilde{G}}(x_j \mid y) - f^{(X)}_{\widetilde{G}}(x_j)\right) f^{(Y)}_{\widetilde{G}}(y)\lambda(\mathrm{d}y) \\
&\quad = \int_{\mathbb{R}} \left(\sum_{i=1}^{J} u_i (f^{(X)}_{\widetilde{G}}(x_i \mid y) - f^{(X)}_{\widetilde{G}}(x_i))\right)^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(\mathrm{d}y) \ge 0.
\end{aligned}$$

For each possible realization G of $\widetilde{G}$, let $\mathbb{G}_G$ be a Gaussian process in $C(\mathbb{R})$, independent of $\widetilde{G}$, and with covariance function as in (13) with G in the place of $\widetilde{G}$. The next theorem provides an upper bound for the fluctuations of $|\mathbb{G}_{\widetilde{G}}|$ on a bounded interval I. Let $\sigma(I)$ and $\psi(z)$ be defined as:

$$\sigma(I) = \left(\sup_{x \in I} \int (f^{(X)}_{\widetilde{G}}(x \mid y) - f^{(X)}_{\widetilde{G}}(x))^2 f^{(Y)}_{\widetilde{G}}(y) \lambda(\mathrm{d}y) \right)^{1/2} \tag{14}$$

and

$$\psi(z) = \sup_{\substack{x_1, x_2 \in I, \\ |x_1 - x_2| < z}} \left(\int (f^{(X)}_{\widetilde{G}}(x_1 \mid y) - f^{(X)}_{\widetilde{G}}(x_2 \mid y))^2 f^{(Y)}_{\widetilde{G}}(y) \lambda(\mathrm{d}y) - (f^{(X)}_{\widetilde{G}}(x_1) - f^{(X)}_{\widetilde{G}}(x_2))^2 \right)^{1/2}. \tag{15}$$

The upper bound for the fluctuations of $|\mathbb{G}_{\widetilde{G}}|$ follows by bounding $E(\sup_{x \in I} \mathbb{G}_{\widetilde{G}}(x) \mid \widetilde{G})$ and then by applying a concentration inequality for Gaussian processes to bound $\mathbb{G}_{\widetilde{G}}(x)$, uniformly in $x \in I$, with respect to its conditional distribution, given $\widetilde{G}$. We refer to B.1.2 for details.

Theorem 3.4. *Let $\psi^{-1}(t) = \inf\{z : \psi(z) > t\}$ denote the inverse of the function ψ in (15). For every $\beta \in (0,1)$,*

$$P\left(\sup_{x \in I} |\mathbb{G}_{\widetilde{G}}(x)| \leq s_{\widetilde{G}}(I, \beta) \,\middle|\, \widetilde{G} \right) \geq 1 - \beta,$$

with

$$s_{\widetilde{G}}(I, \beta) = 12 \int_0^{\sigma(I)} \left(\log\left(1 + \frac{\lambda(I)}{2\psi^{-1}(z/2)} \right) \right)^{1/2} dz + \sigma(I) \sqrt{2 |\log(\beta/2)|}. \tag{16}$$

See B.3.3 for the proof of Theorem 3.4. We now provide an asymptotic credible band for $f^{(X)}_{\widetilde{G}}$. From Theorem 3.3,

$$P\left(\sup_{x \in I} b_n^{1/2} \left| f^{(X)}_{\widetilde{G}}(x) - f^{(X)}_{\widetilde{G}_n}(x) \right| \in \cdot \,\middle|\, \mathcal{G}_n \right) \xrightarrow{w} P\left(\sup_{x \in I} |\mathbb{G}_{\widetilde{G}}(x)| \in \cdot \,\middle|\, \widetilde{G} \right) \qquad P\text{-a.s.},$$

where, from Theorem 3.4, $P\left(\sup_{x \in I} |\mathbb{G}_{\widetilde{G}}(x)| \leq s_{\widetilde{G}}(I, \beta) \,\middle|\, \widetilde{G}\right) \geq 1 - \beta$ for every $\beta \in (0,1)$, with $s_{\widetilde{G}}(I, \beta)$ as in (16). Now, consider the empirical version of $s_{\widetilde{G}}(I, \beta)$, which is defined as

$$s_n(I, \beta) = 12 \int_0^{\sigma_n(I)} \left(\log\left(1 + \frac{\lambda(I)}{2\psi_n^{-1}(z/2)} \right) \right)^{1/2} dz + \sigma_n(I) \sqrt{2 |\log(\beta/2)|}, \tag{17}$$

where we set $\sigma_n(\cdot)$ and $\psi_n(\cdot)$ as $\sigma(\cdot)$ in (14) and $\psi(\cdot)$ in (15), respectively, with $\widetilde{G}_n$ in place of $\widetilde{G}$.

Theorem 3.5. *If $s_{\widetilde{G}}(I, \beta)$ and $s_n(I, \beta)$ are defined as in (16) and (17), respectively, then as $n \to +\infty$*

$$s_n(I, \beta) \to s_{\widetilde{G}}(I, \beta) \qquad P\text{-a.s.}$$

See B.3.4 for the proof of Theorem 3.5. Let $(b_n)_{n \geq 1}$ be a sequence of numbers as defined in Theorem 3.3. Because of the limiting behaviour of $s_n(I, \beta)$ in Theorem 3.5, for every $\epsilon > 0$ and every $\beta \in (0,1)$,

$$\liminf_{n \to +\infty} P\left(\sup_{x \in I} \left| f^{(X)}_{\widetilde{G}}(x) - f^{(X)}_{\widetilde{G}_n}(x) \right| < b_n^{-1/2} \max(s_n(I, \beta), \epsilon) \,\middle|\, \mathcal{G}_n \right) \geq 1 - \beta. \tag{18}$$

See B.3.5 for details on Equation (18). It follows by an application of (18) that, for every $\epsilon > 0$ and every $\beta \in (0,1)$, a sequential asymptotic credible band for $f^{(X)}_{\widetilde{G}}$, on I, with level at least $(1 - \beta)$ is

$$f^{(X)}_{\widetilde{G}_n} \pm b_n^{-1/2} \max(s_n(I, \beta), \epsilon). \tag{19}$$

4 Frequentist properties

4.1 Preliminaries

We establish asymptotic guarantees for the estimate $f^{(X)}_{\widetilde{G}_n}$ under the frequentist modeling assumption that the X_n's are i.i.d. from a mixture model with kernel k and a "true" mixing distribution G^*, i.e.,

$$f^{(X)}_{G^*}(x) = \int_\Theta k(x \mid \theta) G^*(\mathrm{d}\theta) \qquad x \in \mathbb{R}. \tag{20}$$

To underline that the X_n's are i.i.d., we denote by P^* the probability measure on $(\Omega, \mathcal{F})$ under which the X_n's are defined. To establish asymptotic frequentist guarantees we assume A1)–A5) from Section 3.1, together with these assumptions: F1) $\widetilde{G}_0$ is absolutely continuous with respect to a σ-finite measure μ and G^* belongs to the weak closure of the class of probability measures on Θ that are absolutely continuous with respect to μ; F2) Θ is compact; F3) $\int \sup_{\theta_1,\theta_2\in\Theta} \left(\frac{\widetilde{k}(y|\theta_1)}{\widetilde{k}(y|\theta_2)}\right)^2 f^{(Y)}_{G^*}(y)\lambda(\mathrm{d}y) < +\infty$, with $f^{(Y)}_{G^*} = f_Z * f^{(X)}_{G^*}$ and $f^{(X)}_{G^*}$ defined as in (20).

4.2 Consistency and agreement with the direct density estimation problem

The next proposition provides consistency of $f^{(X)}_{\widetilde{G}_n}$ in L^1, under the "true" G^*. This result is an immediate consequence of Martin and Tokdar (2009, Corollary 4.7), where consistency was first established for the direct density estimation problem, in which the X_n's are observed.

Proposition 4.1. *Let $(X_n)_{n\geq 1}$, $(Y_n)_{n\geq 1}$, and $(Z_n)_{n\geq 1}$ be three sequences of random variables defined on a probability space $(\Omega, \mathcal{F}, P^*)$, such that the X_n's are i.i.d. with density $f^{(X)}_{G^*}$ in (20), the Z_n's are i.i.d. with density f_Z and independent of the X_n, and $Y_n = X_n + Z_n$ for all $n \geq 1$. Let $f^{(X)}_{\widetilde{G}_n}$ be defined as in (3), where $\widetilde{G}_n$ satisfies Newton's recursion (4). If A1)–A5) and F1)–F3) hold true, then $\widetilde{G}_n$ converges weakly to G^*, and $f^{(X)}_{\widetilde{G}_n}$ converges to $f^{(X)}_{G^*}$ in L^1, P^*-a.s.*

See B.4.1 for the proof of Proposition 4.1. Martin and Tokdar (2009) also consider model misspecification, proving that when the "true" density lies outside the mixture model with kernel k, the recursive estimate converges almost surely in $L^1(\mathbb{R})$ to the mixture minimizing its Kullback–Leibler divergence from that density. Extending this result to deconvolution would require relating the Kullback–Leibler divergences between observed convolution densities and their latent mixture densities. Such a relationship is generally unavailable because convolution is a smoothing operator: substantially different latent densities may induce very similar convolution densities. Accordingly, we restrict our analysis to the correctly specified setting.

Proposition 4.1 makes it possible to establish that the estimates obtained from the observed Y_n's agree asymptotically with those that would be obtained from the unobserved X_n's. In particular, the quasi-Bayesian learning process on the X_n's is formulated, for $n \geq 0$, as

$$X_{n+1} \mid \mathcal{G}^a_n \sim \int_\Theta k(\cdot \mid \theta)\, G_n(\mathrm{d}\theta), \tag{21}$$

where $\mathcal{G}^a_n$ is the "augmented" σ-algebra generated by $((X_i, Y_i, Z_i))_{i\leq n}$, and G_n is updated recursively by Newton's algorithm, i.e.,

$$G_{n+1}(\mathrm{d}\theta) = (1-\alpha_{n+1})G_n(\mathrm{d}\theta) + \alpha_{n+1}\frac{k(X_{n+1} \mid \theta)\, G_n(\mathrm{d}\theta)}{\int_\Theta k(X_{n+1} \mid \theta')\, G_n(\mathrm{d}\theta')}, \tag{22}$$

given an initial guess G_0 and a learning rate $(\alpha_n)_{n\geq 1}$ with $\sum_{n\geq 1}\alpha_n = +\infty$ and $\sum_{n\geq 1}\alpha_n^2 < +\infty$. Note that α_n in (22) may differ from $\widetilde{\alpha}_n$ in (4), and that k is used in place of the convolution kernel $\widetilde{k}$.

Equations (21) and (22) imply that, for every $n \geq 0$, X_{n+1} is conditionally independent of $((Z_i, Y_i))_{i\leq n}$ given $(X_i)_{i\leq n}$. This reflects the natural assumption that, once $X_1, \ldots, X_n$ are observed, the corresponding noise-contaminated observations provide no additional information for learning. Under the "true" G^*, the next corollary shows that the estimate $f^{(X)}_{\widetilde{G}_n}$ asymptotically agrees with $f^{(X)}_{G_n}$, obtained from the quasi-Bayesian learning process (21)–(22).

Corollary 4.2. *Consider the quasi-Bayesian learning processes in (3)–(4) and (21)–(22). Let the assumptions of Proposition 4.1 hold true, let* $\mathrm{Supp}(G_0) = \Theta$ *and* G_0 *be absolutely continuous with respect to* μ*, and let* $\int \sup_{\theta_1,\theta_2\in\Theta}\left(\frac{k(x|\theta_1)}{k(x|\theta_2)}\right)^2 f^{(X)}_{G^*}(x)\,\lambda(\mathrm{d}x) < +\infty$*. Then, as* $n \to +\infty$

$$\int \left| f^{(X)}_{G_n}(x) - f^{(X)}_{\widetilde{G}_n}(x)\right| \lambda(\mathrm{d}x) \to 0 \qquad P^*\text{-a.s.}$$

See B.4.2 for the proof of Corollary 4.2. Corollary 4.2 may also be used to motivate an optional Monte Carlo simulation-based calibration of the learning rate $\widetilde{\alpha}_n$; we refer to D.1 for details.

4.3 Merging with the Dirichlet process mixture model

Under the "true" G^*, and assuming a Laplace noise distribution, the next proposition provides a rate of convergence of the distribution induced by the estimate $f^{(X)}_{\widetilde{G}_n}$, to the one induced by $f^{(X)}_{G^*}$, in the L^1-Wasserstein metric. This metric is motivated by the inversion inequality of Rousseau and Scricciolo (2024, Theorem 3.1), which relates the L^1 error between $f^{(Y)}_{\widetilde{G}_n}$ and $f^{(Y)}_{G^*}$ to the L^1-Wasserstein distance between the distributions induced by $f^{(X)}_{\widetilde{G}_n}$ and $f^{(X)}_{G^*}$. We write $W_1(F_1, F_2)$ for the Wasserstein distance between the probability measures on $\mathcal{B}(\mathbb{R})$ associated with the distributions F_1 and F_2; it holds $W_1(F_1, F_2) = \int_{\mathbb{R}} |F_1(x) - F_2(x)|\,dx$.

Proposition 4.3. *Let* $(X_n)_{n\geq 1}$*,* $(Y_n)_{n\geq 1}$*, and* $(Z_n)_{n\geq 1}$ *be three sequences of random variables defined on a probability space* $(\Omega, \mathcal{F}, P^*)$*, such that the* X_n *are i.i.d. with density* $f^{(X)}_{G^*}$ *in (20), the* Z_n *are i.i.d. with Laplace distribution and independent of the* X_n*, and* $Y_n = X_n + Z_n$ *for all* $n \geq 1$*. Let* $f^{(X)}_{\widetilde{G}_n}$ *be defined as in (3), where* $\widetilde{G}_n$ *satisfies Newton's recursion (4) with* $\widetilde{\alpha}_n \sim n^{-\gamma}$ *such that* $\gamma \in (2/3, 1)$*. Let A1)–A5) and F1)–F3) hold true and, in addition, assume that:*

(i) there exists $\kappa > 0$ *and* $M_\kappa \in L^1(\mathbb{R})$ *such that*

$$\sup_{\theta\in\Theta}\left|\partial^\ell_x k(x+\delta \mid \theta) - \partial^\ell_x k(x \mid \theta)\right| \leq M_\kappa(x)|\delta|^{\kappa-\ell}, \qquad x, \delta \in \mathbb{R},, \tag{23}$$

where $\ell := \max\{j \in \mathbb{N}_0 : j < \kappa\}$*;*

(ii) there exists $a \in (0, 1)$ *such that*

$$\sup_{\theta\in\Theta}\int_{\mathbb{R}} e^{a|x|}k(x \mid \theta)\,dx < +\infty. \tag{24}$$

Then, denoting by $F^{(X)}_{\widetilde{G}_n}$ *and* $F^{(X)}_{G^*}$ *the distribution functions induced by* $f^{(X)}_{\widetilde{G}_n}$ *and* $f^{(X)}_{G^*}$ *respectively, it holds*

$$W_1\left(F^{(X)}_{\widetilde{G}_n}, F^{(X)}_{G^*}\right) = o_{P^*}\left(n^{-\frac{(1-\gamma)(\kappa+1)}{2(\kappa+2)}}\right). \tag{25}$$

See B.4.3 for the proof of Proposition 4.3. Besides being of independent interest, Proposition 4.3 provides the key ingredient for establishing that, still under the "true" G^*, the quasi-Bayesian distribution function (estimate) $F^{(X)}_{\widetilde{G}_n}$ asymptotically merges, at an explicit rate in the L^1-Wasserstein metric, with the corresponding Bayesian nonparametric posterior mean estimate under a Dirichlet process mixture model (Ferguson, 1973; Lo, 1984), as shown next. To establish this merging result, we consider the Dirichlet process Gaussian-mixture model

$$\begin{aligned} &Y_i = X_i + Z_i, \quad X_i \mid \mu_i, \sigma^2 \overset{\text{ind}}{\sim} \phi(\cdot \mid \mu_i, \sigma^2), \quad Z_i \overset{\text{iid}}{\sim} \text{Laplace}(0, b), \qquad i = 1, \ldots, n, \\ &\mu_i \mid G \overset{\text{iid}}{\sim} G, \quad G \sim \text{DP}(H_0), \quad \sigma \sim \Pi_\sigma. \end{aligned} \tag{26}$$

where $\phi(\cdot \mid \mu, \sigma^2)$ is the Gaussian density with mean μ and variance σ^2, σ is independent of (μ, G), $DP(\cdot)$ is the law of a Dirichlet process with non-atomic (diffuse) base measure H_0, and Π_σ is the prior distribution for σ. Under the model (26), we denote by $\widehat{F}^B_n$ the Bayesian nonparametric posterior mean estimate, i.e. $\widehat{F}^B_n(x) = E_\Pi(F^{(X)}_{G,\sigma}(x) \mid Y_{1:n})$, where $F^{(X)}_{G,\sigma}(x) = P_\Pi(X_i \leq x \mid G, \sigma)$. The posterior contraction rates of Rousseau and Scricciolo (2024, Theorem 4.2 and Theorem 4.4.) do not require the "true" density to admit a mixture representation. In contrast, the merging result in the next corollary relies on Proposition 4.3, whose proof assumes that the "true" density admits a mixture representation, i.e. it is correctly specified.

Corollary 4.4. *Suppose that the assumptions of Proposition 4.3 hold true, and, in addition, assume that:*

(i) the density $f^{(X)}_{G^*}$ satisfies, for some constants $C_0, C_0' > 0$ and $\iota > 1$,

$$f^{(X)}_{G^*}(x) \leq C_0' \exp\{-(1 + C_0)|x|^\iota\}, \qquad x \in \mathbb{R}; \tag{27}$$

(ii) the prior Π_σ has a continuous density π_σ on $(0, +\infty)$ such that, for some constants $D_1, D_1', D_2, D_2' > 0$ and $s_1, s_2, t_1, t_2 \geq 0$,

$$D_1' \sigma^{-s_1} \exp\{-D_1 \sigma^{-1} |\log \sigma|^{t_1}\} \leq \pi_\sigma(\sigma) \leq D_2' \sigma^{-s_2} \exp\{-D_2 \sigma^{-1} |\log \sigma|^{t_2}\} \tag{28}$$

for all sufficiently small $\sigma > 0$. Moreover, for some constants $D_3, D_3' > 0$ and $\varrho > 1$,

$$\Pi_\sigma([\bar{\sigma}, +\infty)) \leq D_3' \exp\{-D_3 \bar{\sigma}^\varrho\}, \qquad \bar{\sigma} \to +\infty; \tag{29}$$

(iii) the base measure H_0 has a continuous and strictly positive density h_0 such that, for some constants $b_0, b_0', c_0, c_0' > 0$,

$$c_0 \exp\{-b_0 |u|^\iota\} \leq h_0(u) \leq c_0' \exp\{-b_0' |u|^\iota\}, \qquad u \in \mathbb{R}, \tag{30}$$

where $\iota > 1$ is as in (27).

Then,

$$W_1\left(F^{(X)}_{\widetilde{G}_n}, \widehat{F}^B_n\right) = o_{P^*}\left(n^{-\frac{(1-\gamma)(\kappa+1)}{2(\kappa+2)}}\right). \tag{31}$$

See B.4.4 for the proof of Corollary 4.4.

5 Synthetic-data analysis

5.1 Unimodal example

For each $n \in \{250; 500; 750; 1,000\}$ we generate random variables $X_1, \ldots, X_n$ i.i.d. according to the density $f^{(X)}(\cdot) = \phi(\cdot \mid 2, 2)$, where $\phi(\cdot \mid \mu, \sigma^2)$ denotes the Gaussian density with mean μ and variance σ^2. Using the mixture representation (1), we write $f^{(X)}$ as $f^{(X)}_{G^*}$, with $G^* = \delta_{(2,2)}$ the true mixing distribution. We consider the Laplace noise distribution for the random variables Z_i's, namely $Z_i \overset{\text{iid}}{\sim} \text{Laplace}(0, b_l)$ with $b_l = \frac{\sigma_l}{\sqrt{2}}$ and $\sigma_l \in \{0.25, 0.50\}$, where σ_l is the standard deviation of the Laplace noise. The Z_i's are independent of the X_i's, and the observations are modeled as follows: $Y_i = X_i + Z_i$, for $i = 1, \ldots, n$. For each n, the same $Y_{1:n} = (Y_1, \ldots, Y_n)$ is used for all the methods under comparison in this section.

We implement Newton's algorithm (4) with $k(\cdot \mid \theta) = \phi(\cdot \mid \mu, \sigma^2)$, where $\theta = (\mu, \sigma^2) \in \Theta = [-20, 20] \times [0.1, 5]$. We use the learning rate $\widetilde{\alpha}_i = (1+i)^{-1}$, for $i \geq 1$, and set $\widetilde{G}_0$ to the Uniform distribution on Θ. Numerical integrals are evaluated using trapezoidal quadrature, and the mixing distribution is renormalized after each update. Since Newton's recursion depends on the order in which the Y_i's are processed, following Tokdar et al. (2009), for each n we repeat it over $R_n = n/10$ random permutations of $Y_{1:n}$. If $\widetilde{G}^{(r)}_n$ denotes Newton's mixing-distribution estimate obtained from the rth permutation, $r = 1, \ldots, R_n$, then we form the order-averaged estimate $\overline{G}_n = \frac{1}{R_n} \sum_{r=1}^{R_n} \widetilde{G}^{(r)}_n$. Based on (3), we define the quasi-Bayes estimate of $f^{(X)}$ as

$$f^{(X)}_{\overline{G}_n}(x) = \int_\Theta \phi(x \mid \mu, \sigma^2) \overline{G}_n(\mathrm{d}\mu, \mathrm{d}\sigma^2). \tag{32}$$

Similarly, based on (12) and (19), we define the quasi-Bayes credible intervals and bands for $f^{(X)}_G$, respectively, using $\overline{G}_n$. For numerical implementation, the parameter space Θ is discretized using increments of 0.1 in both coordinates, giving $401 \times 50 = 20,050$ grid points. Such a discretization imposes no modeling restrictions. Further details are given in C.1.1. In addition, we refer to C.1.1 for a comparison with Monte Carlo credible intervals and bands, showing that the quasi-Bayesian credible intervals closely agree with their Monte Carlo counterparts, whereas the quasi-Bayesian credible bands are wider and thus more conservative, due to the metric-entropy and Gaussian concentration bounds used in their construction.

Figure 1 reports the quasi-Bayes estimate of $f^{(X)}_G$ in (32), and the asymptotic credible intervals and bands. The figure consists of two columns, which correspond to the Laplace noise standard deviations $\sigma_l = 0.25$ and $\sigma_l = 0.50$, respectively. The first row of the figure compares the true density $f^{(X)}_{G^*}$ with the quasi-Bayes density estimates, while the second row displays the credible intervals, and the third row displays the credible bands on the interval $I = [-4, 6]$.

We compare the quasi-Bayesian approach with a Bayesian approach in which the mixing distribution G is endowed with a Dirichlet process prior (Ferguson, 1973; Lo, 1984). Consider the Gaussian kernel $k(\cdot \mid \theta) = \phi(\cdot \mid \mu, \sigma^2)$, where $\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$, and assume the hierarchical model

$$\begin{aligned} &Y_i = X_i + Z_i, \quad X_i \mid \theta_i \overset{\text{ind}}{\sim} k(\cdot \mid \theta_i), \quad Z_i \overset{\text{iid}}{\sim} \text{Laplace}(0, b_l), \qquad i = 1, \ldots, n, \\ &\theta_i \mid G \overset{\text{iid}}{\sim} G, \quad G \mid M \sim \text{DP}(M, H), \quad M \sim \text{Gamma}(1, 1). \end{aligned} \tag{33}$$

where $\text{Gamma}(\cdot, \cdot)$ is the Gamma distribution, and $DP(\cdot, \cdot)$ is the law of a Dirichlet process with strength parameter $M > 0$ and non-atomic (diffuse) base probability measure H (Ferguson, 1973). We set H to be the Uniform distribution on $\Theta = [-20, 20] \times [0.1, 5]$. Posterior inference is performed

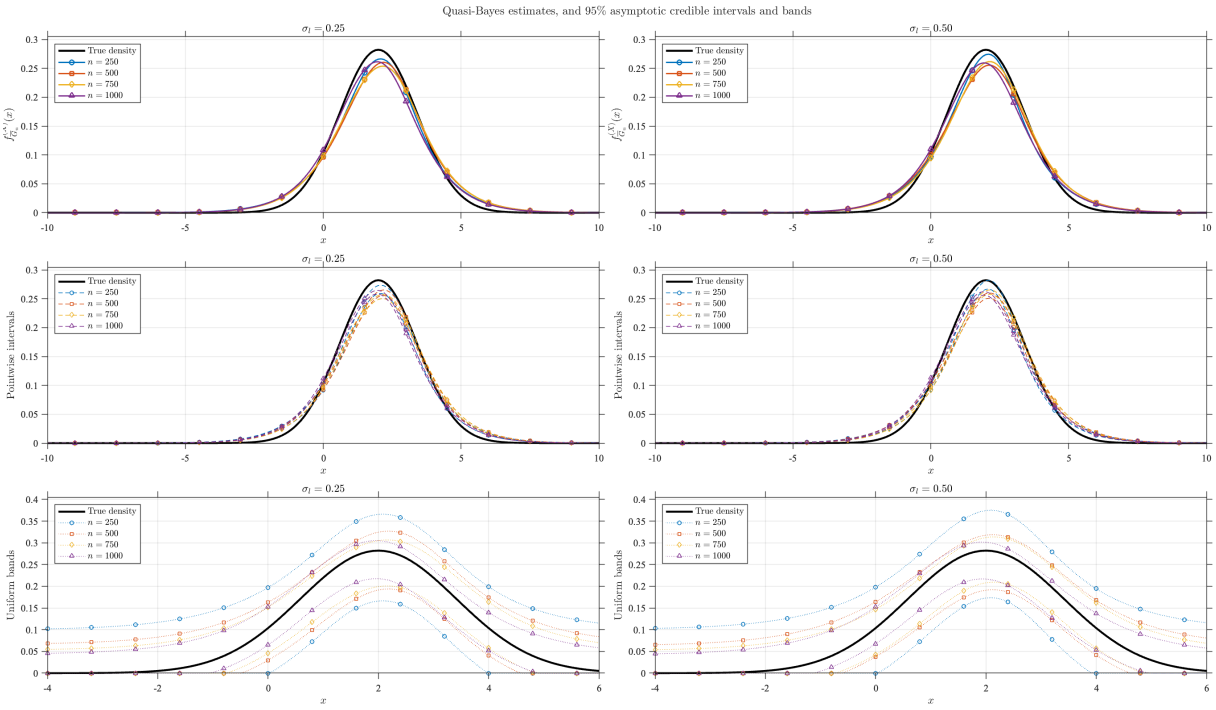


Figure 1: Unimodal example, with Laplace noise: estimates, credible intervals and bands.

both sequentially, using the sequential Monte Carlo (SMC) algorithm of Fearnhead (2004) and in batch, using Markov chain Monte Carlo (MCMC) Algorithm 8 of Neal (2000). Implementation details for the SMC algorithm and Algorithm 8 are given in C.1.1. Newton's and SMC algorithms are averaged over the same R_n permutations, whereas Algorithm 8 is applied once to each complete dataset. Figure 2 summarizes the comparison between the quasi-Bayesian and Bayesian approaches. The first row of the figure displays the estimates of $f_G^{(X)}$ obtained with Newton's algorithm, the SMC algorithm, and Algorithm 8; the second and third rows report total work and total CPU time, respectively.

Figure 2 shows that Newton's algorithm provides estimates that are comparable with those obtained from the SMC algorithm and Algorithm 8. The main differences concern computational efficiency, which is expressed in terms of CPU time. Although Newton's and SMC algorithms process the observations sequentially, the SMC algorithm requires the propagation and possible resampling of particles, whereas Newton's algorithm relies on a single deterministic recursive update. Algorithm 8, instead, requires a complete batch analysis for each sample size. Consequently, Newton's algorithm combines competitive estimation accuracy with a substantially smaller CPU time, particularly relative to Algorithm 8.

We refer to C.1.1 for a more detailed analysis of the unimodal example, also considering the Gaussian noise distribution for the Z_i's, namely $Z_i \overset{\text{iid}}{\sim} N(0, \sigma_g^2)$ with standard deviation $\sigma_g \in \{0.25, 0.50\}$.

5.2 Bimodal example

For each $n \in \{250, 500, 750, 1000\}$, we generate random variables $X_1, \ldots, X_n$ i.i.d. according to the density $f^{(X)}(\cdot) = 0.3\,\phi(\cdot \mid -1, 2) + 0.7\,\phi(\cdot \mid 3, 1.5)$, where $\phi(\cdot \mid \mu, \sigma^2)$ denotes the Gaussian density

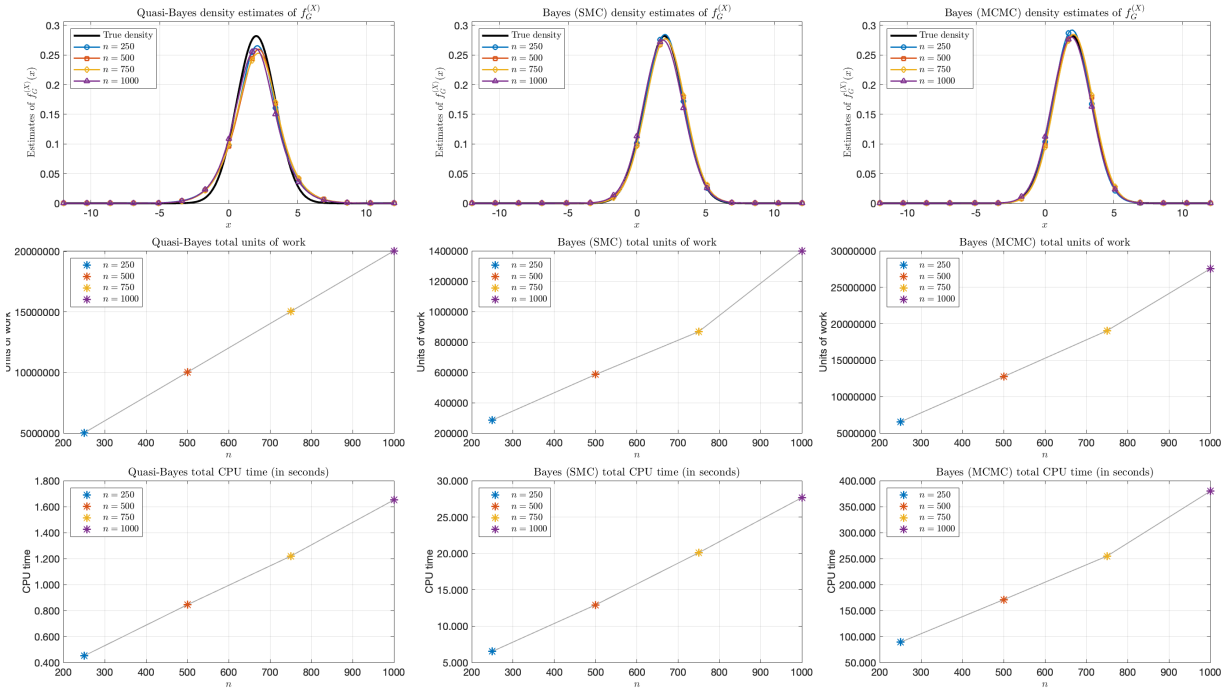


Figure 2: Unimodal example, with Laplace noise ($\sigma_l = 0.25$): comparison between quasi-Bayesian and Bayesian approaches.

with mean μ and variance σ^2. Using the mixture representation (1), we write $f^{(X)}$ as $f_{G^*}^{(X)}$, with $G^* = 0.3\,\delta_{(-1,2)} + 0.7\,\delta_{(3,1.5)}$ the true mixing distribution. We consider the Laplace noise distribution for the random variables Z_i's, namely $Z_i \overset{\text{iid}}{\sim} \text{Laplace}(0, b_l)$ with $b_l = \frac{\sigma_l}{\sqrt{2}}$ and $\sigma_l \in \{0.25, 0.50\}$, where σ_l is the standard deviation of the Laplace noise; this is arguably the most common example of ordinary-smooth noise distribution. The Z_i's are independent of the X_i's, and the observations are modeled as follows: $Y_i = X_i + Z_i$, for $i = 1, \ldots, n$.

We use the same implementation of Newton's algorithm as described in Section 5.1 for the unimodal example: the parameter space, numerical grids, initial distribution, learning rate, trapezoidal quadrature, renormalization, and number of permutations are unchanged. Figure 3 reports the quasi-Bayes estimate of $f_G^{(X)}$, and the asymptotic credible intervals and bands.

We refer to C.1.2 for a more detailed analysis of the bimodal example, including Monte Carlo credible intervals and bands and a comparison between the quasi-Bayesian and Bayesian approaches. A second synthetic-data bimodal example is investigated in C.1.3. The analyses in C.1.2–C.1.3 are conducted under the Laplace and Gaussian noise distributions.

6 Real-data analysis

We consider flow-cytometry data from Torregrosa-Cortés et al. (2023) on the expression of the mesodermal transcription factor Brachyury in differentiating mouse embryonic stem cells. In flow cytometry, cells pass individually through laser beams, and the emitted and scattered light is converted into digital intensity measurements for each detector channel. The fluorescence from a labeled cell contains both the reporter signal of interest and an intrinsic background component, known as autofluorescence, which may obscure low expression levels. The experiment therefore

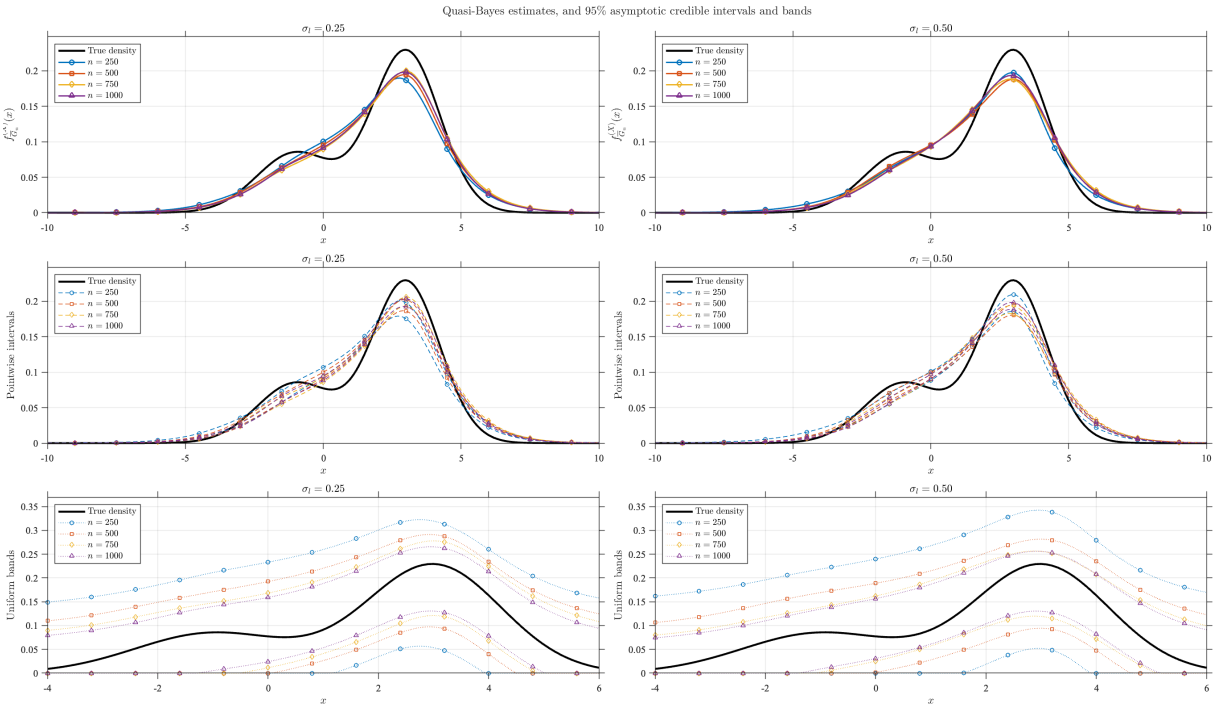


Figure 3: Bimodal example, with Laplace noise: estimates, credible intervals and bands.

includes separate labeled and unlabeled cell populations, the latter characterizing the background distribution (Torregrosa-Cortés et al., 2023). This yields a deconvolution problem aimed at recovering the population distribution of the latent reporter signal. Since the cells are recorded successively, we retain their acquisition order and process the observations sequentially rather than as an arbitrarily ordered static sample. Further details on the experimental setting are provided in C.2.1.

The publicly available data consist of fluorescence intensities and acquisition-time measurements; we refer to `https://github.com/dsb-lab/scBayesDeconv.jl`. Following the organization of the dataset in Torregrosa-Cortés et al. (2023), we use the first column as the sample from the labeled cell population and the sixth as the control sample describing autofluorescence. Since the two columns correspond to distinct cellular populations, their observations are not paired event-by-event. We order the labeled-cell measurements by acquisition time, and then examine the estimates over the nested prefixes corresponding to the first $n \in \{250,\ 1{,}000,\ 5{,}000,\ 10{,}000\}$ events. Retaining the acquisition order does not amount to assuming a time-series dependence structure; it only reflects the order in which the observations become available to the sequential procedure. Let Y_i denote the fluorescence intensity of the i-th labeled cell in acquisition order and assume the additive model $Y_i = X_i + Z_i$ for $i \geq 1$, where X_i is the reporter fluorescence and Z_i is the autofluorescence and background contribution. Further, the sixth column provides $m \geq 1$ independent control samples, say $\widetilde{Z}_1, \ldots, \widetilde{Z}_m$, measured on unlabeled cells and not paired with the Y_i's.

We assume that the random variables X_i's are i.i.d. according to an unknown density f_X, that the noise random variables Z_i's are i.i.d. according to a density f_Z, and that the random variables X_i's and Z_i's are mutually independent and independent of the control random variables $\widetilde{Z}_1, \ldots, \widetilde{Z}_m$. Further, $\widetilde{Z}_1, \ldots, \widetilde{Z}_m$ are i.i.d. with the same distribution as the Z_i's. The inferential target is the density f_X, which is represented as the Gaussian location-scale mixture $f_G^{(X)}(x) =$

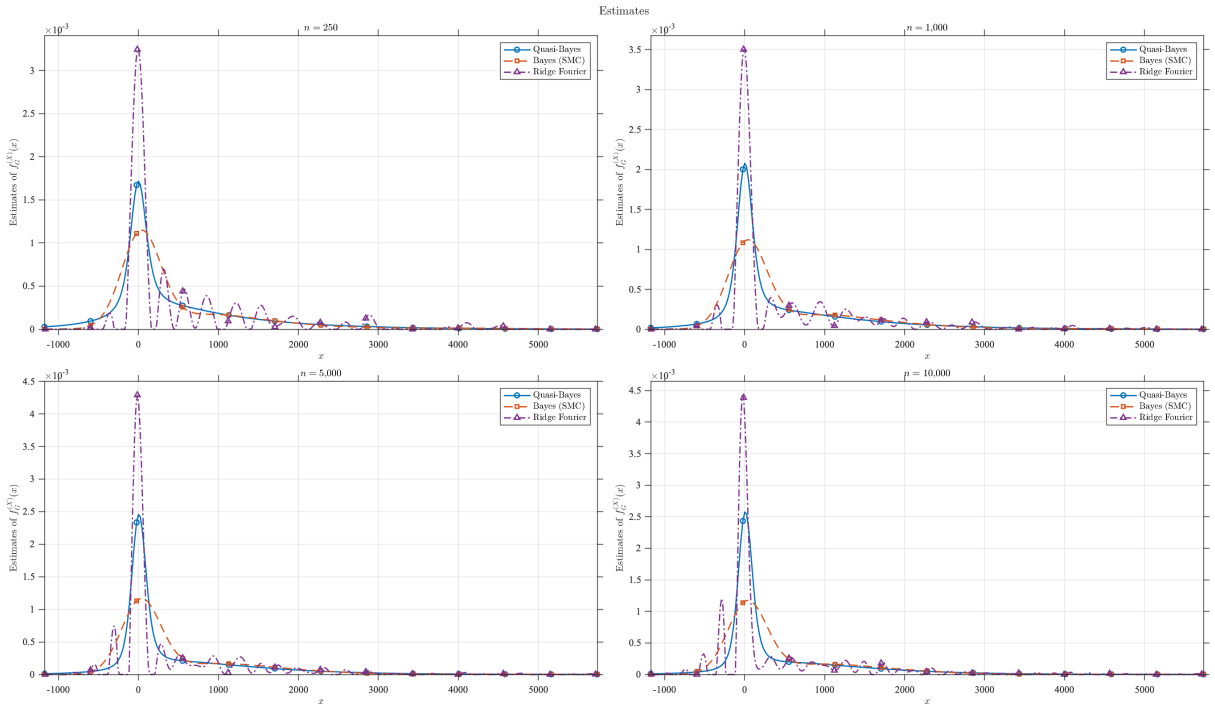


Figure 4: Flow-cytometry data, with Gaussian noise: comparison between quasi-Bayesian, Bayesian and ridge-regularized Fourier deconvolution approaches.

$\int_{\mathbb{R}\times\mathbb{R}^+} \phi(x \mid \mu, \sigma^2)\, G(d\mu, d\sigma^2)$, for $x \in \mathbb{R}$, with G being the unknown mixing distribution. For the noise distribution with density f_Z we consider a Gaussian distribution, i.e. $f_Z(\cdot) = \phi(\cdot \mid \mu_Z, \sigma_Z^2)$. In particular, the density f_Z is estimated once from the control samples $\widetilde{Z}_1, \ldots, \widetilde{Z}_m$ and it is subsequently kept fixed for all sample sizes.

To implement the quasi-Bayesian approach for estimating $f_G^{(X)}$, we apply Newton's algorithm (4), processing the observations Y_i's in acquisition order, with Gaussian kernel $k(\cdot \mid \theta) = \phi(\cdot \mid \mu, \sigma^2)$, where $\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$. On the standardized working scale, the parameter space $\mathbb{R} \times \mathbb{R}^+$ is restricted to $\Theta = [\mu_{\min}, \mu_{\max}] \times [0.01, 5]$, where the bounds $\mu_{\min}$ and $\mu_{\max}$ are determined from the observed and control samples. We use the learning rate $\widetilde{\alpha}_i = (1+i)^{-1}$, for $i \geq 1$, and set $\widetilde{G}_0$ equal to the Uniform distribution on Θ; this is precisely as in the synthetic-data analysis. All integrals with respect to the mixing distribution are evaluated using the two-dimensional trapezoidal rule, and $\widetilde{G}_i$ is numerically renormalized after each update. For numerical implementation, the location and variance coordinates are discretized using, respectively, 121 equally spaced points and 45 logarithmically spaced points, giving 5,445 grid points.

As additional benchmarks, we also consider a Bayesian approach under the Dirichlet-process Gaussian-mixture model (C.27), with posterior inference performed sequentially using the SMC algorithm, and the ridge-regularized Fourier deconvolution approach (Stefanski and Carroll, 1990; Hall and Meister, 2007). We omit further implementation details for these competing procedures, which all apply the same fitted noise distribution. For the nested sample sizes $n \in \{250,\ 1{,}000,\ 5{,}000,\ 10{,}000\}$, Figure 4 reports the estimates of $f_G^{(X)}$ obtained by the quasi-Bayesian, Bayesian, and ridge-regularized Fourier deconvolution approaches.

We refer to C.2.1 for a more detailed analysis of the flow-cytometry data, including a sensitivity analysis with respect to Gaussian, Laplace, and four-component Gaussian-mixture noise

distributions, with the latter being originally proposed by Torregrosa-Cortés et al. (2023); for these analyses, we also report figures augmented with histograms of a reconstructed (proxy) version of the reporter fluorescence samples. Two additional real-data analyses on stellar metallicity data and active power-output data are presented in C.2.2 and C.2.3.

7 Discussion

Although our analysis has focused on univariate observations, the proposed quasi-Bayesian sequential methodology naturally extends to multivariate settings. The structure of Newton's recursive update remains unchanged, though the computational cost per update increases with dimension, primarily due to the greater complexity of numerical integration. We expect local central limit theorems and the associated credible intervals to continue to hold in higher dimensions, along with asymptotic frequentist properties such as consistency and merging. However, uncertainty quantification in the multivariate setting presents additional challenges: in particular, constructing uniform credible bands becomes more delicate, as their width typically increases with dimension, potentially limiting interpretability and coverage accuracy.

While we establish the consistency of the estimate $f^{(X)}_{\widetilde{G}_n}$ in L^1, its rate of convergence in the same norm remains an open problem. Deriving such a rate for quasi-Bayesian sequential deconvolution is likely to be technically challenging, but would provide valuable insight into the statistical efficiency of the method. As a preliminary step, we present in D.2 a convergence rate for finite mixture models, which may provide a foundation for future work.

A natural extension concerns unknown noise distributions. Recent developments on joint estimation of signal and noise in static or batch settings suggest that recursive strategies could be adapted to this scenario. A quasi-Bayesian sequential implementation is an important direction for future research, also related to privacy-preserving inference in streaming settings, where noise is intentionally introduced through a known or partially known perturbation mechanism (Dwork et al., 2010; Cardoso and Rogers, 2022; He et al., 2024).

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Appendix

B Proofs

B.1 Preliminary results

B.1.1 Almost-sure conditional convergence

In this Section, we report some definitions and results, that are critical to Section 3. Given a probability space $(\Omega, \mathcal{F}, P)$ and a Polish space $\mathbb{X}$ with its Borel sigma-algebra $\mathcal{X}$, a kernel on $\mathbb{X}$ is a function $K : \Omega \times \mathcal{X}$ satisfying:

(i) for every $\omega \in \Omega$, $K(\omega, \cdot)$ is a probability measure on $\mathcal{X}$;

(ii) for each $B \in \mathcal{X}$, the function $K(\cdot, B)$ is measurable with respect to $\mathcal{F}$.

A Kernel K on $\mathbb{R}$ is called a Gaussian kernel if $K(\omega, \cdot)$ is a Gaussian distribution with mean $\mu(\omega)$ and variance $\sigma^2(\omega)$, where μ and σ^2 are random variables defined on $(\Omega, \mathcal{F}, P)$. We denote the Gaussian kernel by $\mathcal{N}(\mu, \sigma^2)$ and interpret the Gaussian distribution with zero variance as the degenerate law centered on the mean. A kernel K on the Polish space $C(\mathbb{R})$, with the topology of uniform convergence on compact sets, is called a Gaussian process kernel if $K(\omega, \cdot)$ is the distribution of a Gaussian process in $C(\mathbb{R})$, with mean function $\mu(\omega) \in C(\mathbb{R})$ and covariance function $R(\omega) \in C(\mathbb{R} \times \mathbb{R})$. If $\mu(\omega) = 0$ for all ω, then the Gaussian process kernel is said to be centered.

We now extend the definition of almost sure conditional convergence, given by Crimaldi (2009) for real sequences, to sequences taking values in Polish spaces.

Definition B.1. *Let (X_n) be a sequence of random variables adapted to a filtration $(\mathcal{G}_n)$ and taking values in a Polish space $\mathbb{X}$ with its Borel sigma-algebra $\mathcal{X}$. For every $n \geq 0$, let K_n denote a regular version of the conditional distribution of X_{n+1}, given $\mathcal{G}_n$. If there exists a kernel K such that the sequence $(K_n(\omega, \cdot))_n$ converges weakly to $K(\omega, \cdot)$ for almost every $\omega \in \Omega$, then we say that the sequence (X_n) converges to K in the sense of almost-sure conditional convergence.*

Next results are the main tool in the proof of Theorem 3.2.

Theorem B.2 (Crimaldi (2009), Theorem A.1)**.** *For each $n \geq 1$, let $(M_{n,j})_{j\geq 1}$ be a real valued martingale with respect to the filtration $(\mathcal{F}_{n,j})_{j\geq 1}$, satisfying $M_{n,0} = 0$, and converging in L^1 to a random variable $M_{n,\infty}$. Set*

$$X_{n,j} = M_{n,j} - M_{n,j-1} \text{ for } j \geq 1, \quad U_n = \sum_{j\geq 1} X_{n,j}^2, \quad X_n^* = \sup_{j\geq 1} |X_{n,j}|.$$

Assume that

(a) $(X_n^*)_n$ is dominated in L^1 and converges to zero a.s.

(b) $(U_n)_n$ converges a.s. to a non-negative random variable U.

Then the the sequence $(M_{n,\infty})$ converges to the Gaussian kernel $\mathcal{N}(0, U)$ in the sense of almost-sure conditional convergence.

Remark B.3. *Requesting that the almost sure limit of (U_n) is non-negative guarantees the well-definedness of the kernel $\mathcal{N}(0, U)$ for every $\omega \in \Omega$. Theorem A.1 in Crimaldi (2009) specifies that U should be a "positive random variable". Although not explicitly stated, this requirement should be understood as $U \geq 0$, as becomes evident upon a careful examination of the proof.*

Theorem B.4 (Crimaldi et al. (2016), Lemma 4.1)**.** *Let (Z_n) be a sequence of real valued random variables adapted to the filtration $(\mathcal{G}_n)$ and such that $E(Z_{n+1} \mid \mathcal{G}_n) \to Z$ a.s. for some random variable Z. Moreover, let (a_n) and (b_n) be sequences of real numbers such that*

$$b_n \uparrow +\infty, \quad \sum_{k=1}^{\infty} \frac{E(Z_k^2)}{a_k^2 b_k^2} < +\infty.$$

Then we have:

(a) If $\frac{1}{b_n} \sum_{k=1}^n \frac{1}{a_k} \to c$ for some constant c, then $\frac{1}{b_n} \sum_{k=1}^n \frac{Z_k}{a_k} \to cZ$ a.s.

(b) If $b_n \sum_{k\geq n} \frac{1}{a_k b_k^2} \to c$ for some constant c, then $b_n \sum_{k\geq n} \frac{Z_k}{a_k b_k^2} \to cZ$ a.s.

B.1.2 A concentration inequality for Gaussian processes

Let (T, d) be a totally bounded pseudometric space, i.e. d satisfies the axioms of a metric except that does not necessarily separate points. For every $\delta > 0$, a δ-packing of T is a finite subset T_0 of T such that for every $t_1, t_2 \in T_0$, $d(t_1, t_2) > \delta$. A δ-covering of T is a finite subset T_1 of T whose covering radius r satisfies $r \leq \delta$, where the covering radius of T_1 is defined as

$$r = \inf\{s > 0 : \text{ for every } t \in T \text{ there exists } t_1 \in T_1 \text{ with } d(t, t_1) \leq s\}.$$

The maximal cardinality of a δ-packing of T is called the δ-packing number of T and denoted by $N(\delta, T)$. The minimal cardinality of a δ-covering of T is called the δ-covering number of T, and denoted by $N'(\delta, T)$. It follows from the definitions that

$$N'(\delta, T) \leq N(\delta, T) \leq N'(\delta/2, T).$$

The δ-entropy number of T is defined as

$$H(\delta, T) = \log(N(\delta, T)).$$

The function $H(\cdot, T)$ is called the metric entropy of (T, d). The covariance pseudometric of a centered Gaussian process $(X(t))_{t\in T}$ is defined as

$$d(s,t) = E((X(s) - X(t))^2)^{1/2} = (R(s,s) - 2R(s,t) + R(t,t))^{1/2},$$

where R is the covariance function of $(X(t))_{t\in T}$.

Theorem B.5 (Massart (2007), Theorem 3.18)**.** *Let $(X(t))_{t\in T}$ be a centered Gaussian process and let d be the covariance pseudometric of $(X(t))_{t\in T}$. Assume that (T, d) is totally bounded and denote by $H(\delta, T)$ the δ-entropy number of (T, d), for all positive δ. If $\sqrt{H(\cdot, T)}$ is integrable at 0, then $(X(t))_{t\in T}$ admits a version which is almost surely uniformly continuous on (T, d). Moreover, if $(X(t))_{t\in T}$ is almost surely continuous on (T, d), then*

$$E(\sup_{t\in T} X(t)) \leq 12 \int_0^\sigma \sqrt{H(z, T)} dz,$$

where $\sigma = (\sup_{t\in T} E(X^2(t)))^{1/2}$.

The following concentration inequality allows to give an upper bound in probability for $\sup_{t\in T} X(t)$.

Theorem B.6 (Massart (2007), Proposition 3.19)**.** *Let $(X(t))_{t\in T}$ be an almost surely continuous centered Gaussian process on the totally bounded set (T, d), and let*

$$\sigma = \left(\sup_{t\in T} E(X^2(t))\right)^{1/2}.$$

If $\sigma > 0$, then, for every $z > 0$,

$$P(\sup_{t\in T} X(t) \geq E(\sup_{t\in T} X(t)) + \sigma\sqrt{2z}) \leq e^{-z}.$$

If $\sigma = 0$, then $\sup_{t\in T} X(t) = E(\sup_{t\in T} X(t)) = 0$. Hence the above inequality does not hold. However, in such a case, for every $\epsilon > 0$, $P(\sup_{t\in T} X(t) \geq \epsilon) = 0$. Therefore, whatever $\sigma \geq 0$, we can write that

$$P(\sup_{t\in T} X(t) \geq \max(\epsilon, E(\sup_{t\in T} X(t)) + \sigma\sqrt{2z})) \leq e^{-z},$$

for every $\epsilon > 0$ and $z > 0$.

B.2 Proofs of Section 2

We introduce the following notation: for a filtered probability space $(\Omega, \mathcal{F}, P, (\mathcal{G}_n))$, we denote by $P_n(\cdot) = P(\cdot \mid \mathcal{G}_n)$ and by $E_n(\cdot) = E(\cdot \mid \mathcal{G}_n)$. We also denote by $X_{1:n} = (X_1, \ldots, X_n)$.

B.2.1 Proof of Proposition 2.1

Define recursively the law of (X_n) and (Z_n) for $n \geq 1$ by setting the conditional density of X_{n+1}, given $X_{1:n}, Z_{1:n}, Y_{1,n}$ as in (3), the conditional density of Z_{n+1}, given $X_{1:n+1}, Z_{1:n}, Y_{1,n}$, as f_Z. Then define $Y_{n+1} = X_{n+1} + Z_{n+1}$ and $\widetilde{G}_{n+1}$ as in (4). Let $(\Omega, \mathcal{F}, P)$ be a probability space on which all the random variables (X_n, Y_n, Z_n) are defined. Denoting by E_{n-1} the conditional expectation given $(X_{1:n-1}, Y_{1:n-1}, Z_{1:n-1})$, we can write that

$$\begin{aligned} E_{n-1}(e^{isY_n}) &= \varphi_{f_Z}(s) \int \varphi_k(s \mid \theta) \widetilde{G}_{n-1}(\mathrm{d}\theta) \\ &= \int \varphi_k(s \mid \theta) \varphi_{f_Z}(s) \widetilde{G}_{n-1}(\mathrm{d}\theta) \\ &= \int \varphi_{\widetilde{k}}(s \mid \theta) \widetilde{G}_{n-1}(\mathrm{d}\theta), \end{aligned}$$

where φ_{f_Z} and $\varphi_{\widetilde{k}}(s \mid \theta)$ denote the Fourier transforms of f_Z and of $\widetilde{k}(\cdot \mid \theta) = k(\cdot \mid \theta) * f_Z$, respectively. Hence (5) holds. Moreover, the conditional distribution of X_{n+1}, given $X_{1:n}, Z_{1:n}, Y_{1,n}$ is uniquely identified by

$$\begin{aligned} E_n(e^{isY_{n+1}}) &= E_n(e^{isZ_{n+1}} e^{isX_{n+1}}) \\ &= \varphi_{f_Z}(s) E_n(e^{isX_{n+1}}). \end{aligned}$$

B.2.2 Proof of Theorem 2.2

The convergence of the sequence $(\widetilde{G}_n)$ to a random probability measure $\widetilde{G}$ follows directly from Theorem 1 in Fortini and Petrone (2020). For completeness, and to highlight the role of the filtration $(\mathcal{G}_n)$ in our setting, we briefly outline the argument. For every $B \in \mathcal{B}(\Theta)$, the sequence $(\widetilde{G}_n(B))_{n \geq 0}$ is adapted to the filtration $(\mathcal{G}_n)_{n \geq 0}$. From (5) the conditional distribution of Y_{n+1}, given $\mathcal{G}_n$ has density $f^{(Y)}_{\widetilde{G}_n}(y) = \int_\Theta \widetilde{k}(y \mid \theta) \widetilde{G}_n(\mathrm{d}\theta)$ with respect to λ. Thus, for every $n \geq 0$,

$$\begin{aligned} &E(\widetilde{G}_{n+1}(B) \mid \mathcal{G}_n) \\ &= (1 - \widetilde{\alpha}_{n+1}) \widetilde{G}_n(B) + \widetilde{\alpha}_{n+1} \int_B \int \frac{\widetilde{k}(y \mid \theta)}{\int \widetilde{k}(y \mid \theta') \widetilde{G}_n(\mathrm{d}\theta')} f^{(Y)}_{\widetilde{G}_n}(y) \lambda(dy) \widetilde{G}_n(\mathrm{d}\theta) \\ &= \widetilde{G}_n(B). \end{aligned}$$

Thus, for every $B \in \mathcal{B}(\Theta)$, the sequence $(\widetilde{G}_n(B))_{n \geq 0}$ is a bounded martingale with respect to the filtration $(\mathcal{G}_n)_{n \geq 0}$. By Lemma 7.14 in Aldous (1985), there exists a random probability measure $\widetilde{G}$ on $\mathcal{B}(\Theta)$ such that $\widetilde{G}_n$ converges to $\widetilde{G}$ weakly P-a.s. and for every $n, k \in \mathbb{N}$ and every bounded continuous function h on Θ,

$$\int_\Theta h(\theta) \widetilde{G}_n(\mathrm{d}\theta) = E\left(\int_\Theta h(\theta) \widetilde{G}_{n+k}(\mathrm{d}\theta) \mid \mathcal{G}_n \right) = E\left(\int_\Theta h(\theta) \widetilde{G}(\mathrm{d}\theta) \mid \mathcal{G}_n \right) \quad P\text{-a.s.}$$

We now turn to the latent sequence (X_n), which is not directly covered by Fortini and Petrone (2020). Since $k(x \mid \theta)$ is continuous in θ, then

$$f_{\widetilde{G}_n}(x) = \int_\Theta k(x \mid \theta) \widetilde{G}_n(\mathrm{d}\theta) \to \int_\Theta k(x \mid \theta) \widetilde{G}(\mathrm{d}\theta)$$

P-a.s., for λ-almost all x, as $n \to +\infty$. Moreover, for every $n \geq 0$, $k \geq 1$ and $A \in \mathcal{B}(\mathbb{R})$,

$$\begin{aligned} P(X_{n+k} \in A \mid \mathcal{G}_n) &= E(P(X_{n+k} \in A \mid \mathcal{G}_{n+k-1}) \mid \mathcal{G}_n) \\ &= E\left(\int_A \int_\Theta k(x \mid \theta) \widetilde{G}_{n+k-1}(\mathrm{d}\theta)\lambda(dx) \mid \mathcal{G}_n\right) \\ &= \int_A \int_\Theta k(x \mid \theta) \widetilde{G}_n(\mathrm{d}\theta)\lambda(dx) \\ &= P(X_{n+1} \in A \mid \mathcal{G}_n). \end{aligned} \tag{B.1}$$

We show that the sequence (X_n) is conditionally identically distributed in the sense of Berti et al. (2004). The assumptions of the learning model include the conditional independence of X_{n+1} from $X_1, \ldots, X_n$, given $Y_1, \ldots, Y_n$ and the independence of Z_{n+1} from $X_1, \ldots, X_n, Y_1, \ldots, Y_n$, for every $n \geq 0$ with the convention that a variable with a 0 subscript is omitted. This implies that for every $k \geq 1$, X_{n+k} is conditionally independent of $X_1, \ldots, X_n$, given $Y_1, \ldots, Y_n$. Indeed in a directed acyclic graph representing the joint probability distribution of $(X_1, Y_1, Z_1, \ldots, X_{n+k}, Y_{n+k}, Z_{n+k})$, the random blocks X_{n+k} and $(X_1, Z_1, \ldots, X_n, Z_n)$ are d-separated by Y_n. Let $(\mathcal{G}_n^a)_{n\geq 0}$ denote the natural filtration of the random sequence $(X_n, Y_n, Z_n)_{n\geq 1}$. This enlargement is needed to ensure that (X_n) is adapted. It follows by (B.1) and the conditional independence of X_{n+k} that, for every $A \in \mathcal{B}(\mathbb{R})$,

$$P(X_{n+k} \in A \mid \mathcal{G}_n^a) = P(X_{n+k} \in A \mid \mathcal{G}_n) = P(X_{n+1} \in A \mid \mathcal{G}_n) = P(X_{n+1} \in A \mid \mathcal{G}_n^a).$$

Thus, the sequence $(X_n)_{n\geq 1}$ is adapted to the filtration $(\mathcal{G}_n^a)_n$ and conditionally identically distributed with respect to it (Berti et al. (2004)). Its directing random measure has density $f_{\widetilde{G}}^{(X)}$ with respect to λ, since for every A

$$\begin{aligned} \lim_{n\to+\infty} P(X_{n+1} \in A \mid \mathcal{G}_n^a) &= \lim_{n\to+\infty} P(X_{n+1} \in A \mid \mathcal{G}_n) \\ &= \lim_{n\to+\infty} \int_A \int_\Theta k(x \mid \theta) \widetilde{G}_n(\mathrm{d}\theta)\lambda(dx) \\ &= \int_A \int_\Theta k(x \mid \theta) \widetilde{G}(\mathrm{d}\theta)\lambda(dx) \\ &= \int_A f_{\widetilde{G}}^{(X)}(x)\lambda(dx). \end{aligned}$$

By Lemma 2.1 and Theorem 2.2 in Berti et al. (2004) applied to $f(X_n) = I(X_n \in A)$, we deduce that $\int_A f_{\widetilde{G}}(x)\lambda(dx)$ is the P-a.s. limit of $\frac{1}{n}\sum_{k=1}^n I(X_k \in A)$ for every $A \in \mathcal{B}(\mathbb{R})$. Moreover,

$$\begin{aligned} P(X_{n+k} \in A \mid \mathcal{G}_n^a) &= P(X_{n+k} \in A \mid \mathcal{G}_n) \\ &= E\left(\int_A \int_\Theta k(x \mid \theta) \widetilde{G}_{n+k-1}(\mathrm{d}\theta)\lambda(dx) \,\middle|\, \mathcal{G}_n\right) \\ &= E\left(\int_A \int_\Theta k(x \mid \theta) \widetilde{G}(\mathrm{d}\theta)\lambda(dx) \,\middle|\, \mathcal{G}_n\right) \\ &= E\left(\int_A f_{\widetilde{G}}^{(X)}(x)\lambda(dx) \,\middle|\, \mathcal{G}_n\right). \end{aligned}$$

It follows that

$$P(X_n \in A) = E\left(\int_A f_{\widetilde{G}}^{(X)}(x)\lambda(dx)\right) = \iint_A f_G^{(X)}(x)\lambda(dx)\pi(dG),$$

where π is the probability distribution of $\widetilde{G}$.
To prove that $f^{(X)}_{\widetilde{G}_n}$ converges to $f^{(X)}_{\widetilde{G}}$ in L^1, we can apply the dominated convergence theorem to the sequence of functions $(|f^{(X)}_{\widetilde{G}_n} - f^{(X)}_{\widetilde{G}}|)_{n\geq 0}$, which converges to zero λ-a.e. (P-a.s.) and is dominated by $\sup_{\theta\in\Theta} k(x \mid \theta)$ that is integrable with respect to λ by assumption.

B.3 Proofs of Section 3

B.3.1 Proof of Theorem 3.2

Lemma B.7. *Let $v(x)$ be defined as in (10) and let $v_n(x)$ be defined as $v(x)$ with $\widetilde{G}_n$ in place of $\widetilde{G}$. Then, for every $x \in \mathbb{R}$, $v_n(x)$ is strictly positive and converges to $v(x)$, P-a.s.*

Proof. Let us denote by $k_x(\cdot) = k(x \mid \cdot)$ and $\widetilde{k}_y(\cdot) = \widetilde{k}(y \mid \cdot) = (f_Z * k(\cdot \mid \cdot))(y)$.
To show that $v_n(x) \neq 0$ P-a.s. we proceed by contradiction. If $v_n(x) = 0$, then

$$\int_\Theta k(x \mid \theta) \left(\widetilde{G}_n(\mathrm{d}\theta \mid y) - \widetilde{G}(\mathrm{d}\theta)\right) = 0$$

for λ-almost every y. By the assumption A4), the last equality entails $\widetilde{G}_n(\cdot \mid y) = \widetilde{G}_n(\cdot)$ for λ-almost every y. On the other hand, by the assumptions A1), A2) and A5), $\mathrm{supp}(\widetilde{G}_n) = \Theta$, which implies $\widetilde{k}(\cdot \mid \theta) \neq \int \widetilde{k}(\cdot \mid \theta')\widetilde{G}_n(\mathrm{d}\theta')$, leading to $\widetilde{G}_n(\cdot \mid y) \neq \widetilde{G}_n(\cdot)$ for y in a set of λ-positive measure.

We now prove that $v_n(x)$ converges to $v(x)$, P-a.s. First, notice that, since we have $\int \sup_{\theta\in\Theta} k(x \mid \theta)\lambda(\mathrm{d}x) < +\infty$, then there exists a density function h with respect to λ such that

$$h(x) = \frac{\sup_{\theta\in\Theta} k(x \mid \theta)}{\int \sup_{\theta\in\Theta} k(x \mid \theta)\lambda(\mathrm{d}x)}.$$

Thus, we can write that

$$\begin{aligned}\int \sup_{\theta\in\Theta} \widetilde{k}(y \mid \theta)\lambda(\mathrm{d}y) &= \int \sup_{\theta\in\Theta} \int k(x \mid \theta) f_Z(y - x)\lambda(\mathrm{d}x)\lambda(\mathrm{d}y) \\ &\leq C \iint h(x) f_Z(y - x)\lambda(\mathrm{d}x)\lambda(\mathrm{d}y) < +\infty,\end{aligned}$$

where $C = \int \sup_{\theta\in\Theta} k(x \mid \theta)\lambda(\mathrm{d}x)$. Moreover,

$$\int_\Theta \widetilde{k}_y \mathrm{d}\widetilde{G}_n = \int \int_\Theta k(x \mid \theta) f_Z(y - x)\lambda(\mathrm{d}x)\widetilde{G}_n(\mathrm{d}\theta) \leq C(f_Z * h)(y),$$

which is a function integrable with respect to λ.

By assumption A3) and Theorem 2.2,

$$\int_\Theta k(x \mid \theta) \left(\widetilde{G}_n(\mathrm{d}\theta \mid y) - \widetilde{G}_n(\mathrm{d}\theta)\right) \to \int_\Theta k(x \mid \theta) \left(\widetilde{G}(\mathrm{d}\theta \mid y) - \widetilde{G}(\mathrm{d}\theta)\right) \quad P\text{-a.s.}$$

Since $\left[\int_\Theta k(x \mid \theta) \left(\widetilde{G}(\mathrm{d}\theta \mid y) - \widetilde{G}(\mathrm{d}\theta)\right)\right]^2 \leq \sup_\theta k(x \mid \theta)^2$ and $\int_\Theta \widetilde{k}_y \mathrm{d}\widetilde{G}_n \leq C(f_Z * h)(y)$ which is integrable with respect to λ, then by dominated convergence theorem,

$$\begin{aligned}v_n(x) &\overset{P-a.s.}{\longrightarrow} \int \left[\int_\Theta k(x \mid \theta) \left(\widetilde{G}(\mathrm{d}\theta \mid y) - \widetilde{G}(\mathrm{d}\theta)\right)\right]^2 \left(\int_\Theta \widetilde{k}_y \mathrm{d}\widetilde{G}\right) \lambda(dy) \\ &= \int \left[f^{(X)}_{\widetilde{G}}(x \mid y) - f^{(X)}_{\widetilde{G}}(x)\right]^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(dy) = v(x).\end{aligned}$$

□

Proof of Theorem 3.2. The proof is based on Crimaldi (2009, Theorem A.1) (see Theorem B.2 in Section B). Fix $x \in \mathbb{R}$ and define, for each $n \geq 0$,

$$\mathcal{F}_{n,0} = \mathcal{G}_n, \quad \mathcal{F}_{n,j} = \mathcal{G}_{n+j-1} \text{ for } j \geq 1,$$

$$M_{n,0} = 0, \quad M_{n,j} = b_n^{1/2}(f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}_{n+j-1}}(x)) \text{ for } j \geq 1.$$

For every $n \geq 0$, the sequence $(M_{n,j})_{j\geq 0}$ is a bounded martingale with respect to the filtration $(\mathcal{F}_{n,j})_{j\geq 0}$, converging P-a.s. and in L^1 to

$$M_{n,\infty} = b_n^{1/2}(f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}}(x)).$$

Let

$$X_{n,j} = M_{n,j} - M_{n,j-1} = b_n^{1/2}(f^{(X)}_{\widetilde{G}_{n+j-1}}(x) - f^{(X)}_{\widetilde{G}_{n+j}}(x)),$$

$$X_n^* = \sup_{j\geq 1} |X_{n,j}| = b_n^{1/2} \sup_{j\geq 1} |f^{(X)}_{\widetilde{G}_{n+j}}(x) - f^{(X)}_{\widetilde{G}_{n+j-1}}(x)|,$$

and

$$U_n = \sum_{j=0}^{\infty} X_{n,j}^2 = b_n \sum_{k\geq n} (f^{(X)}_{\widetilde{G}_{k+1}}(x) - f^{(X)}_{\widetilde{G}_k}(x))^2.$$

The thesis follows from Theorem B.2, if we can verify that:

(a) The sequence (X_n^*) is dominated in L^1 and converges to zero P-a.s. as $n \to +\infty$.

(b) $U_n \to v(x)$, P-a.s. as $n \to +\infty$.

Notice that

$$\begin{aligned} f^{(X)}_{\widetilde{G}_k}(x) - f^{(X)}_{\widetilde{G}_{k-1}}(x) &= \int_\Theta k(x \mid \theta)(\widetilde{G}_k(\mathrm{d}\theta) - \widetilde{G}_{k-1}(\mathrm{d}\theta)) \\ &= \widetilde{\alpha}_k \int_\Theta k(x \mid \theta) \left(\widetilde{G}_{k-1}(\mathrm{d}\theta \mid Y_k) - \widetilde{G}_{k-1}(\mathrm{d}\theta)\right), \end{aligned}$$

where $\widetilde{G}_{k-1}(\mathrm{d}\theta \mid Y_k) := \widetilde{k}(Y_k \mid \theta)\widetilde{G}_{k-1}(\mathrm{d}\theta) / \int_\Theta \widetilde{k}(Y_k \mid \theta')\widetilde{G}_{k-1}(\mathrm{d}\theta')$. Thus,

$$X_n^* \leq b_n^{1/2} \sup_{k\geq n} \widetilde{\alpha}_k \int_\Theta k(x \mid \theta) \left|\widetilde{G}_{k-1}(\mathrm{d}\theta \mid Y_k) - \widetilde{G}_{k-1}(\mathrm{d}\theta)\right| \leq \sup_{\theta\in\Theta} k(x \mid \theta) b_n^{1/2} \sup_{k\geq n} \widetilde{\alpha}_k$$

is bounded for λ-almost every x, and converges to zero P-a.s., since $b_n \sup_{k\geq n} \widetilde{\alpha}_k^2 \to 0$. To prove (b), we apply Crimaldi et al. (2016, Lemma 4.1 (b)) (see Theorem B.4 (b)), with

$$Z_n = \widetilde{\alpha}_n^{-2}(f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}_{n-1}}(x))^2,$$

and $a_k = (b_k \widetilde{\alpha}_k)^{-2}$. Defining, for each $n \geq 1$, $f^{(X)}_{\widetilde{G}_n}(x \mid y)$ as $f^{(X)}_{\widetilde{G}}(\cdot \mid y)$ in (8) with $\widetilde{G}_n$ in place of $\widetilde{G}$, we can write that

$$\begin{aligned} E_{n-1}(Z_n) &= E_{n-1}\left(\widetilde{\alpha}_n^{-2}(f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}_{n-1}}(x))^2\right) \\ &= \int \left[\int_\Theta k(x \mid \theta) \left(\widetilde{G}_{n-1}(\mathrm{d}\theta \mid y) - \widetilde{G}_{n-1}(\mathrm{d}\theta)\right)\right]^2 f^{(Y)}_{\widetilde{G}_{n-1}}(y)\lambda(dy) \end{aligned}$$

$$= \int \left[f^{(X)}_{\widetilde{G}_{n-1}}(x \mid y) - f^{(X)}_{\widetilde{G}_{n-1}}(x) \right]^2 f^{(Y)}_{\widetilde{G}_{n-1}}(y)\lambda(dy) = v_n(x).$$

Thus, by Lemma B.7,

$$E_{n-1}(Z_n) \to v(x), \quad P-a.s.$$

Moreover, $b_n \uparrow +\infty$ by assumption, and

$$\sum_{k=1}^{\infty} \frac{E(Z_k^2)}{a_k^2 b_k^2} = \sum_{k=1}^{\infty} \widetilde{\alpha}_k^4 b_k^2 E\left(\left[f^{(X)}_{\widetilde{G}_{k-1}}(x \mid Y_k) - f^{(X)}_{\widetilde{G}_{k-1}}(x) \right]^4 \right) < +\infty.$$

Since, by assumption, $b_n \sum_{k\geq n} a_k^{-1} b_k^{-2} \to 1$, then by Theorem B.4 (b),

$$b_n \sum_{k\geq n} (f^{(X)}_{\widetilde{G}_k}(x) - f^{(X)}_{\widetilde{G}_{k-1}}(x))^2 = b_n \sum_{k\geq n} \frac{Z_k}{a_k b_k^2} \to v(x), \quad P\text{-a.s.}$$

It follows by Theorem B.2 that $M_{n,\infty} = b_n^{1/2}(f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}}(x))$ converges to $\mathcal{N}(0, v(x))$ in the sense of almost-sure conditional convergence.
Now suppose that $\widetilde{\alpha}_n = (\alpha + n)^{-\gamma}$. With $b_n = (2\gamma - 1)n^{2\gamma-1}$, we have $b_n \uparrow +\infty$,

$$\sum_{n=1}^{\infty} \widetilde{\alpha}_n^4 b_n^2 = \sum_{n=1}^{\infty} (\alpha + n)^{-4\gamma} (2\gamma - 1)^2 n^{4\gamma-2} < +\infty,$$

$$b_n \sup_{k\geq n} \widetilde{\alpha}_k^2 = (2\gamma - 1)n^{2\gamma-1}(\alpha + n)^{-2\gamma} \to 0,$$

and

$$b_n \sum_{k\geq n} \widetilde{\alpha}_k^2 = (2\gamma - 1)n^{2\gamma-1} \sum_{k\geq n} (\alpha + k)^{-2\gamma} \to 1.$$

□

B.3.2 Proof of Theorem 3.3

Since a stochastic process taking values in $C(\mathbb{R})$, with the topology of uniform convergence on compact sets, converges in distribution if and only if its restrictions to compact sets converge (see e.g. Kallenberg (2002), Proposition 16.6), then, in this section, we consider, for each $n \geq 1$, the restriction of $b_n^{1/2}(f^{(X)}_{\widetilde{G}_n} - f^{(X)}_{\widetilde{G}})$ to a fixed compact interval $I = [a, b]$. With an abuse of notation, we use the same symbol $b_n^{1/2}(f^{(X)}_{\widetilde{G}_n} - f^{(X)}_{\widetilde{G}})$ irrespective of its domain. Let ν_n denote the (random) conditional distribution of $b_n^{1/2}(f^{(X)}_{\widetilde{G}_n} - f^{(X)}_{\widetilde{G}})$, as an element in $C(I)$, given $\mathcal{G}_n$. To prove the theorem, we need to show that ν_n is tight and that the conditional finite dimensional distributions of ν_n converge almost surely to those of a centered Gaussian process $\mathbb{G}_{\widetilde{G}}$ with covariance function (13) (see Theorem 7.1 in Billingsley (1999)).

Lemma B.8. *Let ν_n denote the (random) conditional distribution of $b_n^{1/2}(f^{(X)}_{\widetilde{G}_n} - f^{(X)}_{\widetilde{G}})$, with $(b_n)_{n\geq 1}$ a sequence of strictly positive numbers such that $b_n \uparrow +\infty$, $\sum_{n\geq 1} \widetilde{\alpha}_n^4 b_n^2 < +\infty$, $b_n \sup_{k\geq n} \widetilde{\alpha}_k^2 \to 0$ and $b_n \sum_{k\geq n} \widetilde{\alpha}_k^2 \to 1$. Then (ν_n) is P-a.s. tight.*

Proof. By Kallenberg (2002, Corollary 16.9), it is sufficient to show that there exists a set N with $P(N) = 0$ such that for every $\omega \notin N$ the following conditions hold:

(i) For each positive η, there exist $k = k(\omega) > 0$ and an $n_0 = n_0(\omega)$ such that

$$\sup_{n \geq n_0} P(b_n^{1/2}(f^{(X)}_{\widetilde{G}_n}(a) - f^{(X)}_{\widetilde{G}}(a))| \geq k \mid \mathcal{G}_n)(\omega) \leq \eta. \tag{B.2}$$

(ii) The exists a constant C such that, for every $s, t \in I$ and for n large enough

$$E\left(b_n(f^{(X)}_{\widetilde{G}_n}(s) - f^{(X)}_{\widetilde{G}}(s) - f^{(X)}_{\widetilde{G}_n}(t) + f^{(X)}_{\widetilde{G}}(t))^2 \mid \mathcal{G}_n\right)(\omega) \leq C(s-t)^2. \tag{B.3}$$

The condition (i) holds, since $b_n^{1/2}(f^{(X)}_{\widetilde{g}_n}(a) - f^{(X)}_{\widetilde{g}}(a))$ converges in the sense of almost-sure conditional convergence. To prove (ii) we can write by martingales properties that

$$\begin{aligned}
&E\left(b_n(f^{(X)}_{\widetilde{G}_n}(s) - f^{(X)}_{\widetilde{G}}(s) - f^{(X)}_{\widetilde{G}_n}(t) + f^{(X)}_{\widetilde{G}}(t))^2 \mid \mathcal{G}_n\right) \\
&= b_n E\left(\left(\sum_{k\geq n} f^{(X)}_{\widetilde{G}_{k+1}}(s) - f^{(X)}_{\widetilde{G}_k}(s) - f^{(X)}_{\widetilde{G}_{k+1}}(t) + f^{(X)}_{\widetilde{G}_k}(t)\right)^2 \mid \mathcal{G}_n\right) \\
&= b_n E\left(\sum_{k\geq n}\left(f^{(X)}_{\widetilde{G}_{k+1}}(s) - f^{(X)}_{\widetilde{G}_k}(s) - f^{(X)}_{\widetilde{G}_{k+1}}(t) + f^{(X)}_{\widetilde{G}_k}(t)\right)^2 \mid \mathcal{G}_n\right) \\
&= b_n \sum_{k\geq n} \alpha_k^2 E\left(E\left(\left[f^{(X)}_{\widetilde{G}_k}(t) - f^{(X)}_{\widetilde{G}_k}(s) - f^{(X)}_{\widetilde{G}_k}(t \mid Y_{k+1}) - f^{(X)}_{\widetilde{G}_k}(s \mid Y_{k+1})\right]^2 \mid \mathcal{G}_k\right) \mid \mathcal{G}_n\right) \\
&= b_n \sum_{k\geq n} \alpha_k^2 E\left(\int \left[f^{(X)}_{\widetilde{G}_k}((t) - f^{(X)}_{\widetilde{G}_k}(s) - f^{(X)}_{\widetilde{G}_k}(t \mid y) - f^{(X)}_{\widetilde{G}_k}(s \mid y)\right]^2 f^{(Y)}_{\widetilde{G}_k}(y)\lambda(\mathrm{d}y) \mid \mathcal{G}_n\right) \\
&\leq 2b_n \sum_{k\geq n} \alpha_k^2 E\left(\int \left[f^{(X)}_{\widetilde{G}_k}((t) - f^{(X)}_{\widetilde{G}_k}(s)\right]^2 f^{(Y)}_{\widetilde{G}_k}(y)\lambda(\mathrm{d}y) \mid \mathcal{G}_n\right) \\
&+ 2b_n \sum_{k\geq n} \alpha_k^2 E\left(\int \left[f^{(X)}_{\widetilde{G}_k}(t \mid y) - f^{(X)}_{\widetilde{G}_k}(s \mid y)\right]^2 f^{(Y)}_{\widetilde{G}_k}(y)\lambda(\mathrm{d}y) \mid \mathcal{G}_n\right) \\
&\leq 2b_n \sum_{k\geq n} \alpha_k^2 E\left(\left[\int_\Theta (k(t \mid \theta) - k(s \mid \theta))\widetilde{G}_k(\mathrm{d}\theta)\right]^2 \mid \mathcal{G}_n\right) \\
&+ 2b_n \sum_{k\geq n} \alpha_k^2 E\left(\int \left[\int_\Theta (k(t \mid \theta) - k(s \mid \theta))\widetilde{G}_k(\mathrm{d}\theta \mid y)\right]^2 f_{\widetilde{G}_k}(y)\lambda(\mathrm{d}y) \mid \mathcal{G}_n\right) \\
&\leq 2b_n \sum_{k\geq n} \alpha_k^2 E\left(\int_\Theta (k(t \mid \theta) - k(s \mid \theta))^2 \widetilde{G}_k(\mathrm{d}\theta) \mid \mathcal{G}_n\right) \\
&+ 2b_n \sum_{k\geq n} \alpha_k^2 E\left(\int \int_\Theta (k(t \mid \theta) - k(s \mid \theta))^2 \widetilde{G}_k(\mathrm{d}\theta \mid y) f_{\widetilde{G}_k}(y)\lambda(\mathrm{d}y) \mid \mathcal{G}_n\right) \\
&\leq 4 \sup_{x,\theta} |k'(x \mid \theta)|^2 (s-t)^2,
\end{aligned}$$

where $k'(x \mid \theta)$ denotes the derivative of $k(x \mid \theta)$ with respect to x. Thus, properties (i) and (ii) hold and the sequence (ν_n) is tight, on a set of probability one. □

Lemma B.9. *Let $(b_n)_{n\geq 1}$ be sequence of strictly positive numbers such that $b_n \uparrow +\infty$, $\sum_{n\geq 1} \widetilde{\alpha}_n^4 b_n^2 < +\infty$, $b_n \sup_{k\geq n} \widetilde{\alpha}_k^2 \to 0$ and $b_n \sum_{k\geq n} \widetilde{\alpha}_k^2 \to 1$. Then, for every $J \geq 1$, $x_1, \ldots, x_J \in I$ and*

$u_1, \dots, u_J \in \mathbb{R}$, $b_n^{1/2} \sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_n}(x_i) - f^{(X)}_{\widetilde{G}}(x_i))$, *converges in the sense of almost sure conditional convergence, to a Gaussian kernel, with zero mean and variance* $\sum_{i,j=1}^J u_i u_j R(x_i, x_j)$, *with* R *as in* (13).

Proof. The proof closely follows the steps of the proof of Theorem 3.2; nevertheless, we provide it here for the reader's convenience.

Without loss of generality, we can assume that $\sum_{j=1}^J u_i^2 \leq 1$. The proof is based on Crimaldi (2009, Theorem A.1) (see Theorem B.2 in Section B). Define, for each $n \geq 0$,

$$\mathcal{F}_{n,0} = \mathcal{G}_n, \quad \mathcal{F}_{n,j} = \mathcal{G}_{n+j-1} \text{ for } j \geq 1,$$

$$M_{n,0} = 0, \quad M_{n,j} = b_n^{1/2} \sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_n}(x_i) - f^{(X)}_{\widetilde{G}_{n+j-1}}(x_i)) \text{ for } j \geq 1.$$

For every $n \geq 0$, the sequence $(M_{n,j})_{j \geq 0}$ is a bounded martingale with respect to the filtration $(\mathcal{F}_{n,j})_{j \geq 0}$, converging in L^1 to

$$M_{n,\infty} = b_n^{1/2} \sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_n}(x_i) - f^{(X)}_{\widetilde{G}}(x_i)).$$

Let

$$X_{n,j} = M_{n,j} - M_{n,j-1} = b_n^{1/2} \sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_{n+j-1}}(x_i) - f^{(X)}_{\widetilde{G}_{n+j}}(x_i)),$$

$$X_n^* = \sup_{j \geq 1} |X_{n,j}| = b_n^{1/2} \sup_{j \geq 1} |\sum_{i=1}^J u_j (f^{(X)}_{\widetilde{G}_{n+j}}(x_i) - f^{(X)}_{\widetilde{G}_{n+j-1}}(x_i))|,$$

and

$$U_n = \sum_{j=0}^{\infty} X_{n,j}^2 = b_n \sum_{k \geq n} (\sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_{k+1}}(x) - f^{(X)}_{\widetilde{G}_k}(x)))^2.$$

The thesis follows from Theorem B.2, if we can verify that the following conditions hold:

(a) The sequence (X_n^*) defined by $X_n^* = b_n^{1/2} \sup_{k \geq n} |\sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_{k+1}}(x_i) - f^{(X)}_{\widetilde{G}_k}(x_i)|$ is bounded in L^1 and converges to zero P-a.s. as $n \to +\infty$.

(b) $U_n = b_n \sum_{k \geq n} (\sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_k}(x_i) - f^{(X)}_{\widetilde{G}_{k-1}}(x_i)))^2$ converges P-a.s., as $n \to +\infty$, to

$$\sum_{i,j=1}^J u_i u_j \int_{\mathbb{R}} \left(f^{(X)}_{\widetilde{G}}(x_i | y) - f^{(X)}_{\widetilde{G}}(x_i) \right) \left(f^{(X)}_{\widetilde{G}}(x_j \mid y) - f^{(X)}_{\widetilde{G}}(x_j) \right) f^{(Y)}_{\widetilde{G}}(y) \lambda(dy).$$

Notice that, for every $i = 1, \dots, J$,

$$\begin{aligned} f^{(X)}_{\widetilde{G}_k}(x_i) - f^{(X)}_{\widetilde{G}_{k-1}}(x_i) &= \int k(x_i \mid \theta)(\widetilde{G}_k(\mathrm{d}\theta) - \widetilde{G}_{k-1}(\mathrm{d}\theta)) \\ &= \widetilde{\alpha}_k \int k(x_i \mid \theta) \left(\widetilde{G}_{k-1}(\mathrm{d}\theta \mid Y_k) - \widetilde{G}_{k-1}(\mathrm{d}\theta) \right), \end{aligned}$$

where $\widetilde{G}_{k-1}(\mathrm{d}\theta \mid Y_k) = \widetilde{k}(Y_k \mid \theta)\widetilde{G}_{k-1}(\mathrm{d}\theta)/\int_\Theta \widetilde{k}(Y_k \mid \theta')\widetilde{G}_{k-1}(\mathrm{d}\theta')$ is a probability measure on Θ. Thus,

$$X_n^* \leq b_n^{1/2} \sup_{k\geq n} \widetilde{\alpha}_k \sum_{i=1}^J |u_i| \int k(x_i \mid \theta) \left|\widetilde{G}_{k-1}(\mathrm{d}\theta \mid Y_k) - \widetilde{G}_{k-1}(\mathrm{d}\theta)\right| \leq \sum_{i=1}^J |u_i| \sup_{\theta\in\Theta} k(x_i \mid \theta) b_n^{1/2} \sup_{k\geq n} \widetilde{\alpha}_k$$

is bounded and converges to zero P-a.s., since $b_n \sup_{k\geq n} \widetilde{\alpha}_k^2 \to 0$ by assumption. To prove (b), we apply Crimaldi et al. (2016, Lemma 4.1 (b)) (see Theorem B.4 (b)), with

$$Z_n = \widetilde{\alpha}_n^{-2} \left(\sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_n}(x_i) - f^{(X)}_{\widetilde{G}_{n-1}}(x_i))\right)^2.$$

We can write that

$$\begin{aligned}
E_{n-1}(Z_n) &= E_{n-1}\left(\widetilde{\alpha}_n^{-2}\left(\sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_n}(x_i) - f^{(X)}_{\widetilde{G}_{n-1}}(x_i))\right)^2\right) \\
&= \int \left[\sum_{i=1}^J u_i \int_\Theta k(x_i \mid \theta)\left(\widetilde{G}_{n-1}(\mathrm{d}\theta \mid y) - \widetilde{G}_{n-1}(\mathrm{d}\theta)\right)\right]^2 f^{(Y)}_{\widetilde{G}_{n-1}}(y)\lambda(dy) \\
&\overset{a.s.}{\to} \sum_{i,j=1}^J u_i u_j \int (f^{(X)}_{\widetilde{G}}(x_i \mid y) - f^{(X)}_{\widetilde{G}}(x_i))(f^{(X)}_{\widetilde{G}}(x_j \mid y) - f^{(X)}_{\widetilde{G}}(x_j)) f^{(Y)}_{\widetilde{G}}(y)\mathrm{d}y.
\end{aligned}$$

By assumption $b_n \uparrow +\infty$, and, defining $a_n = \widetilde{\alpha}_n^{-2} b_n^{-2}$, we obtain

$$\begin{aligned}
\sum_{k=1}^\infty \frac{E(Z_k^2)}{a_k^2 b_k^2} &= \sum_{k=1}^\infty \widetilde{\alpha}_k^4 b_k^2 E\left(\left[\sum_{i=1}^j u_i (f^{(X)}_{\widetilde{G}_{k-1}}(x_i \mid Y_k) - f^{(X)}_{\widetilde{G}_{k-1}}(x_i))\right]^4\right) \\
&\leq \sum_{k=1}^\infty \widetilde{\alpha}_k^4 b_k^2 \left[\sum_{i=1}^j u_i \sup_{\theta\in\Theta} k(x_i \mid \theta)\right]^4 < +\infty.
\end{aligned}$$

Since, by assumption, $b_n \sum_{k\geq n} a_k^{-1} b_k^{-2} \to 1$, then by Theorem B.4 (b),

$$\begin{aligned}
b_n \sum_{k\geq n} \frac{Z_k}{a_k b_k^2} &= b_n \sum_{k\geq n} (\sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_k}(x_i) - f^{(X)}_{\widetilde{G}_{k-1}}(x_i)))^2 \\
&\overset{a.s.}{\to} \sum_{i,j=1}^J u_i u_j \int (f^{(X)}_{\widetilde{G}}(x_i \mid y) - f^{(X)}_{\widetilde{G}}(x_i))(f^{(X)}_{\widetilde{G}}(x_j \mid y) - f^{(X)}_{\widetilde{G}}(x_j)) f^{(Y)}_{\widetilde{G}}(y)\mathrm{d}y.
\end{aligned}$$

It follows by Theorem B.2 that $M_{n,\infty} = b_n^{1/2} \sum_{i=1}^J u_i (f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}}(x))$ converges in the sense of almost-sure conditional convergence to a Gaussian kernel with variance $\sum_{i,j=1}^J u_i u_j \int (f^{(X)}_{\widetilde{G}}(x_i \mid y) - f^{(X)}_{\widetilde{G}}(x_i))(f^{(X)}_{\widetilde{G}}(x_j \mid y) - f^{(X)}_{\widetilde{G}}(x_j)) f^{(Y)}_{\widetilde{G}}(y)\mathrm{d}y$. □

Proof of Theorem 3.3. The proof of Theorem 3.3 is a direct consequence of Lemma B.8, Lemma B.9 and Theorem 7.1 in Billingsley (1999). □

B.3.3 Proof of Theorem 3.4

To prove the theorem, we will first provide a bound for $E(\sup_{x\in I}\mathbb{G}_{\widetilde{G}}(x))$ and then use concentration inequalities to bound $\mathbb{G}_{\widetilde{G}}(x)$ uniformly with respect to $x \in I$

Lemma B.10. *Let $\mathbb{G}_{\widetilde{G}}$ be a centered Gaussian process with random covariance function (13). Then*

$$E(\sup_{x\in I}\mathbb{G}_{\widetilde{G}}(x) \mid \widetilde{G}) \leq 12\int_0^{\sigma(I)}\left(\log\left(1+\frac{\lambda(I)}{2\psi^{-1}(z/2)}\right)\right)^{1/2}dz,$$

where $\sigma(I)$ and ψ are defined as in (14) and (15), respectively.

Proof. The proof is based on Massart (2007, Theorem 3.18) (see Theorem B.5). Consider the pseudometric on I induced by R

$$\begin{aligned}
d(x_1,x_2) &= (R(x_1,x_1)+R(x_2,x_2)-2R(x_1,x_2))^{1/2}\\
&= \left(\int (f^{(X)}_{\widetilde{G}}(x_1\mid y)-f^{(X)}_{\widetilde{G}}(x_1)-f^{(X)}_{\widetilde{G}}(x_2\mid y)+f^{(X)}_{\widetilde{G}}(x_2))^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(dy)\right)^{1/2}\\
&= \left(\int (f^{(X)}_{\widetilde{G}}(x_1\mid y)-f^{(X)}_{\widetilde{G}}(x_2\mid y))^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(dy)-(f^{(X)}_{\widetilde{G}}(x_1)-f^{(X)}_{\widetilde{G}}(x_2))^2\right)^{1/2}.
\end{aligned}$$

Let ψ be defined as in (15) and let $\psi^{-1}(t) = \inf\{z : \psi(z) > t\}$. If $|x_1 - x_2| \leq \psi^{-1}(\epsilon/2)$, then $d(x_1, x_2) \leq \epsilon/2$. Hence a uniform grid on I of step $2\psi^{-1}(\epsilon/2)$ allows to build a $\epsilon/2$-covering of I. It follows that

$$N(\epsilon, I) \leq N'(\epsilon/2, I) \leq \left\lceil \frac{\lambda(I)}{2\psi^{-1}(\epsilon/2)}\right\rceil \leq \frac{\lambda(I)}{2\psi^{-1}(\epsilon/2)}+1.$$

Notice that

$$\begin{aligned}
d(x_1,x_2) &\leq \left(\int (f^{(X)}_{\widetilde{G}}(x_1\mid y)-f^{(X)}_{\widetilde{G}}(x_2\mid y))^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(dy)\right)^{1/2}\\
&\leq \left(\int\int_\Theta (k(x_1\mid\theta)-k(x_2\mid\theta))^2\widetilde{G}(\mathrm{d}\theta\mid y) f^{(Y)}_{\widetilde{G}}(y)\lambda(dy)\right)^{1/2}\\
&\leq k'(I)|x_1-x_2|,
\end{aligned}$$

with $k'(I)$ defined as

$$k'(I) = \sup_{x\in I,\theta\in\Theta}|k'(x\mid\theta)|,$$

where $k'(x \mid \theta)$ is the derivative of $k(x \mid \theta)$ with respect to x. Hence $\psi(z) \leq k'(I)z$, and

$$\psi^{-1}(\epsilon/2) \geq \frac{\epsilon}{2k'(I)}.$$

It follows that

$$\sqrt{H(\epsilon,I)} = \sqrt{\log N(\epsilon,I)} \leq \sqrt{\log\left(1+\frac{\lambda(I)}{2\psi^{-1}(\epsilon/2)}\right)} \leq \sqrt{\frac{\lambda(I)k'(I)}{\epsilon}},$$

which implies that $\sqrt{H(\cdot, I)}$ is integrable in 0. Moreover,

$$E(\sup_{x\in I}\mathbb{G}_{\widetilde{G}}(x) \mid \widetilde{G}) \leq 12\int_0^{\sigma(I)}\sqrt{H(z,I)}dz$$

$$\leq 12 \int_0^{\sigma(I)} \sqrt{\log(N(z,I)}dz$$
$$\leq 12 \int_0^{\sigma(I)} \left(\log\left(1+\frac{\lambda(I)}{2\psi^{-1}(z/2)}\right)\right)^{1/2} dz.$$

□

Proof of Theorem 3.4. The proof is based on Lemma B.10 and on the following concentration inequality (see Theorem B.6): for every $z>0$,

$$P(\sup_{x\in I} \mathbb{G}_{\widetilde{G}}(x) \geq E(\sup_{x\in I} \mathbb{G}_{\widetilde{G}}(x) \mid \widetilde{g}) + \sigma(I)\sqrt{2z} \mid \widetilde{G}) \leq e^{-z},$$

where $\sigma(I)$ is defined as in (14). Taking into account the symmetry of the distribution of $\mathbb{G}_{\widetilde{G}}$, we can write that

$$\begin{aligned} P((&\sup_{x\in I} \mathbb{G}_{\widetilde{G}}(x) \geq E(\sup_{x\in I} \mathbb{G}_{\widetilde{G}}(x) \mid \widetilde{G}) + \sigma(I)\sqrt{2z}) \\ &\cup (\inf_{x\in I} \mathbb{G}_{\widetilde{G}}(x) \leq -E(\sup_{x\in I} \mathbb{G}_{\widetilde{G}}(x)|\widetilde{G}) - \sigma(I)\sqrt{2z})|\widetilde{G}) \\ &\leq 2e^{-z}. \end{aligned}$$

Setting $2e^{-z}=\beta$, we obtain that

$$\begin{aligned} &P(\sup_{x\in I} |\mathbb{G}_{\widetilde{G}}(x)| < 12 \int_0^{\sigma(I)} \left(\log\left(1+\frac{\lambda(I)}{2\psi^{-1}(z/2)}\right)\right)^{1/2} dz + \sigma(I)\sqrt{2|\log(\beta/2)|} \mid \widetilde{G}) \\ &\geq P(\sup_{x\in I} |\mathbb{G}_{\widetilde{G}}(x)| < E(\sup_{x\in I} \mathbb{G}_{\widetilde{G}}(x)|\widetilde{G}) + \sigma(I)\sqrt{2|\log(\beta/2)|} \mid \widetilde{G}) \\ &\geq 1-\beta. \end{aligned}$$

□

B.3.4 Proof of Theorem 3.5

The proof is split into several lemmas.

Lemma B.11. *As $n\to+\infty$, $\sigma_n(I)$ converges to $\sigma(I)$, P-a.s.*

Proof. The functions

$$\int (f^{(X)}_{\widetilde{G}_n}(x|y) - f^{(X)}_{\widetilde{G}_n}(x))^2 f^{(Y)}_{\widetilde{G}_n}(y)\lambda(dy) = \int f^{(X)}_{\widetilde{G}_n}(x|y)^2 f^{(Y)}_{\widetilde{G}_n}(y)\lambda(\mathrm{d}y) - f^{(X)}_{\widetilde{G}_n}(x)^2$$

are continuously differentiable in I, with

$$\left|\frac{\mathrm{d}}{\mathrm{d}x} f^{(X)}_{\widetilde{G}_n}(x)^2\right| = 2f^{(X)}_{\widetilde{G}_n}(x)\left|\int_\Theta k'(x\mid\theta)\widetilde{G}_n(\mathrm{d}\theta)\right| \leq 2\sup_{x,\theta} k(x\mid\theta)\sup_{x,\theta}|k'(x\mid\theta)|,$$

and

$$\left|\frac{\mathrm{d}}{\mathrm{d}x}\int f^{(X)}_{\widetilde{G}_n}(x|y)^2 f^{(Y)}_{\widetilde{G}_n}(y)\lambda(\mathrm{d}y)\right| \leq 2\left|\int\int_\Theta k'(x\mid\theta)\widetilde{G}_n(\mathrm{d}\theta\mid y) f^{(X)}_{\widetilde{G}}(x|y) f^{(Y)}_{\widetilde{G}}(y)\lambda(\mathrm{d}y)\right|$$

$$\leq 2 \sup_{x,\theta} k(x \mid \theta) \sup_{x,\theta} |k'(x \mid \theta)|.$$

Thus, they have uniformly bounded derivative, and therefore are equicontinuous. On the other hand they converge pointwise to

$$\int (f^{(X)}_{\widetilde{G}}(x|y) - f^{(X)}_{\widetilde{G}}(x))^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(dy)$$

P-a.s. By Ascoli-Arzelà theorem the convergence is uniform in I, and therefore

$$\begin{aligned}\sup_{x\in I} &\int (f^{(X)}_{\widetilde{G}_n}(x|y) - f^{(X)}_{\widetilde{G}_n}(x))^2 f^{(Y)}_{\widetilde{G}_n}(y)\lambda(dy) \\ &\to \sup_{x\in I} \int (f^{(X)}_{\widetilde{G}}(x|y) - f^{(X)}_{\widetilde{G}}(x))^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(dy)\end{aligned}$$

P-a.s. as $n \to +\infty$. □

Lemma B.12. *For every $z \in [0, \lambda(I)]$, $\psi_n(z)$ converges to $\psi(z)$, P-a.s. as $n \to +\infty$,*

Proof. The functions

$$(x_1, x_2) \mapsto \int (f^{(X)}_{\widetilde{G}_n}(x_1 \mid y) - f^{(X)}_{\widetilde{G}_n}(x_2 \mid y))^2 f^{(Y)}_{\widetilde{G}_n}(y)\lambda(\mathrm{d}y) - (f^{(X)}_{\widetilde{G}_n}(x_1) - f^{(X)}_{\widetilde{G}_n}(x_2))^2$$

are continuously differentiable in $I \times I$, with

$$\begin{aligned}\left| \frac{\partial}{\partial x_1} (f^{(X)}_{\widetilde{G}_n}(x_1) - f^{(X)}_{\widetilde{G}_n}(x_2))^2 \right| &\leq 2(f^{(X)}_{\widetilde{G}_n}(x_1) + f^{(X)}_{\widetilde{G}_n}(x_2)) \int_\Theta |k'(x_1 \mid \theta)| \widetilde{G}_n(\mathrm{d}\theta) \\ &\leq 4 \sup_{x,\theta} k(x \mid \theta) \sup_{x,\theta} |k'(x \mid \theta)|,\end{aligned}$$

and

$$\begin{aligned}&\left| \frac{\partial}{\partial x_1} \int (f^{(X)}_{\widetilde{G}_n}(x_1 \mid y) - f^{(X)}_{\widetilde{G}_n}(x_2 \mid y))^2 f^{(Y)}_{\widetilde{G}_n}(y)\lambda(\mathrm{d}y) \right| \\ &\qquad \leq 2 \int (f^{(X)}_{\widetilde{G}_n}(x_1 \mid y) + f^{(X)}_{\widetilde{G}_n}(x_2 \mid y)) \int |k'(x_1 \mid \theta)| \widetilde{G}_n(\mathrm{d}\theta \mid y) f^{(Y)}_{\widetilde{G}_n}(y)\lambda(\mathrm{d}y) \\ &\qquad \leq 4 \sup_{x,\theta} k(x \mid \theta) \sup_{x,\theta} |k'(x \mid \theta)|,\end{aligned}$$

and analogously for the derivatives with respect to x_2. Therefore, they are equicontinuous in $I \times I$, and converge pointwise to

$$\int (f^{(X)}_{\widetilde{G}}(x_1 \mid y) - f^{(X)}_{\widetilde{G}}(x_2 \mid y))^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(\mathrm{d}y) - (f^{(X)}_{\widetilde{G}}(x_1) - f^{(X)}_{\widetilde{G}}(x_2))^2,$$

P-a.s. By Ascoli-Arzelà Theorem, the convergence is uniform, and therefore

$$\sup_{\substack{x_1, x_2 \in I, \\ |x_1 - x_2| < z}} \left(\int (f^{(X)}_{\widetilde{G}_n}(x_1 \mid y) - f^{(X)}_{\widetilde{G}_n}(x_2 \mid y))^2 f^{(Y)}_{\widetilde{G}_n}(y)\lambda(\mathrm{d}y) - (f^{(X)}_{\widetilde{G}_n}(x_1) - f^{(X)}_{\widetilde{G}_n}(x_2))^2 \right)^{1/2}$$

$$= \Big(\sup_{\substack{x_1,x_2\in I,\\ |x_1-x_2|<z}} \int (f^{(X)}_{\widetilde{G}_n}(x_1\mid y)-f^{(X)}_{\widetilde{G}_n}(x_2\mid y))^2 f^{(Y)}_{\widetilde{G}_n}(y)\lambda(\mathrm{d}y) - (f^{(X)}_{\widetilde{G}_n}(x_1)-f^{(X)}_{\widetilde{G}_n}(x_2))^2\Big)^{1/2}$$

$$\to \Big(\sup_{\substack{x_1,x_2\in I,\\ |x_1-x_2|<z}} \int (f^{(X)}_{\widetilde{G}}(x_1\mid y)-f^{(X)}_{\widetilde{G}}(x_2\mid y))^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(\mathrm{d}y) - (f^{(X)}_{\widetilde{G}}(x_1)-f^{(X)}_{\widetilde{G}}(x_2))^2\Big)^{1/2}$$

$$= \sup_{\substack{x_1,x_2\in I,\\ |x_1-x_2|<z}} \Big(\int (f^{(X)}_{\widetilde{G}}(x_1\mid y)-f^{(X)}_{\widetilde{G}}(x_2\mid y))^2 f^{(Y)}_{\widetilde{G}}(y)\lambda(\mathrm{d}y) - (f^{(X)}_{\widetilde{G}}(x_1)-f^{(X)}_{\widetilde{G}}(x_2))^2\Big)^{1/2}.$$

□

Lemma B.13. *As $n \to +\infty$,*

$$\int_0^{\sigma_n(I)} \left(\log\left(1+\frac{\lambda(I)}{2\psi_n^{-1}(z/2)}\right)\right)^{1/2} dz \to \int_0^{\sigma(I)} \left(\log\left(1+\frac{\lambda(I)}{2\psi^{-1}(z/2)}\right)\right)^{1/2} dz$$

P-a.s.

Proof. For P-a.s. every $\omega\in\Omega$, the functions $\psi_n(\cdot)(\omega)$ and $\psi(\cdot)(\omega)$ are continuous and monotone non decreasing on the closed interval $[0,\lambda(I)]$. Hence, by Lemma B.12, we can find a set N with $P(N)=0$ such that for every $\omega\in N^c$, $\psi_n(\cdot)(\omega)$ converges to $\psi(\cdot)(\omega)$ in $C([0,\lambda(I)])$. In the rest of the proof we restrict to N^c. It can be proved as in Lemma B.10 that

$$\left(\log\left(1+\frac{\lambda(I)}{2\psi_n^{-1}(z/2)}\right)\right)^{1/2} \text{ and } \left(\log\left(1+\frac{\lambda(I)}{2\psi^{-1}(z/2)}\right)\right)^{1/2}$$

are integrable in a right neighborhood of the origin. Moreover

$$\int_0^{\sigma_n(I)} \left(\log\left(1+\frac{\lambda(I)}{2\psi_n^{-1}(z/2)}\right)\right)^{1/2} dz = 2\int_0^{\psi_n^{-1}(\sigma_n(I)/2)} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi_n(t)$$

and

$$\int_0^{\sigma(I)} \left(\log\left(1+\frac{\lambda(I)}{2\psi^{-1}(z/2)}\right)\right)^{1/2} dz = 2\int_0^{\psi^{-1}(\sigma(I)/2)} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi(t),$$

where $d\psi_n$ and $d\psi$ denote the measure associated to the monotone non-decrasing and continuous function ψ_n and ψ, respectively.

Let $u_1,u_2\in[0,\lambda(I)]$ be such $u_1\le u_2$ and $\psi(u)=\sigma(I)/2$ for every $u\in[u_1,u_2]$. If $\underline{u}<u_1$ and $\overline{u}>u_2$, then $\psi(\underline{u})<\sigma(I)/2$ and $\psi(\overline{u})>\sigma(I)/2$, which implies that $\psi_n(\underline{u})<\sigma(I)/2$ and $\psi_n(\overline{u})>\sigma(I)/2$ for n large enough. Since ψ_n converges weakly to ψ, then, for n large enough,

$$\int_0^{\underline{u}} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi(t) \le \int_0^{\psi_n^{-1}(\sigma_n(I)/2)} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi_n(t)$$

and

$$\int_0^{\overline{u}} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi(t) \ge \int_0^{\psi_n^{-1}(\sigma_n(I)/2)} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi_n(t).$$

Hence, for every $\underline{u} < u_1 \le u_2 < \overline{u}$,

$$\int_0^{\underline{u}} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi(t) \le \liminf_n \int_0^{\psi_n^{-1}(\sigma_n(I)/2)} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi_n(t)$$
$$\le \limsup_n \int_0^{\psi_n^{-1}(\sigma_n(I)/2)} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi(t) \le \int_0^{\overline{u}} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi(t).$$

Since the integrals are continuous with respect to the endpoints,

$$\int_0^{u_1} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi(t) \le \liminf_n \int_0^{\psi_n^{-1}(\sigma_n(I)/2)} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi_n(t)$$
$$\le \limsup_n \int_0^{\psi_n^{-1}(\sigma_n(I)/2)} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi(t) \le \int_0^{u_2} \left(\log\left(1+\frac{\lambda(I)}{2t}\right)\right)^{1/2} d\psi(t).$$

Since ψ is constant over the interval $[u_1, u_2]$, then the first and last terms of the above equation are equal. Moreover, by ψ^{-1} and continuity of ψ, $u_2 = \psi^{-1}(\sigma(I)/2)$. □

Proof of Theorem 3.5. The proof is an immediate consequence of Lemma B.11 and Lemma B.13. □

B.3.5 Proof of Equation 18

Since $s_n(I,\beta)$ is measurable with respect to $Y_{1:n}$, $s_{\widetilde{G}}(I,\beta)$ is measurable with respect to $\widetilde{G}$, and $s_n(I,\beta)$ converges to $s_{\widetilde{G}}(I,\beta)$ P-a.s., then, for every $t_1, t_2 \in \mathbb{R}$,

$$E(\exp(i(t_1 s_n(I,\beta) + t_2 \sup_{x\in I} b_n^{1/2} |f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}}(x)|)) \mid \mathcal{G}_n)$$
$$\xrightarrow{P-a.s.} E(\exp(i(t_1 s_{\widetilde{G}}(I,\beta) + t_2 \sup_{x\in I} |\mathbb{G}_{\widetilde{G}}(x))|))|\widetilde{G}).$$

In other words, $(s_n(I,\beta), \sup_{x\in I} b_n^{1/2} |f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}}(x)|)$ converges to $s_{\widetilde{G}}(I,\beta), \sup_{x\in I} |\mathbb{G}_{\widetilde{G}}(x)|)$ in the sense of almost-sure conditional convergence. By the Portmanteau theorem,

$$\liminf_{n\to+\infty} P(\sup_{x\in I} |b_n^{1/2}(f^{(X)}_{\widetilde{G}_n}(x) - f^{(X)}_{\widetilde{G}}(x))| < \max(s_n(I,\beta),\epsilon) \mid \mathcal{G}_n)$$
$$\ge P(\sup_{x\in I} |\mathbb{G}_{\widetilde{G}(\omega)}| < \max(s_{\widetilde{G}}(I,\beta),\epsilon) \mid \widetilde{G})$$
$$\ge 1-\beta.$$

B.4 Proofs for Section 4

B.4.1 Proof of Proposition 4.1

Proof. Under P^*, the random variables Y_n are i.i.d. with density function $f^{(Y)}_{G^*}(y) = \int_\Theta \widetilde{k}(y \mid \theta) G^*(\mathrm{d}\theta)$, where $\widetilde{k}(y \mid \theta)$ identifies the mixing distribution by the assumption A4). The sequence $\widetilde{G}_n$ in (4) represents the recursive estimates of the true mixing distribution G^* obtained via Newton's algorithm for the mixture model $\int_\Theta \widetilde{k}(y \mid \theta)\, G(\mathrm{d}\theta)$. Let $\mathbb{F}$ denote the class of mixing distributions that are absolutely continuous with respect to μ, and let $\overline{\mathbb{F}}$ be its weak closure. We first prove that $\widetilde{G}_n$ converges weakly to G^*, P-a.s.

The proof is based on Corollary 4.7 of Martin and Tokdar (2009). The assumptions in Martin and Tokdar (2009) differ slightly from A1)–A5), F1)–F3). In particular, Martin and Tokdar (2009) requires that $\mathbb{F}$ is precompact, which holds in our setting since Θ is compact, and also that $\sum_n (\sum_{i=1}^n \widetilde{\alpha}_i)\, \widetilde{\alpha}_n^2 < +\infty$, but a careful inspection of the proofs reveals that the latter condition is actually necessary only for the rate of convergence. Thus, we can apply Corollary 4.7 in Martin and Tokdar (2009), and deduce that the sequence $\widetilde{G}_n$ converges weakly to the unique minimizer $G^\dagger$ in $\overline{\mathbb{F}}$ of the Kullback–Leibler divergence between $f_{G^*}^{(X)}$ and $f_G^{(X)}$. Since, by assumption F1), $G^* \in \overline{\mathbb{F}}$, then $G^\dagger = G^*$, and thus G_n converges weakly to G^*.

We now prove that $f_{\widetilde{G}_n}^{(X)}$ converges to $f_{G^*}^{(X)}$ pointwise and in L^1. Since $k(x \mid \theta)$ is bounded and continuous in θ by the assumption A3), then, for every x, $f_{\widetilde{G}_n}^{(X)}(x) = \int_\Theta k(x \mid \theta)\widetilde{G}_n(\mathrm{d}\theta)$ converges to $f_{G^*}^{(X)}(x) = \int_\Theta k(x \mid \theta) G^*(\mathrm{d}\theta)$, P^*-a.s. Since $f_{\widetilde{G}_n}^{(X)}$ is dominated by $\sup_\theta k(x \mid \theta)$ which is integrable by the assumption A3), then $f_{\widetilde{G}_n}^{(X)}$ converges to $f_{G^*}^{(X)}$ in L^1, P-a.s. □

B.4.2 Proof of Corollary 4.2

Proof. By Proposition 4.1, we have

$$\int \left| f_{\widetilde{G}_n}^{(X)}(x) - f_{G^*}^{(X)}(x) \right| \lambda(dx) \to 0, \qquad P^*\text{-a.s.}$$

Moreover, by Corollary 4.6 in Martin and Tokdar (2009), applied to the direct mixture model with kernel k, the recursive estimate based on the latent variables (X_n) satisfies

$$\int \left| f_{G_n}^{(X)}(x) - f_{G^*}^{(X)}(x) \right| \lambda(dx) \to 0, \qquad P^*\text{-a.s.}$$

The result follows immediately by the triangle inequality. □

B.4.3 Proof of Proposition 4.3

For a function f, we denote by $f^{(j)}$ its j-th derivative, with $f^{(0)} := f$. For $\kappa > 0$, let

$$\ell = \max\{j \in \mathbb{N}_0 : j < \kappa\}. \tag{B.4}$$

A kernel K is said to be of order $\ell + 1$ if

$$\int_{\mathbb{R}} K(z)\, dz = 1, \qquad \int_{\mathbb{R}} z^j K(z)\, dz = 0, \quad j = 1, \ldots, \ell + 1.$$

For $h > 0$, let

$$K_h(x) := h^{-1} K(x/h)$$

denote the corresponding rescaled kernel.

The proof of Proposition 4.3 combines the inversion inequality of Theorem 3.1 in Rousseau and Scricciolo (2024) with the convergence-rate result of Corollary 4.10 in Martin and Tokdar (2009). The following two lemmas verify the assumptions required to apply the inversion inequality.

Lemma B.14. *Let Assumption (i) of Proposition 4.3 hold. Let K be a kernel of order $\ell + 1$ satisfying $\int_{\mathbb{R}} |z|^{\kappa+1} |K(z)|\, dz < +\infty$. Then there exists a constant C such that, for every probability measure G on Θ,*

$$||F_G^{(X)} - F_G^{(X)} * K_h||_1 \leq C h^{\kappa+1}, \qquad 0 < h \leq 1, \tag{B.5}$$

where $||\cdot||_1$ *denotes the* $L^1(\mathbb{R})$ *norm, and* C *is a constant that does not depend on* G*. Moreover, the derivative* $f_{G^*}^{(X)^{(\ell)}}$ *satisfies*

$$\left| f_{G^*}^{(X)^{(\ell)}}(x+\delta) - f_{G^*}^{(X)^{(\ell)}}(x) \right| \leq L_0(x)|\delta|^{\kappa-\ell}, \qquad x, \delta \in \mathbb{R}, \tag{B.6}$$

with ℓ *as in* (B.4) *and* $L_0 = M_\kappa \in L^1(\mathbb{R})$*.*

Proof. Differentiation under the integral sign gives

$$f_G^{(X)^{(\ell)}}(x) = \int_\Theta \partial_x^\ell k(x \mid \theta)\, G(d\theta).$$

Hence, by condition (i) of Proposition 4.3,

$$\left| f_G^{(X)^{(\ell)}}(x+\delta) - f_G^{(X)^{(\ell)}}(x) \right| \leq M_\kappa(x)|\delta|^{\kappa-\ell}. \tag{B.7}$$

Integrating with respect to x yields

$$\int_{\mathbb{R}} \left| f_G^{(X)^{(\ell)}}(x+\delta) - f_G^{(X)^{(\ell)}}(x) \right| dx \leq \|M_\kappa\|_1 |\delta|^{\kappa-\ell}, \tag{B.8}$$

uniformly in G. Now, we consider

$$\begin{aligned} ||F_G^{(X)} - F_G^{(X)} * K_h||_1 &= \int_{\mathbb{R}} \left| F_G^{(X)}(x) - \int_{\mathbb{R}} F_G^{(X)}(x-u) K_h(u)\, du \right| dx \\ &= \int_{\mathbb{R}} \left| \int_{\mathbb{R}} \left(F_G^{(X)}(x) - F_G^{(X)}(x-hz) \right) K(z) dz \right| dx, \end{aligned}$$

where the last equality comes from $\int_{\mathbb{R}} K(z)dz = 1$. Moreover,

$$\int_{\mathbb{R}} F_G^{(X)}(x) K(z) dz = \int_{\mathbb{R}} \left(\sum_{j=0}^{\ell+1} \frac{(-hz)^j}{j!} F_G^{(X)^{(j)}}(x) \right) K(z)\, dz,$$

where we have used $\int_{\mathbb{R}} z^j K(z)dz = 0$ for $j = 1, \ldots, \ell+1$.

Now we consider $F_G^{(X)}(x-hz)$. Since $F_G^{(X)(\ell+1)} = f_G^{(X)(\ell)}$, Taylor's formula with integral remainder yields for every $u \in \mathbb{R}$

$$\begin{aligned} F_G^{(X)}(x-u) - \sum_{j=0}^{\ell+1} \frac{(-u)^j}{j!} F_G^{(X)(j)}(x) &= \frac{(-u)^{\ell+2}}{(\ell+1)!} \int_0^1 (1-t)^{\ell+1} F_G^{(X)(\ell+2)}(x-tu)\, dt \\ &= \frac{(-u)^{\ell+2}}{(\ell+1)!} \int_0^1 (1-t)^{\ell+1} f_G^{(X)(\ell+1)}(x-tu)\, dt. \end{aligned} \tag{B.9}$$

Since $\frac{d}{dt} f_G^{(X)(\ell)}(x-tu) = -u\, f_G^{(X)(\ell+1)}(x-tu)$, then integration by parts gives

$$\begin{aligned} &\frac{(-u)^{\ell+2}}{(\ell+1)!} \int_0^1 (1-t)^{\ell+1} f_G^{(X)(\ell+1)}(x-tu)\, dt \\ &= \frac{(-u)^{\ell+1}}{(\ell+1)!} f_G^{(X)(\ell)}(x) - \frac{(-u)^{\ell+1}}{\ell!} \int_0^1 (1-t)^{\ell} f_G^{(X)(\ell)}(x-tu)\, dt \end{aligned}$$

$$= \frac{(-u)^{\ell+1}}{\ell!} \int_0^1 (1-t)^\ell \left(f_G^{(X)(\ell)}(x-tu) - f_G^{(X)(\ell)}(x) \right) dt.$$

Substituting this expression into (B.9) gives

$$F_G^{(X)}(x-u) - \sum_{j=0}^{\ell+1} \frac{(-u)^j}{j!} F_G^{(X)(j)}(x) = \frac{(-u)^{\ell+1}}{\ell!} \int_0^1 (1-t)^\ell \left[f_G^{(X)(\ell)}(x-tu) - f_G^{(X)(\ell)}(x) \right] dt.$$

Thus, by Assumption (i),

$$\begin{aligned} \|F_G^{(X)} - F_G^{(X)} * K_h\|_1 &\leq \int_{\mathbb{R}} \left| \int_{\mathbb{R}} \frac{(-hz)^{\ell+1}}{\ell!} \int_0^1 (1-t)^\ell \left[f_G^{(X)(\ell)}(x-thz) - f_G^{(X)(\ell)}(x) \right] dt K(z)\, dz \right| dx \\ &\leq C_\kappa \|M_\kappa\|_1 h^{\kappa+1} \int_{\mathbb{R}} |z|^{\kappa+1} |K(z)|\, dz, \end{aligned}$$

which proves (B.5). To prove (B.6), take $G = G^*$ in (B.7). This yields

$$\left| f_{G^*}^{(X)^{(\ell)}}(x+\delta) - f_{G^*}^{(X)^{(\ell)}}(x) \right| \leq M_\kappa(x) |\delta|^{\kappa-\ell}.$$

Thus (B.6) holds with $L_0 = M_\kappa \in L^1(\mathbb{R})$. □

Lemma B.15. *Assume that Z has Laplace distribution and that, for some $a \in (0,1)$,*

$$\sup_{\theta \in \Theta} \int_{\mathbb{R}} e^{a|x|} k(x \mid \theta)\, dx < +\infty. \tag{B.10}$$

If, for some sequence $\widetilde{\epsilon}_n \to 0$,

$$\left\| f_{\widetilde{G}_n}^{(Y)} - f_{G^*}^{(Y)} \right\|_1 \leq \widetilde{\epsilon}_n,$$

then

$$W_1\left(F_{\widetilde{G}_n}^{(Y)}, F_{G^*}^{(Y)} \right) \leq C\, \widetilde{\epsilon}_n \log\left(\frac{1}{\widetilde{\epsilon}_n} \right), \tag{B.11}$$

for a constant C not depending on $\widetilde{G}_n$.

Proof. By equation (B.10),

$$\sup_G \int_{\mathbb{R}} e^{a|x|} f_G^{(X)}(x)\, dx \leq \sup_{\theta \in \Theta} \int_{\mathbb{R}} e^{a|x|} k(x \mid \theta)\, dx < +\infty.$$

Since $a < 1$ and Z has Laplace distribution, then $E([e^{a|Z|}) < +\infty$. Therefore,

$$\sup_G \int_{\mathbb{R}} e^{a|y|} f_G^{(Y)}(y)\, dy \leq \sup_G E_G(e^{a|X|}) E(e^{a|Z|}) < +\infty.$$

Let

$$C_0 := \sup_G \int_{\mathbb{R}} e^{a|y|} f_G^{(Y)}(y)\, dy < +\infty.$$

For every $R > 0$,

$$\sup_G \int_{|y|>R} |y| f_G^{(Y)}(y)\, dy = \sup_G \int_{|y|>R} (|y| e^{-a|y|}) e^{a|y|} f_G^{(Y)}(y)\, dy$$

$$\leq \sup_{y>R}(ye^{-ay}) \sup_{G} \int_{|y|>R} e^{a|y|} f_G^{(Y)}(y)\, dy$$
$$\leq C_0 \sup_{y>R}(ye^{-ay}).$$

Defining $y = R + s$ with $s \geq 0$, we can write

$$\begin{aligned}
\sup_{y>R}(ye^{-ay}) &= e^{-aR} \sup_{s>0}((R+s)e^{-as}) \\
&\leq e^{-aR}\left(R + \sup_{s>0} se^{-as}\right) \\
&= e^{-aR}\left(R + \frac{1}{ae}\right) \\
&\leq C_a(1+R)e^{-aR},
\end{aligned}$$

where $C_a = \max(1, 1/(ae))$, and we have used $\sup_{s>0} se^{-as} = \frac{1}{ae}$. Thus,

$$\begin{aligned}
\sup_{G} \int_{|y|>R} |y| f_G^{(Y)}(y)\, dy &\leq C_0 C_a (1+R)e^{-aR} \\
&= C_1(1+R)e^{-aR}, \qquad \text{(B.12)}
\end{aligned}$$

Since $W_1\left(F_G^{(Y)}, F_{G^*}^{(Y)}(y)\right) = \int_{\mathbb{R}} \left|F_G^{(Y)}(y) - F_{G^*}^{(Y)}(y)\right| dy$, then there exists a constant C_2 such that

$$W_1\left(F_G^{(Y)}(y), F_{G^*}^{(Y)}\right) \leq C_2 R \left\|f_G^{(Y)} - f_{G^*}^{(Y)}\right\|_1 + C_1(1+R)e^{-aR},$$

where we have used (B.12) and

$$\begin{aligned}
\int_{-R}^{R} \left|F_G^{(Y)}(y) - F_{G^*}^{(Y)}(y)\right| dy &\leq 2R \sup_{y\in\mathbb{R}} \left|F_G^{(Y)}(y) - F_{G^*}^{(Y)}(y)\right| \\
&= 2R \sup_{y\in\mathbb{R}} \left|\int_{-\infty}^{y} \left(f_G^{(Y)}(u) - f_{G^*}^{(Y)}(u)\right) du\right| \\
&\leq 2R \int_{\mathbb{R}} \left|f_G^{(Y)}(u) - f_{G^*}^{(Y)}(u)\right| du \\
&= 2R \left\|f_G^{(Y)} - f_{G^*}^{(Y)}\right\|_1.
\end{aligned}$$

Choose

$$R = R_n := \frac{1}{a} \log\left(\frac{1}{\widetilde{\epsilon}_n}\right).$$

Since $\widetilde{\epsilon}_n \to 0$, we have $R_n > 0$ for all sufficiently large n. Moreover,

$$e^{-aR_n} = \exp\left(-\log\left(\frac{1}{\widetilde{\epsilon}_n}\right)\right) = \widetilde{\epsilon}_n.$$

Therefore,

$$\begin{aligned}
W_1\left(F_G^{(Y)}, F_{G^*}^{(Y)}\right) &\leq C_2 R_n \left\|f_G^{(Y)} - f_{G^*}^{(Y)}\right\|_1 + C_1(1+R_n)e^{-aR_n} \\
&\leq C_2 R_n \widetilde{\epsilon}_n + C_1(1+R_n)e^{-aR_n}
\end{aligned}$$

$$= \frac{C_2}{a}\widetilde{\epsilon}_n \log\left(\frac{1}{\widetilde{\epsilon}_n}\right) + C_1\left[1 + \frac{1}{a}\log\left(\frac{1}{\widetilde{\epsilon}_n}\right)\right]\widetilde{\epsilon}_n$$
$$= C_1\widetilde{\epsilon}_n + \frac{C_1 + C_2}{a}\widetilde{\epsilon}_n \log\left(\frac{1}{\widetilde{\epsilon}_n}\right).$$

Since $\widetilde{\epsilon}_n \to 0$, then for n sufficiently large, $\widetilde{\epsilon}_n \leq e^{-1}$. Thus,

$$W_1\left(F_G^{(Y)}, F_{G^*}^{(Y)}\right) \leq \left(C_1 + \frac{C_1 + C_2}{a}\right)\widetilde{\epsilon}_n \log\left(\frac{1}{\widetilde{\epsilon}_n}\right)$$
$$\leq C\,\widetilde{\epsilon}_n \log\left(\frac{1}{\widetilde{\epsilon}_n}\right),$$

where C is a constant that does not depend on G. □

Proof of Proposition 4.3. By Corollary 4.10 in Martin and Tokdar (2009),

$$\rho(f_{\widetilde{G}_n}^{(Y)}, f_{G^*}^{(Y)}) = o_{P^*}\left(n^{-\frac{1-\gamma}{2}}\right),$$

where ρ denotes the Hellinger distance. Since the L^1 distance is dominated by the Hellinger distance, we can write

$$\left\| f_{\widetilde{G}_n}^{(Y)} - f_{G^*}^{(Y)} \right\|_1 = O_{P^*}(\widetilde{\epsilon}_n). \tag{B.13}$$

with $\widetilde{\epsilon}_n = o(n^{-\frac{1-\gamma}{2}})$.

By Lemma B.14, the condition (23) implies that $f_{G^*}^{(X)}$ satisfies Assumption 3.3 and Equation (3.6) of Rousseau and Scricciolo (2024). The Laplace distribution is ordinary smooth of order $\beta = 2$. Therefore, the inversion inequality in Theorem 3.1 of Rousseau and Scricciolo (2024) yields, for every $h \in (0, 1]$,

$$W_1\left(F_{\widetilde{G}_n}^{(X)}, F_{G^*}^{(X)}\right) \leq C_1\left(h^{\kappa+1} + W_1\left(F_{\widetilde{G}_n}^{(Y)}, F_{G^*}^{(Y)}\right) + h^{-1}||f_{\widetilde{G}_n}^{(Y)} - f_{G^*}^{(Y)}||_1\right), \tag{B.14}$$

for a constant C_1. By Lemma B.15, conditions (24) and (B.13) imply that

$$W_1\left(F_{\widetilde{G}_n}^{(Y)}, F_{G^*}^{(Y)}\right) \leq C\left(\widetilde{\epsilon}_n \log\frac{1}{\widetilde{\epsilon}_n}\right), \tag{B.15}$$

for a constant C not depending on G_n. Combining (B.13), (B.15), and (B.14), we obtain for every $h \in (0, 1]$

$$W_1\left(F_{\widetilde{G}_n}^{(X)}, F_{G^*}^{(X)}\right) \leq C_2\left(h^{\kappa+1} + \widetilde{\epsilon}_n \log\frac{1}{\widetilde{\epsilon}_n} + h^{-1}\widetilde{\epsilon}_n\right),$$

for a constant C_2. Setting $h = h_n = \widetilde{\epsilon}_n^{1/(\kappa+2)}$, we get

$$h_n^{\kappa+1} = h_n^{-1}\widetilde{\epsilon}_n = \widetilde{\epsilon}_n^{\frac{\kappa+1}{\kappa+2}},$$

so that

$$W_1\left(F_{\widetilde{G}_n}^{(X)}, F_{G^*}^{(X)}\right) = O_{P^*}\left(\widetilde{\epsilon}_n^{\frac{\kappa+1}{\kappa+2}} + \widetilde{\epsilon}_n \log\frac{1}{\widetilde{\epsilon}_n}\right).$$

Since

$$\frac{\widetilde{\epsilon}_n \log(1/\widetilde{\epsilon}_n)}{\widetilde{\epsilon}_n^{(\kappa+1)/(\kappa+2)}} = \widetilde{\epsilon}_n^{1/(\kappa+2)} \log\frac{1}{\widetilde{\epsilon}_n} \longrightarrow 0,$$

as $n \to +\infty$, then $\widetilde{\epsilon}_n \log(1/\widetilde{\epsilon}_n)$ is asymptotically negligible with respect to $\widetilde{\epsilon}_n^{(\kappa+1)/(\kappa+2)}$. It follows that

$$W_1\left(F^{(X)}_{\widetilde{G}_n}, F^{(X)}_{G^*}\right) = O_{P^*}\left(\widetilde{\epsilon}_n^{\frac{\kappa+1}{\kappa+2}}\right) = o_{P^*}\left(n^{-\frac{(1-\gamma)(\kappa+1)}{2(\kappa+2)}}\right).$$

□

B.4.4 Proof of Corollary 4.4

Proof. Set

$$r_n := n^{-\frac{(1-\gamma)(\kappa+1)}{2(\kappa+2)}}.$$

By Proposition 4.3,

$$W_1\left(F^{(X)}_{\widetilde{G}_n}, F^{(X)}_{G^*}\right) = o_{P^*}(r_n). \tag{B.16}$$

Under conditions (27)–(30), Theorem 4.3 and Corollary 4.1 of Rousseau and Scricciolo (2024) imply that there exist constants $M_B > 0$ and $q_B > 0$ such that

$$W_1\left(\widehat{F}^B_n, F^{(X)}_{G^*}\right) = O_{P^*}\left(n^{-1/5}(\log n)^{q_B}\right). \tag{B.17}$$

Since $\gamma \in (2/3, 1)$ and $\kappa > 0$,

$$\frac{(1-\gamma)(\kappa+1)}{2(\kappa+2)} < \frac{1}{6} < \frac{1}{5}.$$

Consequently, for every fixed $q_B > 0$,

$$\frac{n^{-1/5}(\log n)^{q_B}}{r_n} = n^{-\frac{1}{5}+\frac{(1-\gamma)(\kappa+1)}{2(\kappa+2)}}(\log n)^{q_B} \to 0. \tag{B.18}$$

It follows from (B.17) and (B.18) that

$$W_1\left(\widehat{F}^B_n, F^{(X)}_{G^*}\right) = o_{P^*}(r_n). \tag{B.19}$$

Finally, by the triangle inequality,

$$W_1\left(F^{(X)}_{\widetilde{G}_n}, \widehat{F}^B_n\right) \le W_1\left(F^{(X)}_{\widetilde{G}_n}, F^{(X)}_{G^*}\right) + W_1\left(\widehat{F}^B_n, F^{(X)}_{G^*}\right).$$

Combining (B.16) and (B.19) proves (31). □

C Other numerical experiments and applications

C.1 Synthetic-data analysis

C.1.1 Unimodal example

Synthetic-data generation For each $n \in \{250; 500; 750; 1,000\}$ we generate random variables $X_1, \ldots, X_n$ i.i.d. according to the density $f^{(X)}(\cdot) = \phi(\cdot \mid 2, 2)$, where $\phi(\cdot \mid \mu, \sigma^2)$ denotes the Gaussian density with mean μ and variance σ^2. Using the mixture representation (1), we write $f^{(X)}$ as $f^{(X)}_{G^*}$, with $G^* = \delta_{(2,2)}$ the true mixing distribution. We consider ordinary-smooth and super-smooth distributions for the noise random variables Z_i's. In the ordinary-smooth case, $Z_i \overset{\text{iid}}{\sim} \text{Laplace}(0, b_l)$ with $b_l = \frac{\sigma_l}{\sqrt{2}}$ and $\sigma_l \in \{0.25, 0.50\}$, namely σ_l is the standard deviation of the

Laplace noise; in the super-smooth case, $Z_i \overset{\text{iid}}{\sim} N(0, \sigma_g^2)$ with standard deviation $\sigma_g \in \{0.25, 0.50\}$. The Z_i's are independent of the X_i's, and the observations are modeled as follows:

$$Y_i = X_i + Z_i, \qquad i = 1, \ldots, n.$$

For each n, the same $Y_{1:n} = (Y_1, \ldots, Y_n)$ is used for all the methods under comparison in this section.

Quasi-Bayesian estimation and uncertainty quantification To implement the quasi-Bayesian approach for estimating $f_G^{(X)}$, we consider Newton's algorithm (4) with the Gaussian kernel $k(\cdot \mid \theta) = \phi(\cdot \mid \mu, \sigma^2)$, where $\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$. The parameter space $\mathbb{R} \times \mathbb{R}^+$ is restricted to $\Theta = [-20, 20] \times [0.1, 5]$, and it is discretized using increments of 0.1 in both coordinates, giving $401 \times 50 = 20,050$ grid points. We use the learning rate $\widetilde{\alpha}_i = (1+i)^{-1}$, for $i \geq 1$, and set $\widetilde{G}_0$ as the Uniform distribution on Θ. All integrals with respect to the mixing distribution are evaluated by the two-dimensional trapezoidal quadrature, and the mixing distribution $\widetilde{G}_n$ is numerically renormalized after each update. The updates use the closed-form Gaussian–Gaussian or Gaussian–Laplace convolution kernel $\widetilde{k}(\cdot \mid \theta) = (f_Z * k(\cdot \mid \theta))(\cdot)$, depending on the choice of the noise distribution with density f_Z. The estimate of $f_G^{(X)}$ is computed on the grid $\mathcal{X} = \{-12, -11.9, \ldots, 12\}$ as

$$f^{(X)}_{\widetilde{G}_n}(x) = \int_\Theta \phi(x \mid \mu, \sigma^2) \widetilde{G}_n(\mathrm{d}\mu, \mathrm{d}\sigma^2), \qquad x \in \mathcal{X}.$$

The discretization of the parameter space Θ is used only for numerical evaluation, through the implementation of the two-dimensional trapezoidal quadrature, and it imposes no modeling restrictions. Newton's algorithm depends on the order in which the observations Y_i's are processed. We therefore repeat the estimation procedure over $R_n = n/10$ random permutations of $Y_{1:n}$, which are generated using random seed 123. If $\widetilde{G}_n^{(r)}$ denotes the Newton's mixing-distribution estimate obtained from the r-th permutation, $r = 1, \ldots, R_n$, then we form the order-averaged estimate

$$\overline{G}_n = \frac{1}{R_n} \sum_{r=1}^{R_n} \widetilde{G}_n^{(r)}, \tag{C.20}$$

which is renormalized (numerically) after averaging. Therefore, based on (3), the quasi-Bayes estimate of $f_G^{(X)}$ is

$$f^{(X)}_{\overline{G}_n}(x) = \int_\Theta \phi(x \mid \mu, \sigma^2) \overline{G}_n(\mathrm{d}\mu, \mathrm{d}\sigma^2), \qquad x \in \mathcal{X}. \tag{C.21}$$

The quasi-Bayes asymptotic credible intervals and bands for $f_G^{(X)}$ are also computed from the order-averaged estimate $\overline{G}_n$, based on (12) and (19), respectively. In particular, let $\overline{v}_n(x) = v_n(x; \overline{G}_n)$ denote the plug-in version of the variance in (10), which is evaluated by replacing the unknown mixing distribution with the estimate $\overline{G}_n$. Since $\gamma = 1$, we have $b_n = n$, and the lower and upper limits of the 95% asymptotic credible interval are, at a given $x \in \mathcal{X}$,

$$L_{n,\mathrm{as}}(x) = \max\left\{0,\ f^{(X)}_{\overline{G}_n}(x) - \frac{z_{0.975}}{\sqrt{n}} \left[\max\left\{\overline{v}_n(x), 10^{-8}\right\}\right]^{1/2}\right\}$$

and

$$U_{n,\mathrm{as}}(x) = f^{(X)}_{\overline{G}_n}(x) + \frac{z_{0.975}}{\sqrt{n}} \left[\max\left\{\overline{v}_n(x), 10^{-8}\right\}\right]^{1/2},$$

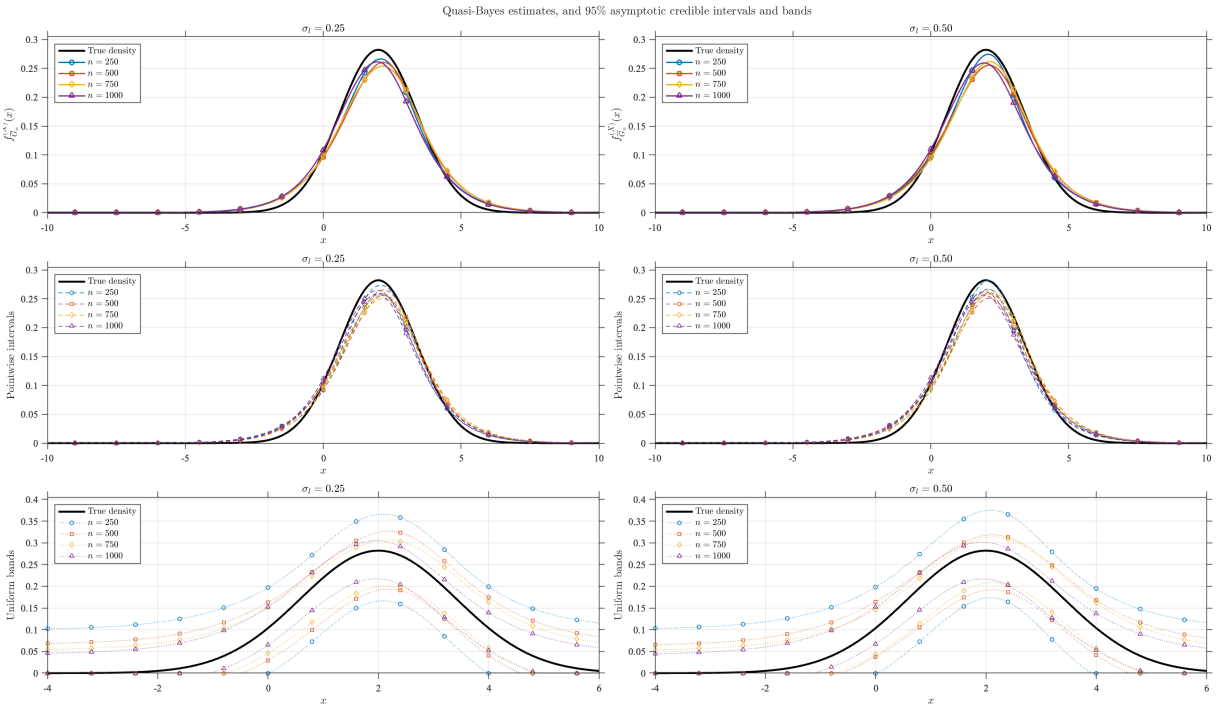


Figure 5: Unimodal example, with Laplace noise: estimates, credible intervals and bands.

respectively. From these lower and upper limits, the 95% asymptotic credible interval (12), at a given $x \in \mathcal{X}$, is

$$[L_{n,\mathrm{as}}(x), U_{n,\mathrm{as}}(x)]\,. \tag{C.22}$$

Similarly, we construct the 95% asymptotic credible band on $I = [-4, 6]$. Let $\overline{s}_n(I, 0.05) = s_n(I, 0.05; \overline{G}_n)$ be the plug-in version of (17), evaluated using $\overline{G}_n$, and define $\overline{\rho}_n(I) = \frac{1}{\sqrt{n}} \max\{\overline{s}_n(I, 0.05), 10^{-8}\}$. The lower and upper functions of the 95% asymptotic credible band are, for every $x \in I$,

$$L^I_{n,\mathrm{as}}(x) = \max\left\{0,\ f^{(X)}_{\overline{G}_n}(x) - \overline{\rho}_n(I)\right\}$$

and

$$U^I_{n,\mathrm{as}}(x) = f^{(X)}_{\overline{G}_n}(x) + \overline{\rho}_n(I),$$

respectively. From these lower and upper functions, the 95% asymptotic credible band (19), for every $x \in I$, is

$$\left\{h \in C(I) : L^I_{n,\mathrm{as}}(x) \le h(x) \le U^I_{n,\mathrm{as}}(x) \text{ for every } x \in I\right\}. \tag{C.23}$$

All integrals entering $\overline{v}_n(x)$ and $\overline{s}_n(I, 0.05)$ are approximated by means of the trapezoidal quadrature. In particular, all integrals are evaluated on the same grid that is used for the estimate $f^{(X)}_{\overline{G}_n}(x)$, whereas all quantities entering the credible bands are restricted to $\mathcal{X}_I = \mathcal{X} \cap I$.

Figures 5–6 report the quasi-Bayes estimates of $f^{(X)}_G$ in (C.21), and the asymptotic credible intervals and bands in (C.22)–(C.23). The figures consist of two columns, which correspond to the noise standard deviations 0.25 and 0.50, respectively. The first row of each figure compares the true density $f^{(X)}_{G^*}$ with the quasi-Bayes density estimates, while the second row displays the credible intervals, and the third row displays the credible bands on the interval $I = [-4, 6]$.

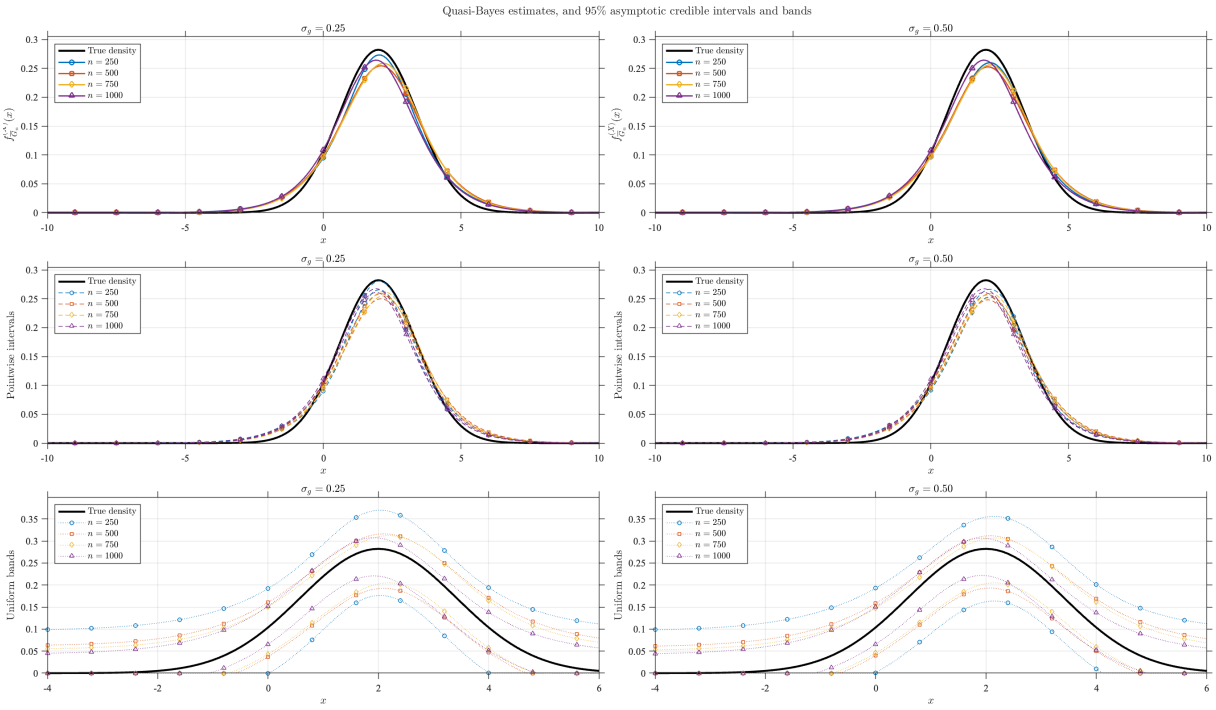


Figure 6: Unimodal example, with Gaussian noise: estimates, credible intervals and bands.

Monte Carlo uncertainty quantification We approximate the conditional distribution of $f^{(X)}_{\widetilde{G}}$, given $Y_{1:n}$, by continuing the quasi-Bayesian learning process (2)–(4), with the Gaussian kernel $k(\cdot \mid \theta) = \phi(\cdot \mid \mu, \sigma^2)$, where $\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$. The continuations start from the same order-averaged estimate $\overline{G}_n$ in (C.20). For each $b = 1, \ldots, B$, we simulate a continuation of length $N_{\text{Future}} = 10,000$. More precisely, at the continuation step s, we draw

$$\theta^{(b)}_{n+s} \sim \widetilde{G}^{(b)}_{n+s-1}, \qquad X^{(b)}_{n+s} \mid \theta^{(b)}_{n+s} \sim k(\cdot \mid \theta^{(b)}_{n+s}),$$

generate $Z^{(b)}_{n+s}$ from the noise distribution under consideration, set $Y^{(b)}_{n+s} = X^{(b)}_{n+s} + Z^{(b)}_{n+s}$, and update $\widetilde{G}^{(b)}_{n+s-1}$ using (4) with learning rate $(1+n+s)^{-1}$. The resulring $\widetilde{G}^{(b)}_{n+N_{\text{Future}}}$ is then used to compute

$$f_b(x) := \int_\Theta \phi(x \mid \theta) \widetilde{G}^{(b)}_{n+N_{\text{Future}}}(\mathrm{d}\theta), \qquad x \in \mathcal{X},$$

with the integral being evaluated numerically using trapezoidal quadrature. Therefore, the b-th Monte Carlo replication produces the density $f_b(x)$, for $x \in \mathcal{X}$. We make use of $B = 1000$ Monte Carlo replications for both the noise distributions, generated using random seed 123.

According to the above Monte Carlo sampling scheme, a Monte Carlo estimate of $f^{(X)}_G$ is

$$f^{(X)}_{\text{MC},n}(x) = \frac{1}{B} \sum_{b=1}^{B} f_b(x), \qquad x \in \mathcal{X}. \tag{C.24}$$

For $\tau \in (0,1)$, and for a given $x \in \mathcal{X}$, we define $q^{\text{MC}}_{\tau,n}(x) = \text{Quantile}_\tau \{f_b(x) : b = 1, \ldots, B\}$, namely the empirical τ-quantile of $f_1(x), \ldots, f_B(x)$. The 95% Monte Carlo credible interval at a given $x \in \mathcal{X}$ is

$$\left[q^{\text{MC}}_{0.025,n}(x), q^{\text{MC}}_{0.975,n}(x)\right]. \tag{C.25}$$

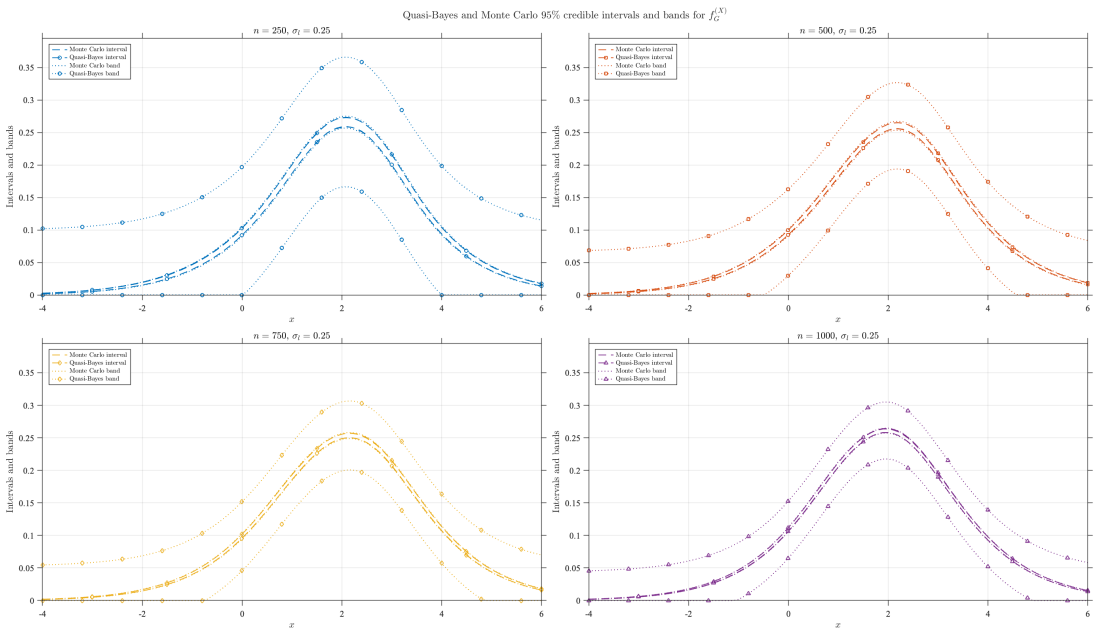


Figure 7: Unimodal example, with Laplace noise ($\sigma_l = 0.25$): quasi-Bayes versus Monte Carlo credible intervals and bands.

Similarly, to construct the Monte Carlo credible band on $I = [-4, 6]$, for every $x \in \mathcal{X}_I$ let $\overline{s}_{\mathrm{MC},n}(x)$ denote the empirical standard deviation of $f_1(x), \ldots, f_B(x)$ and define $T_b = \max_{x \in \mathcal{X}_I} \frac{\left| f_b(x) - f^{(X)}_{\mathrm{MC},n}(x) \right|}{\max\left\{ \overline{s}_{\mathrm{MC},n}(x), 10^{-10} \right\}}$, for $b = 1, \ldots, B$. Let $c_{0.95}$ be the empirical 0.95-quantile of $T_1, \ldots, T_B$. The lower and upper functions of the 95% Monte Carlo credible band are, for every $x \in I$,

$$L^I_{\mathrm{MC},n}(x) = \max\left\{0,\ f^{(X)}_{\mathrm{MC},n}(x) - c_{0.95}\overline{s}_{\mathrm{MC},n}(x)\right\}$$

and

$$U^I_{\mathrm{MC},n}(x) = f^{(X)}_{\mathrm{MC},n}(x) + c_{0.95}\overline{s}_{\mathrm{MC},n}(x),$$

respectively. From these lower and upper functions, the 95% Monte Carlo credible band, for every $x \in I$ is

$$\left[L^I_{\mathrm{MC},n}(x), U^I_{\mathrm{MC},n}(x)\right]. \tag{C.26}$$

Figures 7–10 display Monte Carlo credible intervals and bands in (C.25)–(C.26), which are compared with the corresponding quasi-Bayes asymptotic credible intervals and bands in (C.22)–(C.23). The quasi-Bayesian credible intervals closely agree with their Monte Carlo counterparts, whereas the quasi-Bayesian credible bands are wider and thus more conservative, due to the metric-entropy and Gaussian concentration bounds used in their construction.

Dirichlet process mixture model benchmarks We compare the quasi-Bayesian approach with a Bayesian approach in which the mixing distribution G is endowed with a Dirichlet process prior (Ferguson, 1973; Lo, 1984). Consider the Gaussian kernel $k(\cdot \mid \theta) = \phi(\cdot \mid \mu, \sigma^2)$, where

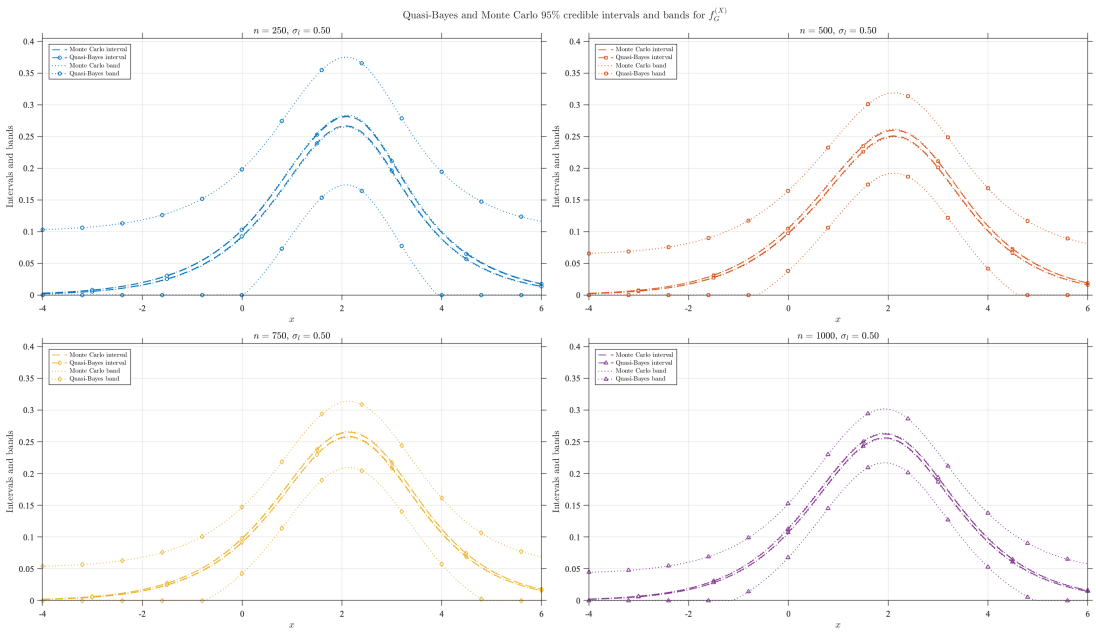


Figure 8: Unimodal example, with Laplace noise ($\sigma_l = 0.50$): quasi-Bayes versus Monte Carlo credible intervals and bands.

$\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$, and assume the hierarchical model

$$\begin{aligned} Y_i &= X_i + Z_i, \qquad i = 1, \ldots, n \\ X_i \mid \theta_i &\sim k(\cdot \mid \theta_i), \\ \theta_i \mid G &\overset{\text{iid}}{\sim} G, \\ G \mid M &\sim \mathrm{DP}(M, H), \\ M &\sim \mathrm{Gamma}(1, 1), \end{aligned} \tag{C.27}$$

where $\mathrm{Gamma}(\cdot, \cdot)$ is the Gamma distribution, and $DP(\cdot, \cdot)$ is the law of a Dirichlet process with strength parameter $M > 0$ and non-atomic (diffuse) base probability measure H (Ferguson, 1973). We set H to be the Uniform distribution on $\Theta = [-20, 20] \times [0.1, 5]$. Inference is based on the closed-form Gaussian–Gaussian or Gaussian–Laplace convolution kernel $\widetilde{k}(\cdot \mid \theta_i) = (f_Z * k(\cdot \mid \theta_i))(\cdot)$, depending on the choice of the noise distribution with density f_Z.

For the Dirichlet process Gaussian-mixture model (C.27), we perform sequential posterior inference using the sequential Monte Carlo (SMC) algorithm of Fearnhead (2004); see also Griffin (2017, Algorithm 1) and references therein. The algorithm uses $P = 500$ particles. At time i, the p-th particle is characterized by $K_{i-1}^{(p)}$ occupied atoms, their corresponding counts, the strength parameter $M_{i-1}^{(p)}$, and the normalized weight $W_{i-1}^{(p)}$. The particle first draws a candidate atom $\theta_{\text{new},i}^{(p)} \sim H$, such that the allocation weights for the occupied components are $a_{i,j}^{(p)} = n_{j,i-1}^{(p)} \widetilde{k}(Y_i \mid \theta_{j,i-1}^{(p)})$, for $j = 1, \ldots, K_{i-1}^{(p)}$, and the allocation weight for the candidate is $a_{i,0}^{(p)} = M_{i-1}^{(p)} \widetilde{k}(Y_i \mid \theta_{\text{new},i}^{(p)})$. The observation is assigned to one of these components with probabilities proportional to the corresponding weights. Selecting the proposed atom creates a new occupied component with count one. Letting $A_i^{(p)} = a_{i,0}^{(p)} + \sum_{j=1}^{K_{i-1}^{(p)}} a_{i,j}^{(p)}$ the normalizing constant, the incremental importance-weight update is $\widetilde{W}_i^{(p)} = W_{i-1}^{(p)} \frac{A_i^{(p)}}{M_{i-1}^{(p)} + i - 1}$ after which the particle weights are normalized. Systematic resampling is performed whenever $\mathrm{ESS}_i = \{\sum_{p=1}^{P} (W_i^{(p)})^2\}^{-1} < 250$. Following resampling, the weights

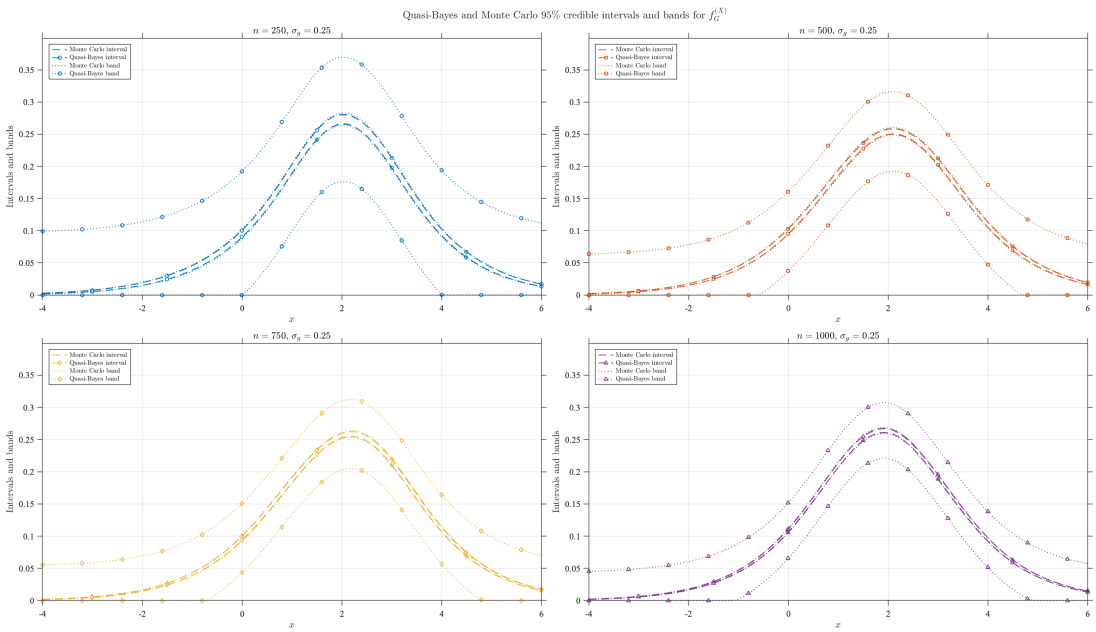


Figure 9: Unimodal example, with Gaussian noise ($\sigma_g = 0.25$): quasi-Bayes versus Monte Carlo credible intervals and bands.

are reset to $1/P$ and $M_i^{(p)}$ is updated once in every particle using the Escobar–West step (Escobar and West, 1995). One additional Escobar–West step is performed at the end of the sequential pass. For the p-th terminal particle, the posterior estimate of the mixing distribution G is

$$\widehat{G}_n^{(p)}(\cdot) = \frac{M_n^{(p)} H(\cdot) + \sum_{j=1}^{K_n^{(p)}} n_{j,n}^{(p)} \delta_{\theta_{j,n}^{(p)}}(\cdot)}{M_n^{(p)} + n},$$

such that, by letting $m_H(x) = \int_\Theta k(x \mid \theta) H(\mathrm{d}\theta)$, the induced posterior estimate of $f_G^{(X)}$ is given by

$$f_{p,n}^{(X)}(x) = \int_\Theta k(x \mid \theta) \widehat{G}_n^{(p)}(\mathrm{d}\theta) = \frac{M_n^{(p)} m_H(x) + \sum_{j=1}^{K_n^{(p)}} n_{j,n}^{(p)} k(x \mid \theta_{k,n}^{(p)})}{M_n^{(p)} + n}.$$

Therefore, the reported estimate of $f_G^{(X)}$ is the average $f_{\mathrm{SMC},n}^{(X)}(x) = \sum_{p=1}^{P} W_n^{(p)} f_{p,n}^{(X)}(x)$, for $x \in \mathcal{X}$.

For the hierarchical model (C.27), we also perform Markov chain Monte Carlo (MCMC) posterior inference using the batch implementation of Algorithm 8 in Neal (2000). At each MCMC sweep, the observations $Y_{1:n}$ are grouped into K currently occupied mixture components, with distinct atoms $\theta_1, \ldots, \theta_K$. Use $c_i = j$ to indicate that Y_i is assigned to component j, whose size is $n_j = \sum_{i=1}^{n} I(c_i = j)$. To update c_i, the i-th observation is temporarily removed and reassigned either to an occupied component, with weight $n_{-i,j} \widetilde{k}(Y_i \mid \theta_j)$, or to one of $m = 2$ auxiliary components, each with weight $\frac{M}{2} \widetilde{k}(Y_i \mid \theta)$. If removing i eliminates a singleton component, its former atom is used as one of the two auxiliary atoms; otherwise, both auxiliary atoms are drawn from H. After updating all allocations, each occupied atom is updated by one random-walk Metropolis–Hastings step, using proposal standard deviations 0.20 for the location and 0.10 for the variance; proposals outside Θ are rejected. Finally, the strength parameter M is updated by one Escobar–West step. We run 5,000 MCMC sweeps, discard the first 2,500, and retain every tenth subsequent

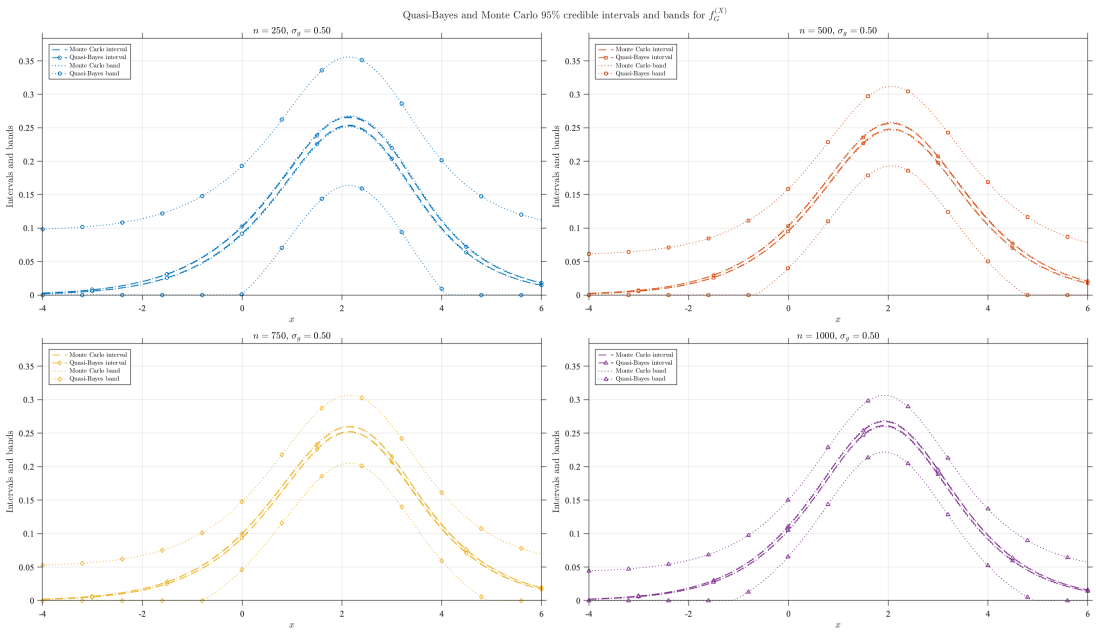


Figure 10: Unimodal example, with Gaussian noise ($\sigma_g = 0.50$): quasi-Bayes versus Monte Carlo credible intervals and bands.

sweep, obtaining $S = 250$ states. For the s-th retained state, the posterior estimate of the mixing distribution G is

$$\widehat{G}_n^{(s)}(\cdot) = \frac{M^{(s)} H(\cdot) + \sum_{j=1}^{K^{(s)}} n_j^{(s)} \delta_{\theta_j^{(s)}}(\cdot)}{M^{(s)} + n},$$

such that, by letting $m_H(x) = \int_\Theta k(x \mid \theta) H(\mathrm{d}\theta)$, the induced posterior estimate of $f_G^{(X)}$ is given by

$$f_{s,n}^{(X)}(x) = \int_\Theta k(x \mid \theta) \widehat{G}_n^{(s)}(\mathrm{d}\theta) = \frac{M^{(s)} m_H(x) + \sum_{j=1}^{K^{(s)}} n_j^{(s)} k(x \mid \theta_j^{(s)})}{M^{(s)} + n}.$$

Therefore, the reported estimate of $f_G^{(X)}$ is the average $f_{\mathrm{MCMC},n}^{(X)}(x) = \frac{1}{S} \sum_{s=1}^{S} f_{s,n}^{(X)}(x)$, for $x \in \mathcal{X}$.

The comparison between the quasi-Bayesian and Bayesian approaches is performed for sample size $n \in \{250, 500, 750, 1000\}$, separately for Laplace and Gaussian noise distributions with $\sigma_l = \sigma_g = 0.25$. To account for order dependence in the observations Y_i's, results obtained with Newton's and SMC algorithms are averaged over the same $R_n = n/10$ permutations. Instead Algorithm 8 is run once on each full dataset, as it is designed to work in batch.

In addition to quasi-Bayes and Bayes estimates, we record total computational work and CPU time. A unit of work is one evaluation of the convolution kernel. Newton's algorithm evaluates the kernel at all $20,050$ grid points for every incoming observation, and therefore has exactly $20,050n$ work units per pass and constant work per update. For the SMC algorithm, the work required by the i-th observation is $\sum_{p=1}^{500} \big(K_{i-1}^{(p)} + 1\big)$, where $K_{i-1}^{(p)}$ is the number of occupied components in particle p before processing the i-th observation. For Algorithm 8, the work count includes the convolved-likelihood evaluations used in both the allocation updates and the Metropolis–Hastings updates, accumulated over all $5,000$ sweeps. For Newton's and SMC algorithms the plotted total work and CPU time correspond to one sequential pass, which is averaged over all the permutations considered, rather than to the aggregate cost of computing the order-averaged estimate; for

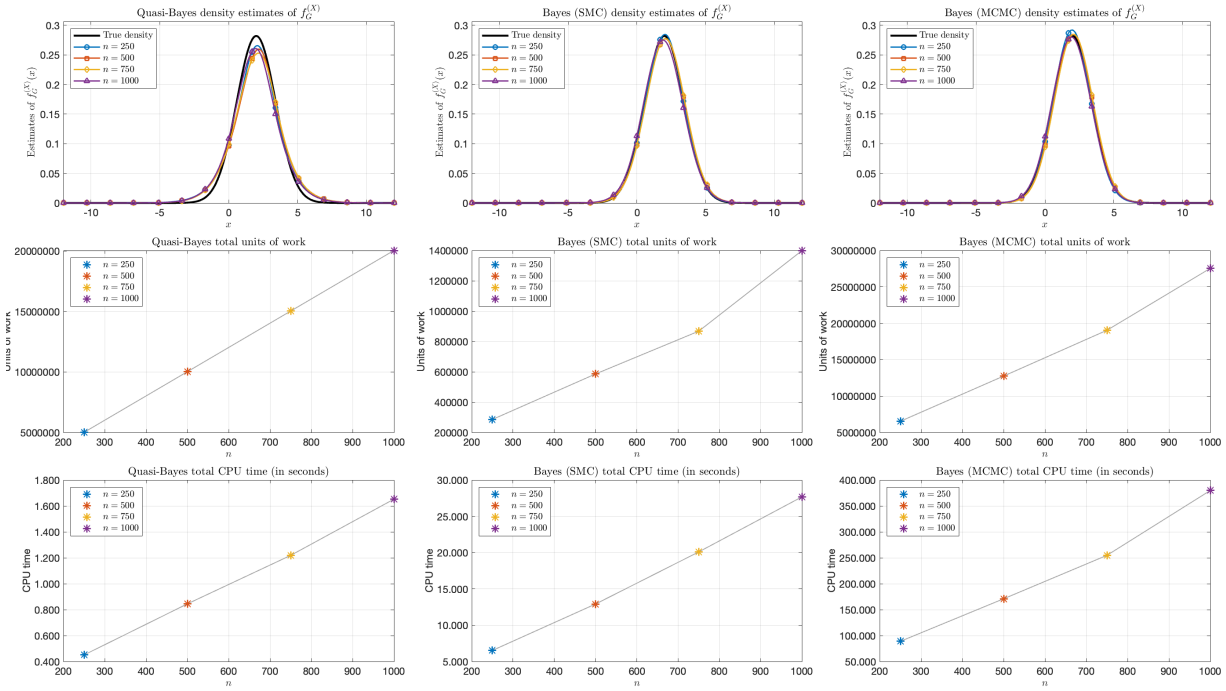


Figure 11: Unimodal example, with Laplace noise ($\sigma_l = 0.25$): comparison between quasi-Bayesian and Bayesian approaches.

Algorithm 8 they correspond to the complete batch run of the algorithm. CPU time is measured around the updating operations, excluding file input/output and figure creation. Figures 11–12 summarize the comparison between the quasi-Bayesian and Bayesian approaches. The first row of these figures displays the estimates of $f_G^{(X)}$ obtained with Newton's algorithm, the SMC algorithm, and Algorithm 8; the second and third rows report total work and total CPU time, respectively.

C.1.2 Bimodal example I

Synthetic-data generation. For each $n \in \{250, 500, 750, 1000\}$, we generate random variables $X_1, \ldots, X_n$ i.i.d. according to the density $f^{(X)}(\cdot) = 0.3\,\phi(\cdot \mid -1, 2) + 0.7\,\phi(\cdot \mid 3, 1.5)$, where $\phi(\cdot \mid \mu, \sigma^2)$ denotes the Gaussian density with mean μ and variance σ^2. Using the mixture representation (1), we write $f^{(X)}$ as $f_{G^*}^{(X)}$, with $G^* = 0.3\,\delta_{(-1,2)} + 0.7\,\delta_{(3,1.5)}$ the true mixing distribution. We consider ordinary-smooth and super-smooth distributions for the noise random variables Z_i's. In the ordinary-smooth case, $Z_i \overset{\text{iid}}{\sim} \text{Laplace}(0, b_l)$ with $b_l = \frac{\sigma_l}{\sqrt{2}}$ and standard deviation $\sigma_l \in \{0.25, 0.50\}$; in the super-smooth case, $Z_i \overset{\text{iid}}{\sim} N(0, \sigma_g^2)$ with standard deviation $\sigma_g \in \{0.25, 0.50\}$. The Z_i's are independent of the X_i's, and the observations are modeled as follows:

$$Y_i = X_i + Z_i, \qquad i = 1, \ldots, n.$$

For each n, the same realization of $Y_{1:n} = (Y_1, \ldots, Y_n)$ is used for all methods under comparison.

Quasi-Bayesian estimation and uncertainty quantification We use the same implementation of Section C.1.1: the parameter space, numerical grids, initial distribution, learning rate, trapezoidal quadrature, renormalization, number of permutations, and random seed are unchanged. For the r-th permutation, let $\widetilde{G}_n^{(r)}$ denote the resulting Newton's mixing-distribution

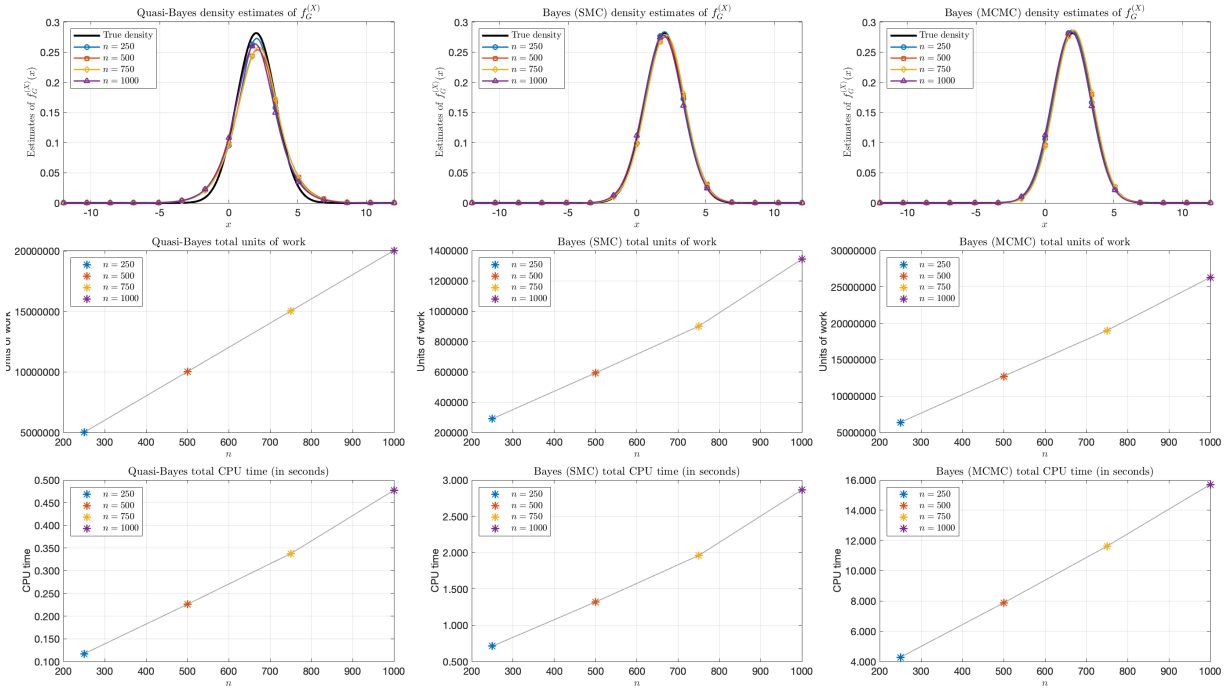


Figure 12: Unimodal example, with Gaussian noise ($\sigma_g = 0.25$): comparison between quasi-Bayesian and Bayesian approaches.

estimate, $r = 1, \ldots, R_n$. Then, the order-averaged estimate is defined as

$$\overline{G}_n = \frac{1}{R_n} \sum_{r=1}^{R_n} \widetilde{G}_n^{(r)}, \tag{C.28}$$

which, following (C.21), induces the quasi-Bayes estimate $f_{\overline{G}_n}^{(X)}$ of $f_G^{(X)}$. The quasi-Bayes asymptotic credible intervals and bands for $f_G^{(X)}$ are also computed directly from $\overline{G}_n$, following (C.22)–(C.23). Figures 13–14 report the quasi-Bayes estimates of $f_G^{(X)}$, and the asymptotic credible intervals and bands. The figures consist of two columns, which correspond to the noise standard deviations 0.25 and 0.50, respectively. The first row of each figure compares the true density $f_{G^*}^{(X)}$ with the quasi-Bayes density estimates, while the second row displays the credible intervals, and the third row displays the credible bands on the interval $I = [-4, 6]$.

Monte Carlo uncertainty quantification The Monte Carlo procedure is the same as the procedure presented in Section C.1.1; it relies on the continuation of the quasi-Bayesian learning process (2)–(4), initialized at the order-averaged mixing distribution $\overline{G}_n$ in (C.28). The continuation length, number of Monte Carlo replications, and random seed are unchanged. Further, the Monte Carlo estimates, credible intervals and bands are constructed as in (C.24)–(C.26). Figures 15–18 display Monte Carlo credible intervals and bands, which are compared with the corresponding quasi-Bayes asymptotic credible intervals and bands.

Dirichlet process mixture model benchmarks The Dirichlet process Gaussian-mixture model (C.27), the SMC algorithm, Algorithm 8, tuning parameters, and computational-work definitions are identical to those described in Section C.1.1. Furthermore, as in the unimodal example,

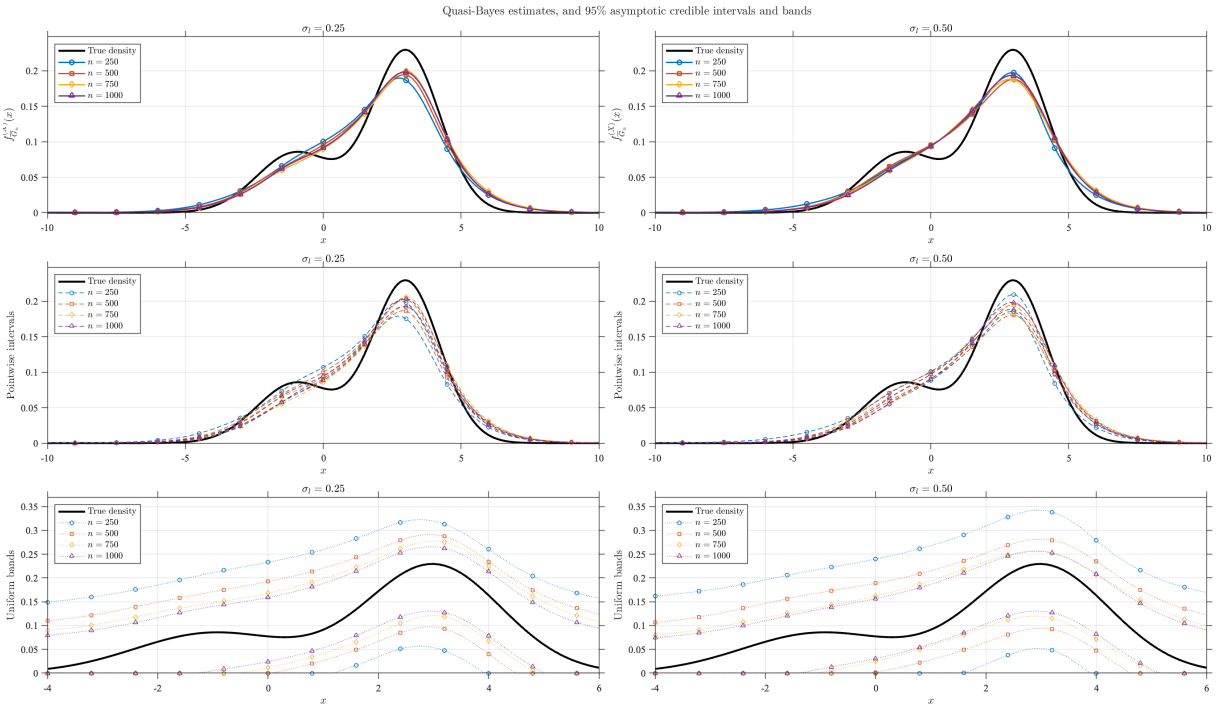


Figure 13: Bimodal example I, with Laplace noise: estimates, credible intervals and bands.

Newton's and SMC algorithms are averaged over the same R_n permutations, whereas Neal's Algorithm 8 is applied once to each complete dataset. Figures 19–20 summarize the comparison between the quasi-Bayesian and Bayesian approaches. Their first row displays the estimates of $f_G^{(X)}$ obtained with Newton's algorithm, the SMC algorithm, and Algorithm 8; the second and third rows report total work and total CPU time, respectively.

C.1.3 Bimodal example II

Synthetic-data generation For each $n \in \{250, 500, 750, 1000\}$, we generate random variables $X_1, \ldots, X_n$ i.i.d. according to the density $f^{(X)}(\cdot) = 0.5\,\phi(\cdot \mid -2, 1) + 0.5\,\phi(\cdot \mid 4, 1)$, where $\phi(\cdot \mid \mu, \sigma^2)$ denotes the Gaussian density with mean μ and variance σ^2. Using the mixture representation (1), we write $f^{(X)}$ as $f_{G^*}^{(X)}$, with $G^* = 0.5\,\delta_{(-2,1)} + 0.5\,\delta_{(4,1)}$ the true mixing distribution. We consider ordinary-smooth and super-smooth distributions for the noise random variables Z_i's. In the ordinary-smooth case, $Z_i \overset{\text{iid}}{\sim} \text{Laplace}(0, b_l)$ with $b_l = \frac{\sigma_l}{\sqrt{2}}$ and standard deviation $\sigma_l \in \{0.25, 0.50\}$; in the super-smooth case, $Z_i \overset{\text{iid}}{\sim} N(0, \sigma_g^2)$ with standard deviation $\sigma_g \in \{0.25, 0.50\}$. The Z_i's are independent of the X_i's, and the observations are modeled as follows:

$$Y_i = X_i + Z_i, \qquad i = 1, \ldots, n.$$

For each n, the same realization of $Y_{1:n} = (Y_1, \ldots, Y_n)$ is used for all methods under comparison.

Quasi-Bayesian estimation and uncertainty quantification We use the same implementation of Section C.1.1: the parameter space, numerical grids, initial distribution, learning rate, trapezoidal quadrature, renormalization, number of permutations, and random seed are unchanged. For the r-th permutation, let $\widetilde{G}_n^{(r)}$ denote the resulting Newton's mixing-distribution

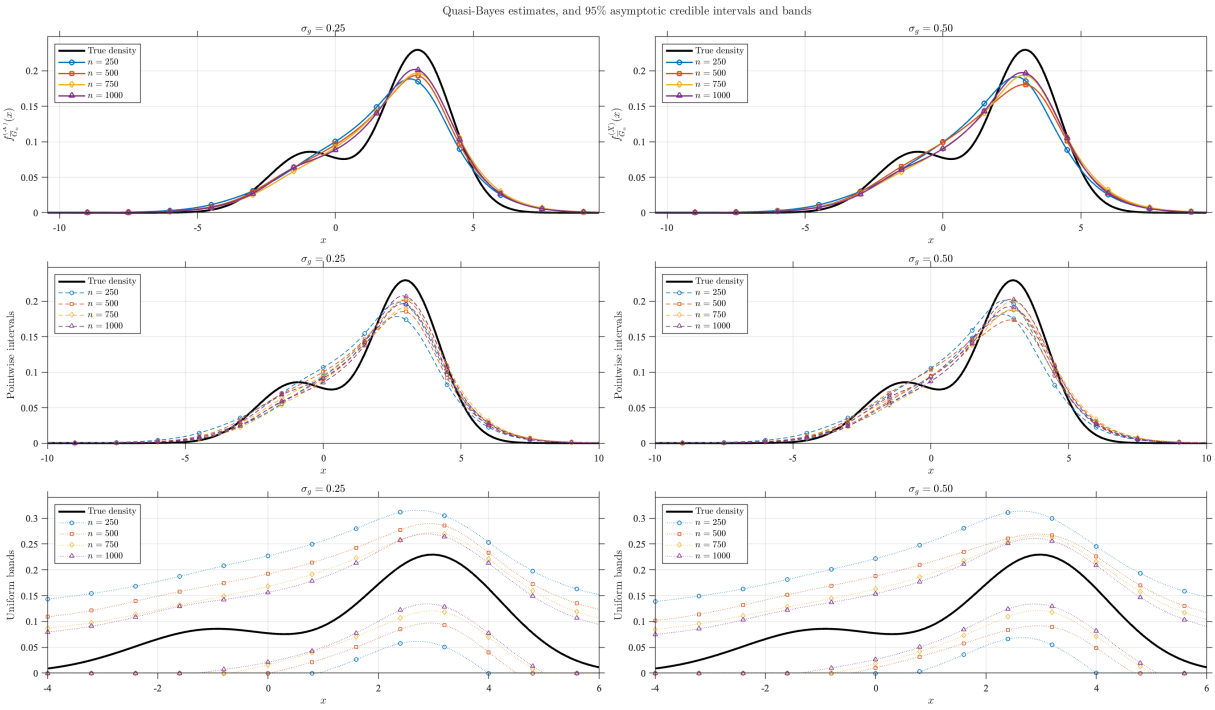


Figure 14: Bimodal example I, with Gaussian noise: estimates, credible intervals and bands.

estimate, $r = 1, \ldots, R_n$. Then, the order-averaged estimate is defined as

$$\overline{G}_n = \frac{1}{R_n} \sum_{r=1}^{R_n} \widetilde{G}_n^{(r)}, \tag{C.29}$$

which, following (C.21), induces the quasi-Bayes estimate $f^{(X)}_{\overline{G}_n}$ of $f^{(X)}_G$. The quasi-Bayes asymptotic credible intervals and bands for $f^{(X)}_G$ are also computed directly from $\overline{G}_n$, following (C.22)–(C.23). Figures 21–22 report the quasi-Bayes estimates of $f^{(X)}_G$, and the asymptotic credible intervals and bands. The figures consist of two columns, which correspond to the noise standard deviations 0.25 and 0.50, respectively. The first row of each figure compares the true density $f^{(X)}_{G^*}$ with the quasi-Bayes density estimates, while the second row displays the credible intervals, and the third row displays the credible bands on the interval $I = [-4, 6]$.

Monte Carlo uncertainty quantification The Monte Carlo procedure is the same as the procedure presented in Section C.1.1; it relies on the continuation of the quasi-Bayesian learning process (2)–(4), initialized at the order-averaged mixing distribution $\overline{G}_n$ in (C.29). The continuation length, number of Monte Carlo replications, and random seed are unchanged. Further, the Monte Carlo estimates, credible intervals and bands are constructed as in (C.24)–(C.26). Figures 23–26 display Monte Carlo credible intervals and bands, which are compared with the corresponding quasi-Bayes asymptotic credible intervals and bands.

Dirichlet process mixture model benchmarks The Dirichlet process Gaussian-mixture model (C.27), the SMC algorithm, Algorithm 8, tuning parameters, and computational-work definitions are identical to those described in Section C.1.1. Furthermore, as in the unimodal example,

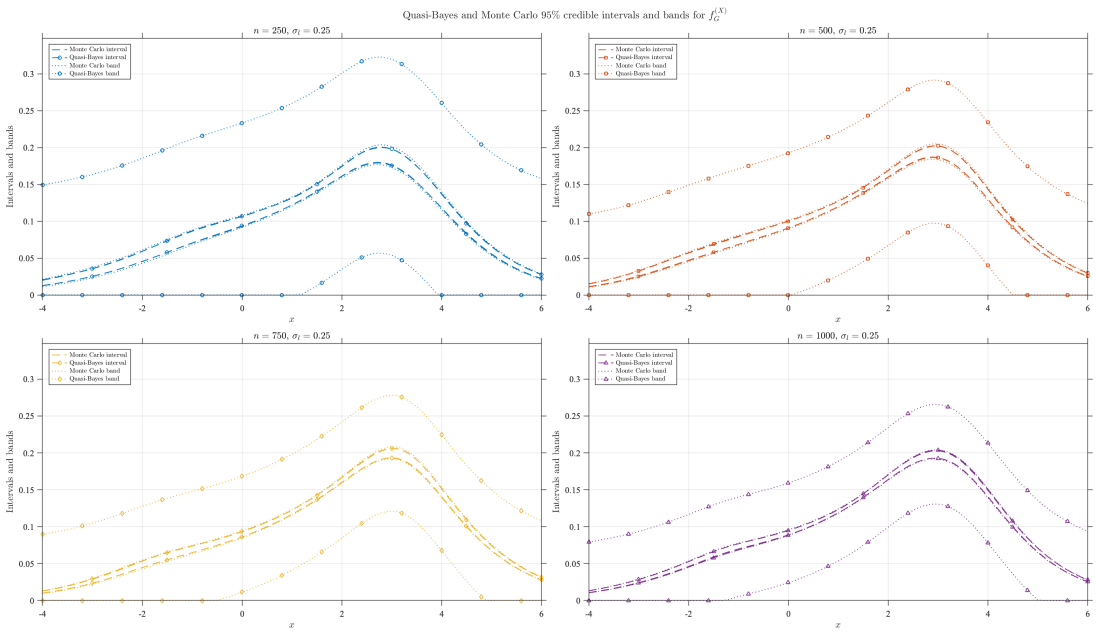


Figure 15: Bimodal example I, with Laplace noise ($\sigma_l = 0.25$): quasi-Bayes versus Monte Carlo credible intervals and bands.

Newton's and SMC algorithms are averaged over the same R_n permutations, whereas Neal's Algorithm 8 is applied once to each complete dataset. Figures 27–28 summarize the comparison between the quasi-Bayesian and Bayesian approaches. Their first row displays the estimates of $f_G^{(X)}$ obtained with Newton's algorithm, the SMC algorithm, and Algorithm 8; the second and third rows report total work and total CPU time, respectively.

C.2 Real-data analysis

C.2.1 Flow-cytometry data

In Section 6 we analyzed flow-cytometry data from the study of Torregrosa-Cortés et al. (2023) on the expression of Brachyury in differentiating mouse embryonic stem cells. Brachyury is a mesodermal transcription factor whose expression varies across cells during differentiation. Flow cytometry provides a large number of single-cell fluorescence measurements by recording the optical signal generated as cells pass individually through the instrument. Each recorded event therefore corresponds to a distinct cell and is associated with both a fluorescence intensity and an acquisition time. The fluorescence measured on a labeled cell contains the reported signal associated with Brachyury expression together with autofluorescence and other background contributions. To characterize the latter, the experiment also includes a separate population of unlabeled cells. The labeled and unlabeled measurements are consequently obtained from different cells and cannot be matched event-by-event. In particular, an observation from the unlabeled population does not represent the specific background realization affecting any individual labeled-cell measurement; rather, the unlabeled population provides an independent sample from the common background distribution.

The data are available at `https://github.com/dsb-lab/scBayesDeconv.jl` and include fluorescence intensities and acquisition-time measurements. Following the organization of the dataset in Torregrosa-Cortés et al. (2023), the first column is used as the sample from the labeled population, whereas the sixth column is used as the autofluorescence control sample. The acquisition-time mea-

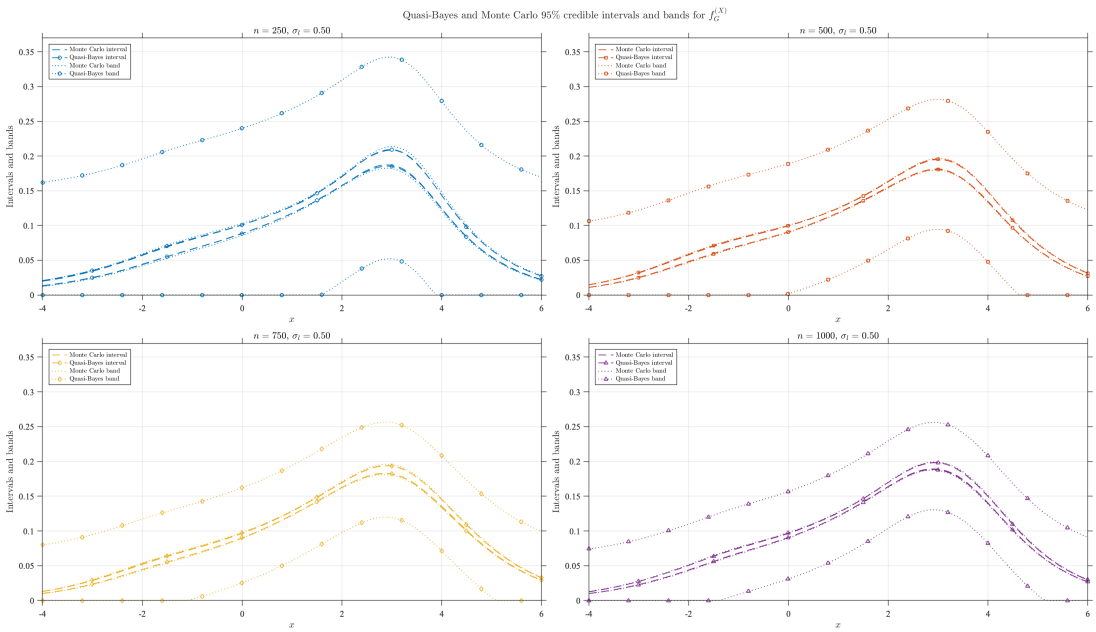


Figure 16: Bimodal example I, with Laplace noise ($\sigma_l = 0.50$): quasi-Bayes versus Monte Carlo credible intervals and bands.

surements are used to restore the order in which the labeled cells were recorded. We then process the observations sequentially and examine the estimates over the nested prefixes corresponding to the first $n \in \{250,\ 1{,}000,\ 5{,}000,\ 10{,}000\}$ events. Let Y_i denote the fluorescence intensity of the i-th labeled cell in acquisition order and assume the additive model $Y_i = X_i + Z_i$ for $i \geq 1$, where X_i is the reporter fluorescence and Z_i is the autofluorescence and background contribution. The control measurements consist of $m \geq 1$ independent samples, say $\widetilde{Z}_1, \ldots, \widetilde{Z}_m$, measured on unlabeled cells and not paired with the Y_i's.

The distributional assumptions on the X_i's and Z_i's, as well as on $\widetilde{Z}_1, \ldots, \widetilde{Z}_m$, are those specified in Section 6. In particular, in the analysis of Section 6, the random variables Z_i's are i.i.d. according to a Gaussian distribution, i.e. $f_Z(z) = \phi(z \mid \mu_Z, \sigma_Z^2)$. Here, we investigate the sensitivity of the results to this specification of f_Z, by considering two additional working models for the noise distributions. More precisely, we consider a Laplace noise distribution, namely

$$f_Z(z) = \frac{1}{2b_Z} \exp\left\{-\frac{|z - \mu_Z|}{b_Z}\right\},$$

and the four-component Gaussian-mixture noise distribution (Torregrosa-Cortés et al., 2023), namely

$$f_Z(z) = \sum_{\ell=1}^{4} q_\ell \, \phi(z \mid \eta_\ell, \tau_\ell^2).$$

With regards to the use of the control samples $\widetilde{Z}_1, \ldots, \widetilde{Z}_m$, within each noise distribution, the parameters of f_Z are estimated once from the same control samples, and the resulting fitted density is kept fixed over the nested sample sizes $n \in \{250,\ 1{,}000,\ 5{,}000,\ 10{,}000\}$. Moreover, the quasi-Bayesian, Bayesian, and ridge-regularized Fourier deconvolution approaches use exactly the same fitted density. Figures 29–30 report the estimates of $f_G^{(X)}$ obtained by the quasi-Bayesian, Bayesian, and ridge-regularized Fourier deconvolution approaches.

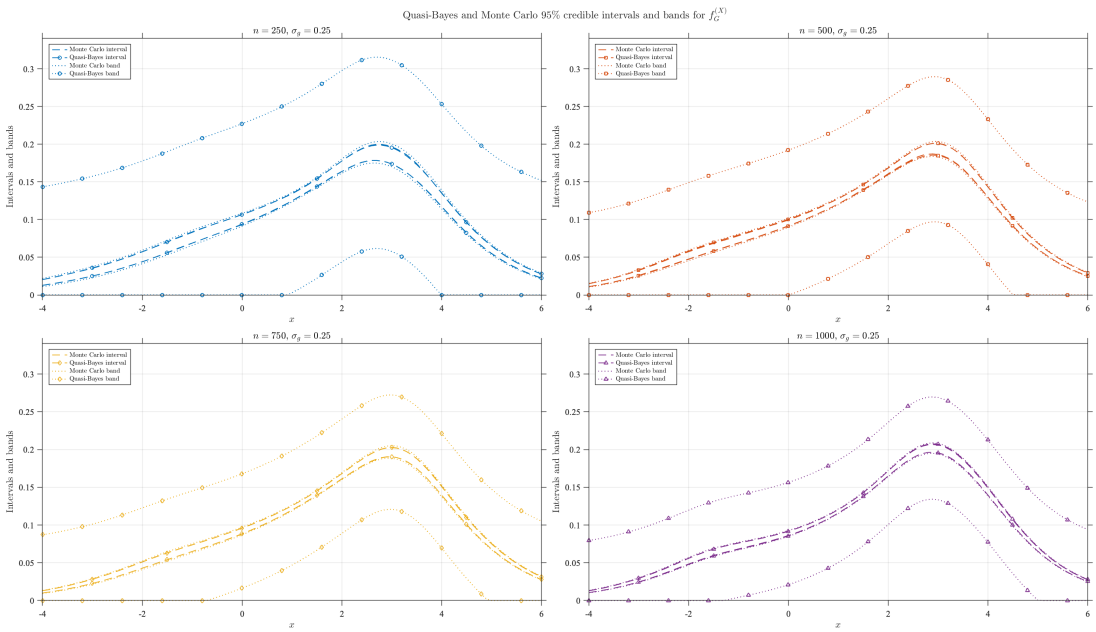


Figure 17: Bimodal example I, with Gaussian noise ($\sigma_g = 0.25$): quasi-Bayes versus Monte Carlo credible intervals and bands.

To provide an additional graphical representation, we also augment the preceding figures with histograms based on a reconstructed (proxy) version of the reporter fluorescence samples. Since the X_i's are not observed, for each $n \in \{250,\ 1{,}000,\ 5{,}000,\ 10{,}000\}$ and each fitted noise specification, we make use the quasi-Bayes estimate $f^{(X)}_{\widetilde{G}_n}$ to define the conditional density

$$\widehat{p}_n(x \mid Y_i) = \frac{f^{(X)}_{\widetilde{G}_n}(x)\, \widehat{f}_Z(Y_i - x)}{\displaystyle\int_{\mathbb{R}} f^{(X)}_{\widetilde{G}_n}(u)\, \widehat{f}_Z(Y_i - u)\, du}, \qquad i = 1, \ldots, n,$$

where $\widehat{f}_Z$ denotes the density of the noise distribution estimated from the control samples $\widetilde{Z}_1, \ldots, \widetilde{Z}_m$. We then generate one reconstructed value $\widehat{X}_i^{(n)}$ from $\widehat{p}_n(\cdot \mid Y_i)$ for each observation and display the normalized histogram of $\widehat{X}_1^{(n)}, \ldots, \widehat{X}_n^{(n)}$. These reconstructed values should not be interpreted as observations of the X_i's. Their distribution depends on both the quasi-Bayes estimate and the selected noise specification, and the resulting histogram is therefore intended only as a descriptive visualization, rather than as an independent benchmark for comparing the estimates. The corresponding plots are reported in Figures 31–33.

C.2.2 Stellar metallicity data

We consider a dataset of 16,377 stellar metallicity measurements obtained from the Geneva–Copenhagen Survey, which provides kinematic properties, ages, and metallicities for F- and G-type dwarf stars in the solar neighborhood; see Nördstrom, et al. (2004) for details. Data from the same survey were previously analyzed by Bissantz et al. (2007) using kernel-based deconvolution methods (Stefanski and Carroll, 1990; Hall and Meister, 2007). Metallicity, expressed as [Fe/H], is the logarithmic abundance of iron relative to hydrogen and provides a measure of the fraction of heavy elements in a star, which is important for understanding stellar-formation mechanisms. Stellar

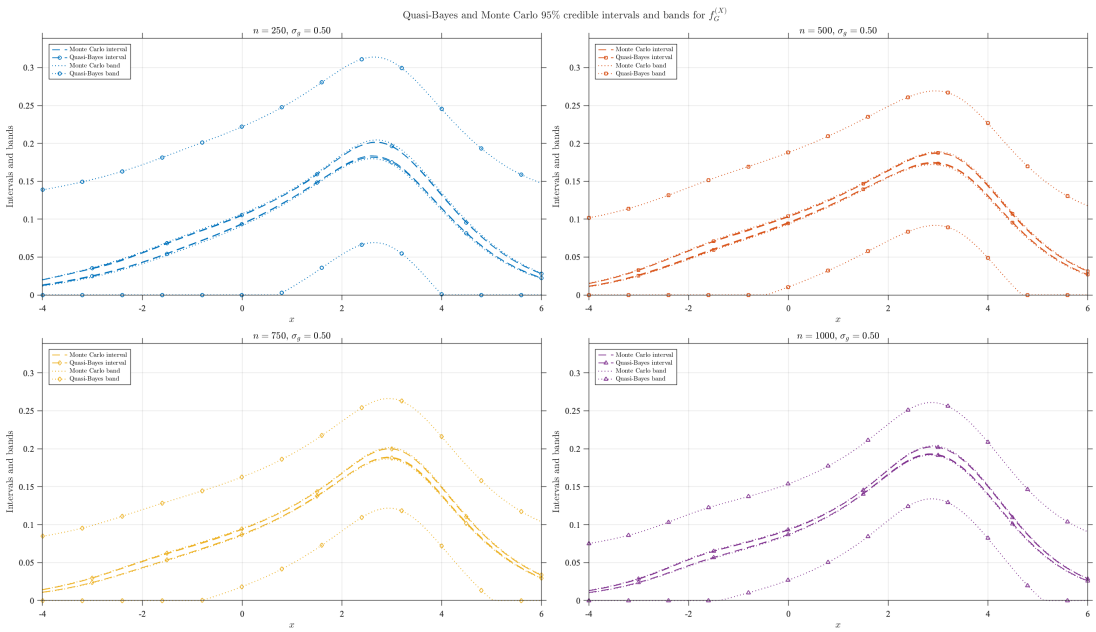


Figure 18: Bimodal example I, with Gaussian noise ($\sigma_g = 0.50$): quasi-Bayes versus Monte Carlo credible intervals and bands.

metallicity cannot be measured directly, but is inferred from a star's brightness in selected spectral bands through suitable calibration procedures. The resulting calibration error is assumed to be largely stochastic and can be assessed from the dispersion between metallicity estimates obtained for the same star using different calibrations. For these data, a typical value of this dispersion is approximately 0.1 (Nördstrom, et al., 2004).

More precisely, let Y_i denote the observed metallicity of the i-th star and assume the additive model $Y_i = X_i + Z_i$, for $i = 1, \ldots, n$ where X_i is the physical metallicity [Fe/H] and Z_i is the corresponding calibration error. We assume that the random variables X_i's are i.i.d. according to an unknown density f_X, that the noise variables Z_i's are i.i.d. according to a density f_Z, and that the random variables X_i's and Z_i's are mutually independent. The inferential target is the density f_X, which is represented as the Gaussian location–scale mixture

$$f_G^{(X)}(x) = \int_{\mathbb{R}\times\mathbb{R}^+} \phi(x \mid \mu, \sigma^2)\, G(d\mu, d\sigma^2), \qquad x \in \mathbb{R},$$

where G is the unknown mixing distribution. Following the analysis of Bissantz et al. (2007), for the noise distribution with density f_Z we consider a centered (zero-mean) Laplace distribution with standard deviation $\sigma_l \in \{0.08, 0.10, 0.12\}$. As a sensitivity specification, we also consider a centered (zero-mean) Gaussian distribution with standard deviation $\sigma_g = 0.10$.

To implement the quasi-Bayesian approach for estimating $f_G^{(X)}$, we apply Newton's algorithm (4) with Gaussian kernel $k(\cdot \mid \theta) = \phi(\cdot \mid \mu, \sigma^2)$, where $\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$. The parameter space $\mathbb{R}\times\mathbb{R}^+$ is restricted to $\Theta = [-3, 2] \times [0.1, 0.5]$. We use the learning rate $\widetilde{\alpha}_i = (1+i)^{-1}$, for $i \geq 1$, and set $\widetilde{G}_0$ equal to the Uniform distribution on Θ. All integrals with respect to the mixing distribution are evaluated using the two-dimensional trapezoidal quadrature, and $\widetilde{G}_i$ is numerically renormalized after each update. For numerical implementation, Θ is discretized on a regular 100×100 grid.

As additional benchmarks, we also consider a Bayesian approach under the Dirichlet-process Gaussian-mixture model (C.27), with posterior inference performed sequentially using the SMC

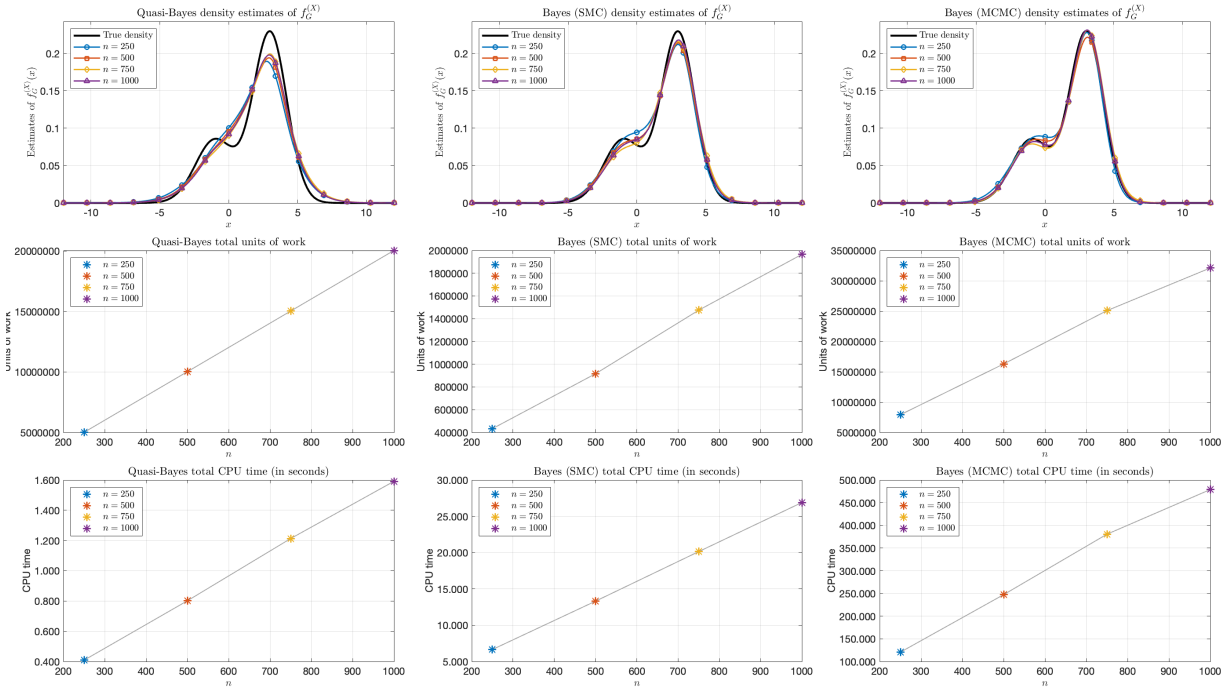


Figure 19: Bimodal example I, with Laplace noise ($\sigma_l = 0.25$): comparison between quasi-Bayesian and Bayesian approaches.

algorithm, and the ridge-regularized Fourier deconvolution approach (Stefanski and Carroll, 1990; Hall and Meister, 2007). Figure 34 reports the estimates of $f_G^{(X)}$ obtained by the quasi-Bayesian, Bayesian, and ridge-regularized Fourier deconvolution approaches.

C.2.3 Active power-output data

We consider a semi-synthetic dataset of active power outputs, expressed in megawatts, from a 68-bus power network with 16 generators. Using GridSTAGE, a platform for emulating power-system data, we obtain a sequence of 4,000 noise-free outputs at a sampling rate of 50 observations per second; see Nandanoori et al. (2020) for details. The observations are retained in their original acquisition order, which, however, does not amount to assuming a time-series dependence structure; it only reflects the order in which the observations become available to the sequential procedure. Since the outputs generated by GridSTAGE are noise-free, we construct a semi-synthetic deconvolution experiment by adding independent random perturbations. This allows us to evaluate the procedures under controlled noise distributions and levels, while retaining the original GridSTAGE outputs as a benchmark for the latent signal.

More precisely, let X_i denote the noise-free active power output at the i-th acquisition time and define the perturbed observation $Y_i = X_i + Z_i$, for $i = 1, \ldots, n$, where Z_i is the artificially added measurement error. We assume that the random variables X_i's are i.i.d. according to an unknown density f_X, that the noise random variables Z_i's are i.i.d. according to a density f_Z, and that the random variables X_i's and Z_i's are mutually independent. Then, the inferential target is the density f_X, which is represented as the Gaussian location-scale mixture,

$$f_G^{(X)}(x) = \int_{\mathbb{R}\times\mathbb{R}^+} \phi(x \mid \mu, \sigma^2)\, G(d\mu, d\sigma^2), \qquad x \in \mathbb{R},$$

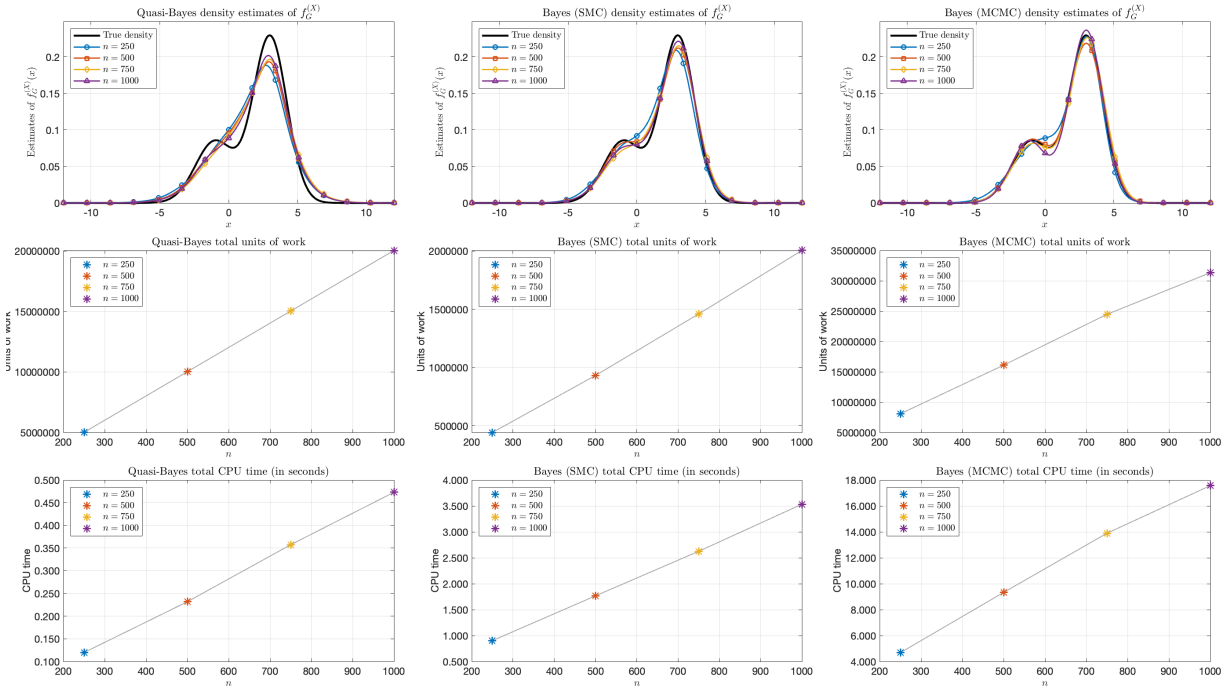


Figure 20: Bimodal example I, with Gaussian noise ($\sigma_g = 0.25$): comparison between quasi-Bayesian and Bayesian approaches.

where G is the unknown mixing distribution. For the noise distribution with density f_Z, which is fully controlled in the experiment, we consider either a centered (zero-mean) Laplace distribution with standard deviation $\sigma_l \in \{0.5, 1, 2, 4\}$ or a centered (zero-mean) Gaussian distribution with standard deviation $\sigma_g \in \{0.5, 1, 2, 4\}$. Since the experiment is semi-synthetic, the values $X_1, \ldots, X_n$ are available and can be used as a benchmark to assess the estimated $f_G^{(X)}$.

To implement the quasi-Bayesian approach for estimating $f_G^{(X)}$, we apply Newton's algorithm (4) with Gaussian kernel $k(\cdot \mid \theta) = \phi(\cdot \mid \mu, \sigma^2)$, where $\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}^+$. The parameter space $\mathbb{R} \times \mathbb{R}^+$ is restricted to $\Theta = [50, 150] \times [1, 5]$. We use the learning rate $\widetilde{\alpha}_i = (1+i)^{-1}$, for $i \geq 1$, and set $\widetilde{G}_0$ equal to the Uniform distribution on Θ. All integrals with respect to the mixing distribution are evaluated using the two-dimensional trapezoidal quadrature, and $\widetilde{G}_i$ is numerically renormalized after each update. For numerical implementation Θ is discretized on a regular 100×100 grid.

As additional benchmarks, we also consider a Bayesian approach under the Dirichlet-process Gaussian-mixture model (C.27), with posterior inference performed sequentially using the SMC algorithm, and the ridge-regularized Fourier deconvolution approach (Stefanski and Carroll, 1990; Hall and Meister, 2007). Figures 35–36 report the estimates of $f_G^{(X)}$ obtained by the quasi-Bayesian, Bayesian, and ridge-regularized Fourier deconvolution approaches, together with the (benchmark) histogram of the noise-free power outputs $X_1, \ldots, X_n$.

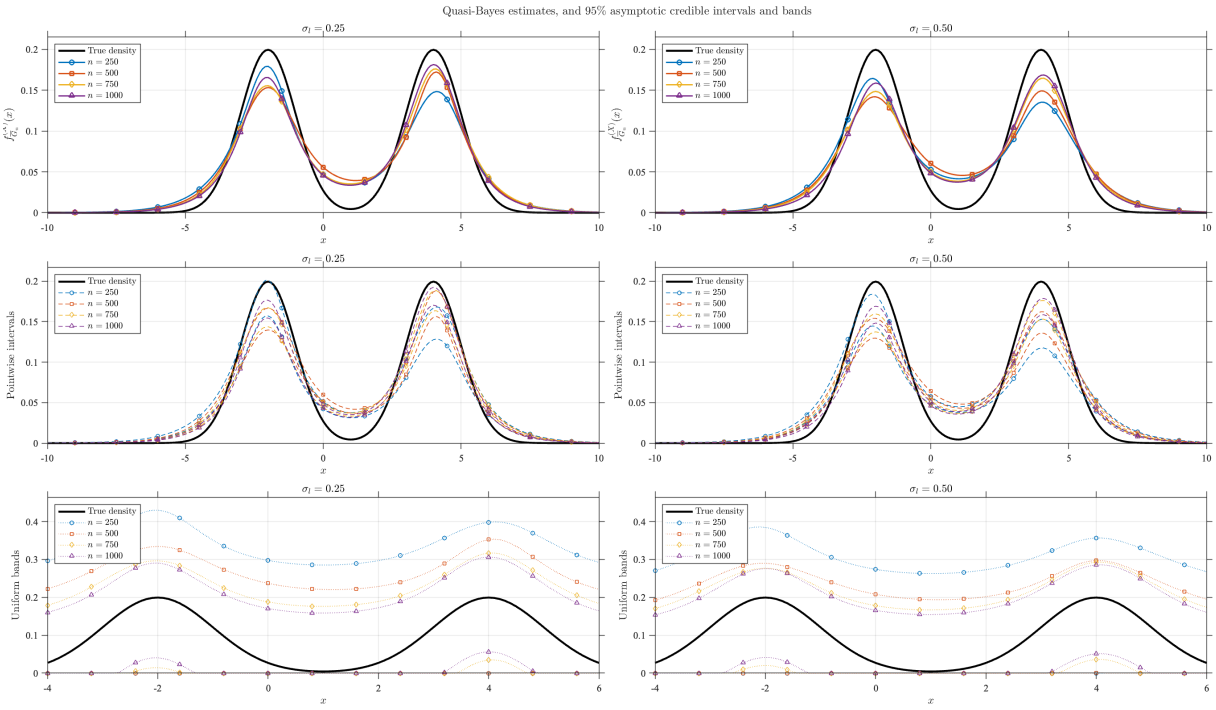


Figure 21: Bimodal example II, with Laplace noise: estimates, credible intervals and bands.

D Auxiliary results

D.1 Learning-rate calibration

In this section we propose a data-driven calibration of the learning rates $\widetilde{\alpha}_n$ in the quasi-Bayesian learning model (2)–(4). The proposed calibration is based on comparing the evolution of the recursive estimate based on the noisy observations (Y_n) with the evolution that would be obtained if Newton's algorithm were applied directly to the latent variables (X_n), that is, in the absence of measurement noise. The frequentist merging result in Corollary 4.2 provides the theoretical justification for calibrating $\widetilde{\alpha}_n$ by matching the updates of the two recursive procedures.

For Newton's algorithm based on the X_n's, we assume the standard learning rate $\alpha_n = (\alpha+n)^{-1}$, for $\alpha > 0$. Denote by $g_n(\theta)$ and $\widetilde{g}_n(\theta)$ the densities of G_n and $\widetilde{G}_n$ with respect to the dominating measure μ. The updates or innovations of $(g_n)_{n\geq 0}$ are

$$I_i(\theta) = g_i(\theta) - g_{i-1}(\theta) = \frac{1}{\alpha + i}(g_{i-1}(\theta \mid X_i) - g_{i-1}(\theta)) \qquad i \geq 1, \tag{D.30}$$

where $g_{i-1}(\theta \mid X_i)$ is the density with respect to μ and at the point θ of $G_{i-1}(\cdot \mid X_i)$. Instead, for Newton's algorithm based on the Y_n's, we assume that $\widetilde{\alpha}_n$ is in a class of standard learning rates defined as $\{(\alpha + n)^{-\gamma} : 2^{-1} < \gamma \leq 1\}$. Then, the updates or innovations of $(\widetilde{g}_n)_{n\geq 0}$ are of the form

$$\widetilde{I}_i(\theta, \gamma) = \widetilde{g}_i(\theta) - \widetilde{g}_{i-1}(\theta) = (\alpha + i)^{-\gamma}(\widetilde{g}_{i-1,\gamma}(\theta \mid Y_i) - \widetilde{g}_{i-1,\gamma}(\theta)) \qquad i \geq 1. \tag{D.31}$$

Other choices for α_n and $\widetilde{\alpha}_n$ are possible; we refer to Fortini and Petrone (2020) and references therein for details.

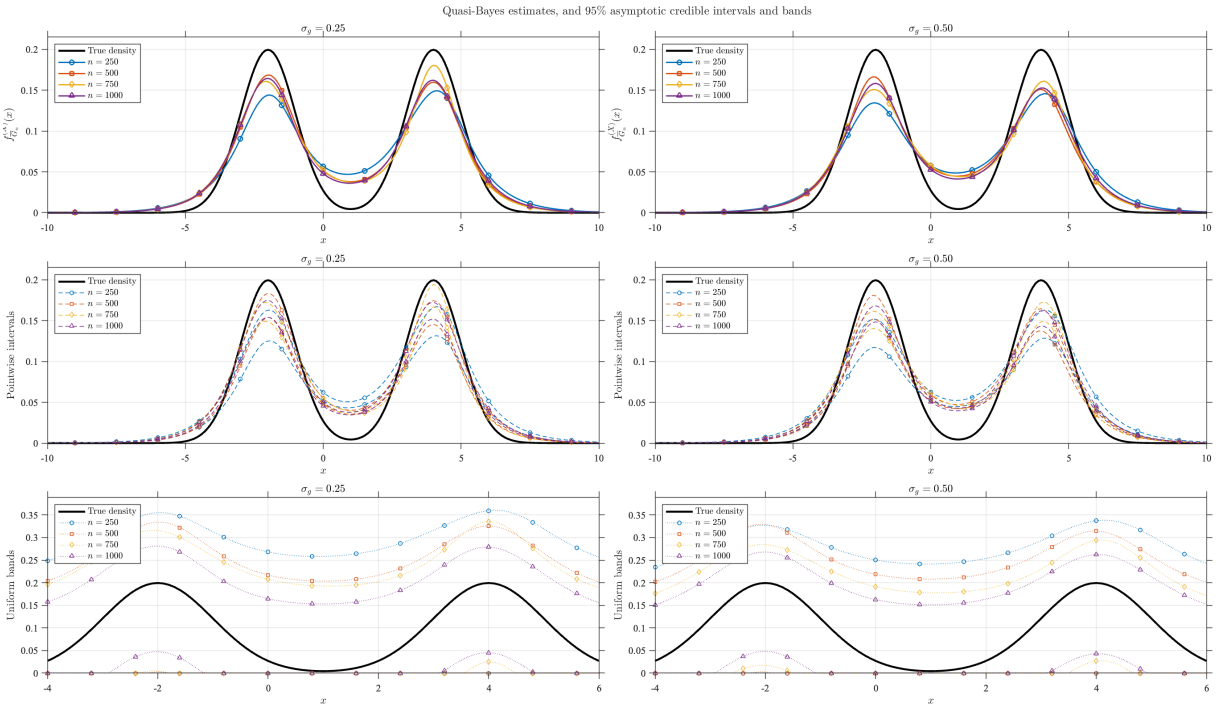


Figure 22: Bimodal example II, with Gaussian noise: estimates, credible intervals and bands.

Based on (D.30)-(D.31), we calibrate the parameter γ by matching the sizes of the updates. For $i \geq 1$, we set

$$\Delta_i(\theta, \gamma) = \log |g_{i-1}(\theta \mid X_i) - g_{i-1}(\theta)| - \log |\widetilde{g}_{i-1,\gamma}(\theta \mid Y_i) - \widetilde{g}_{i-1,\gamma}(\theta)|,$$

such that $\log |\widetilde{I}_i(\theta, \gamma)| - \log |I_i(\theta)| = \Delta_i(\theta, \gamma) - (1-\gamma)\log(\alpha + i)$. In particular, to achieve the matching we need to find the value of the parameter $\gamma \in (\frac{1}{2}, 1]$ that minimizes, for some choice of $M \in \mathbb{N}$,

$$\sum_{i=1}^{M} \int_{\Theta} E\left((\Delta_i(\theta, \gamma) - (1-\gamma)\log(\alpha + i))^2 \,\middle|\, \mathcal{G}_{i-1} \right) \widetilde{g}_{i-1,\gamma}(\theta)\mu(\mathrm{d}\theta).$$

We solve the problem through Monte Carlo simulation, for γ in a grid, by means of random sampling $(\theta_0^{(\gamma)}, \ldots, \theta_M^{(\gamma)})$. See Algorithm 1 for the details on the algorithm for the calibration of the learning rate.

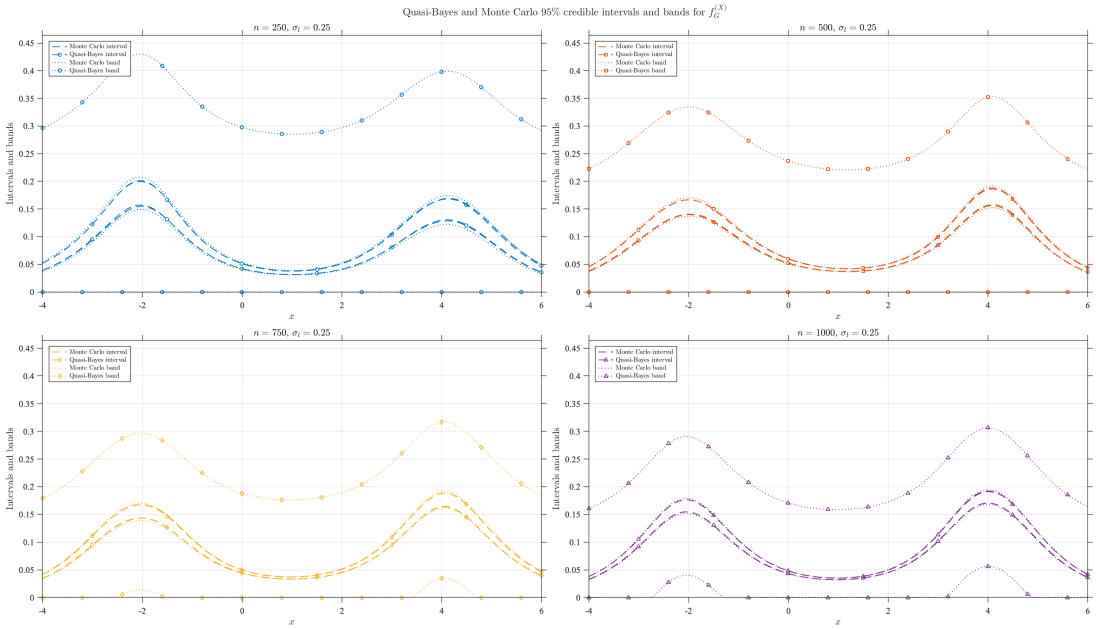


Figure 23: Bimodal example II, with Laplace noise ($\sigma_l = 0.25$): quasi-Bayes versus Monte Carlo credible intervals and bands.

Algorithm 1: Calibrating the learning rate $\widetilde{\alpha}_n = (\alpha + n)^{-\gamma}$.

Input: N, L; M; $B = \{b_1, \ldots, b_L\}$; g_0; k; $\widetilde{k}$;
for $l = 1$ *to* L **do**
  Set $\widetilde{g}_0 = g_0$
  for $i = 1$ *to* M **do**
    Sample $\theta_i \sim \widetilde{g}_{i-1}(\cdot)$; $X_i \sim k(\cdot \mid \theta_i)$; $Z_i \sim f_Z(z)$
    Compute $Y_i = X_i + Z_i$
    Compute $\widetilde{g}_{i-1}(\mathrm{d}\theta \mid Y_i) = \frac{\widetilde{k}(Y_i|\theta)\widetilde{g}_{i-1}(\mathrm{d}\theta)}{\int_\Theta \widetilde{k}(Y_i|\theta)\widetilde{g}_{i-1}(\mathrm{d}\theta)}$
    Compute $g_{i-1}(\mathrm{d}\theta \mid X_i) = \frac{k(X_i|\theta)g_{i-1}(\mathrm{d}\theta)}{\int_\Theta k(X_i|\theta)g_{i-1}(\mathrm{d}\theta)}$
    Compute $\widetilde{g}_i(\mathrm{d}\theta) = \widetilde{g}_{i-1}(\mathrm{d}\theta) + \frac{(\widetilde{g}_{i-1}(\mathrm{d}\theta|Y_i) - \widetilde{g}_{i-1}(\mathrm{d}\theta))}{\left(1+\frac{i}{\alpha}\right)^{\gamma_l}}$
    Compute $g_i(\theta) = g_{i-1}(\theta) + \frac{(g_{i-1}(\theta|X_i) - g_{i-1}(\theta))}{1+\frac{i}{\alpha}}$
    Return $\Delta_i(\theta, \gamma_l) = \log \frac{|g_{i-1}(\theta|X_i) - g_{i-1}(\theta)|}{|\widetilde{g}_{i-1}(\theta|Y_i) - \widetilde{g}_{i-1}(\theta)|}$
  end
end
Output: $\hat{\gamma} = \mathrm{argmin}_{l=1,\ldots,L} \sum_{i=1}^{M} \left(\Delta_i(\theta, \gamma_l) - (1 - \gamma_l)\log(\alpha + i)\right)^2$

D.2 Rate of convergence for finite mixtures

In Section 4 we proved that, under regularity assumptions, the quasi-Bayesian sequential learning estimate of $f_{G^*}^{(X)}$ is consistent. In this section, we derive an upper bound on the rate of convergence in case of finite mixtures with known support points.

Let Θ be a known finite set and let μ denote the counting measure on it. Assume that the X_n

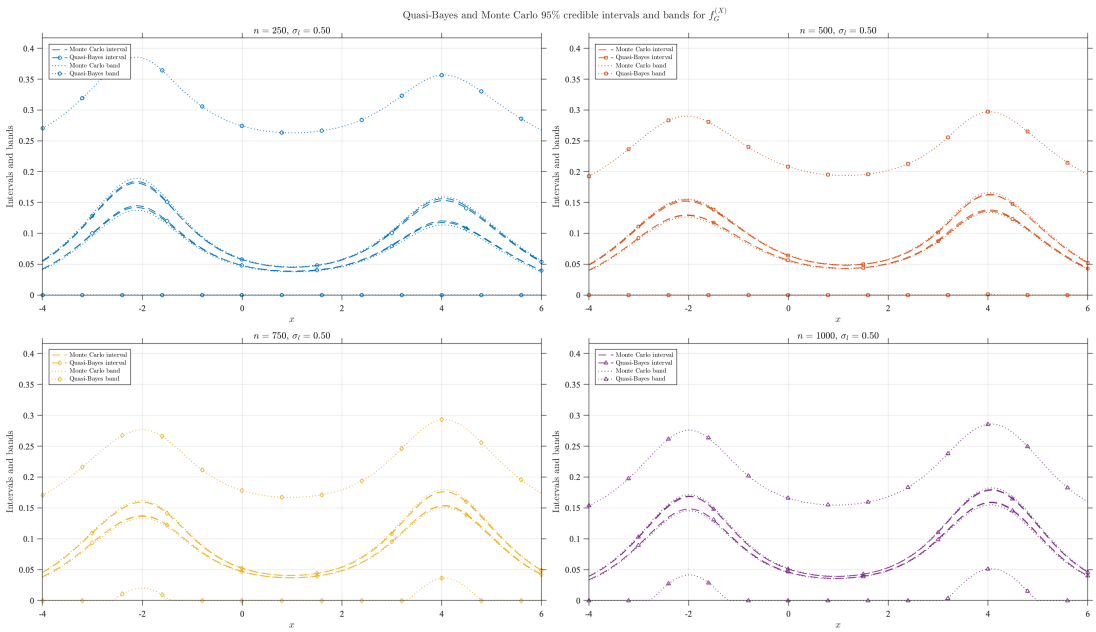


Figure 24: Bimodal example II, with Laplace noise ($\sigma_l = 0.50$): quasi-Bayes versus Monte Carlo credible intervals and bands.

are i.i.d. with density function

$$f_{g^*}^{(X)}(x) = \int k(x \mid \theta) g^*(\theta) \mu(d\theta), \tag{D.32}$$

with respect to a dominating measure λ. The quasi-Bayesian learning process for the probability distribution of the X_n, based on the observation of the Y_n is given by

$$f_{\widetilde{g}_n}^{(X)}(x) = \int k(x \mid \theta) \widetilde{g}_n(\theta) \mu(\mathrm{d}\theta) \tag{D.33}$$

where

$$\widetilde{g}_{n+1}(\theta) = (1 - \widetilde{\alpha}_{n+1}) \widetilde{g}_n(\theta) + \frac{(f_Z * k)(Y_{n+1} \mid \theta)}{\int_\Theta (f_Z * k)(Y_{n+1} \mid \theta) \widetilde{g}_n(\theta) \mu(\mathrm{d}\theta)} \widetilde{g}_n(\theta), \tag{D.34}$$

given an initial guess $\widetilde{g}_0$ and a sequence $(\widetilde{\alpha}_n)_{n \geq 1}$ of real numbers in $(0, 1)$ with $\sum_{n \geq 1} \widetilde{\alpha}_n = +\infty$ and $\sum_{n \geq 1} \widetilde{\alpha}_n^2 < +\infty$. To underline that the X_n's are i.i.d., we denote the probability measure on $(\Omega, \mathcal{F})$ by P^*.

Theorem D.1. *Let $(X_n)_{n \geq 1}$, $(Y_n)_{n \geq 1}$ and $(Z_n)_{n \geq 1}$ be sequences of random variables defined on $(\Omega, \mathcal{F}, P^*)$, such that the X_n's are i.i.d. with density $f_{g^*}^{(X)}$ in (D.32), the Z_n's are i.i.d. with density f_Z and independent of the X_n's, and $Y_n = X_n + Z_n$ for $n \geq 1$. Let $f_{\widetilde{g}_n}^{(X)}$ be in (D.33), where $\widetilde{g}_n$ satisfies (D.34) with $\widetilde{\alpha}_n = (\alpha + n)^{-\gamma}$ for $\alpha > 0$ and $\gamma \in (1/2, 1)$ and $g_0(\theta) > 0$ for every $\theta \in \Theta$. Under the assumptions A1)–A5) and F3) it holds P^*-a.s. that, for $\delta < 1 - 1/(2\gamma)$,*

$$|f_{\widetilde{g}_n}^{(X)}(x) - f_{g^*}^{(X)}(x)| = o(n^{-\delta}) \qquad x \in \mathbb{R},$$

and

$$\int_{\mathbb{R}} |f_{\widetilde{g}_n}^{(X)}(x) - f_{g^*}^{(X)}(x)| \lambda(dx) = o(n^{-\delta}).$$

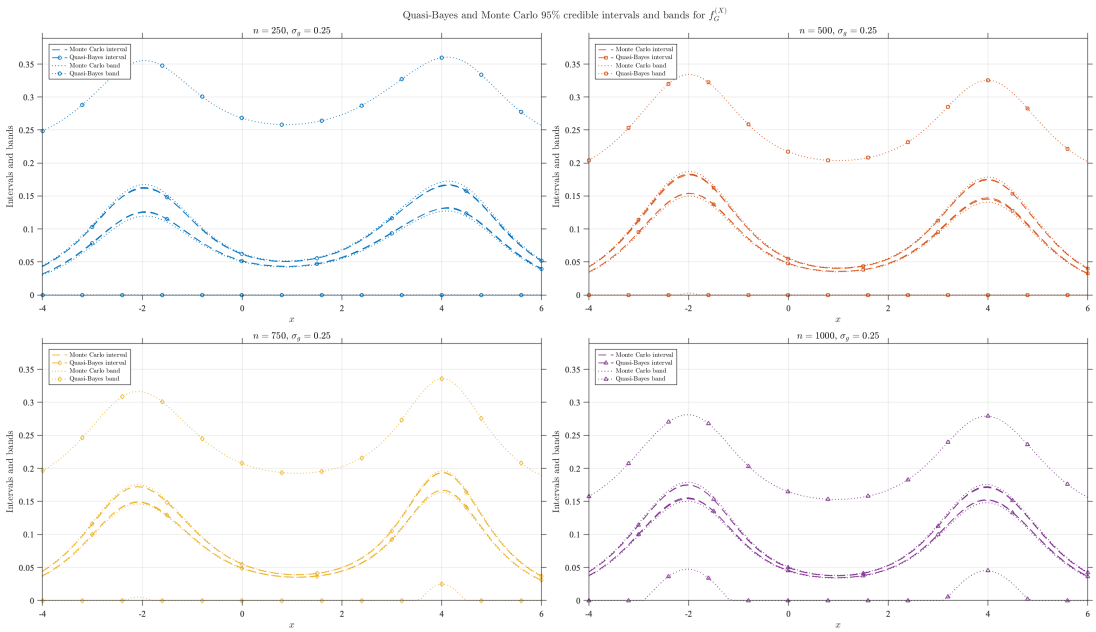


Figure 25: Bimodal example II, with Gaussian noise ($\sigma_g = 0.25$): quasi-Bayes versus Monte Carlo credible intervals and bands.

Note that Theorem D.1 specifies a learning rate of the form $\widetilde{\alpha}_n = (\alpha + n)^{-\gamma}$ with $\gamma \in (1/2, 1)$. This is a standard choice for the learning rate, and the exclusion of the value $\gamma = 1$ is not particularly restrictive; indeed calibrating $\widetilde{\alpha}_n$ as in Section D.1 typically results in $\gamma < 1$.

Arguing as in the proof of Lemma B.15, one can show that

$$W_1\left(F^{(X)}_{\widetilde{G}_n}, F^{(X)}_{G^\star}\right) = o_{\mathbb{P}^\star}\left(n^{-\delta}\right),$$

for every $\delta < 1 - \frac{1}{2\gamma}$. Under the assumptions of Proposition 4.3, with $\gamma \in (2/3, 1)$, one also has

$$W_1\left(F^{(X)}_{\widetilde{G}_n}, F^{(X)}_{G^\star}\right) = o_{\mathbb{P}^\star}\left(n^{-\frac{(1-\gamma)(\alpha+1)}{2(\alpha+2)}}\right).$$

Since

$$1 - \frac{1}{2\gamma} > \frac{(1-\gamma)(\alpha+1)}{2(\alpha+2)}, \qquad \gamma \in (2/3, 1),$$

the convergence rate established in Theorem D.1 for finite mixtures is strictly faster than that of Proposition 4.3 for general mixtures under Laplace error distribution.

D.2.1 Proof of Theorem D.1

The proof of Theorem D.1 is based on the following Lemma D.2 and on the rate of convergence of $\widetilde{g}_n$ to g^*. This last result is known in the literature (Martin, 2012), but we argue that the proof is not complete. We provide for completeness the proof in Lemma D.3.

Let d be the cardinality of Θ. We can identify each density function with respect to the counting measure on Θ with a vector in $\mathbb{R}^d$. More precisely, let $\Theta = \{\vartheta_1, \ldots, \vartheta_d\}$. Then $g = [g(\vartheta_1), \ldots, g(\vartheta_d)]^T$. We also write $\widetilde{k}_y$ for the d-dimensional vector $\widetilde{k}(y \mid \theta)$ with $\theta \in \Theta$, and define, for each density function g with respect to μ, $f^{(Y)}_g(y) = \int_\Theta \widetilde{k}(y \mid \theta) g(\theta) \mu(d\theta)$.

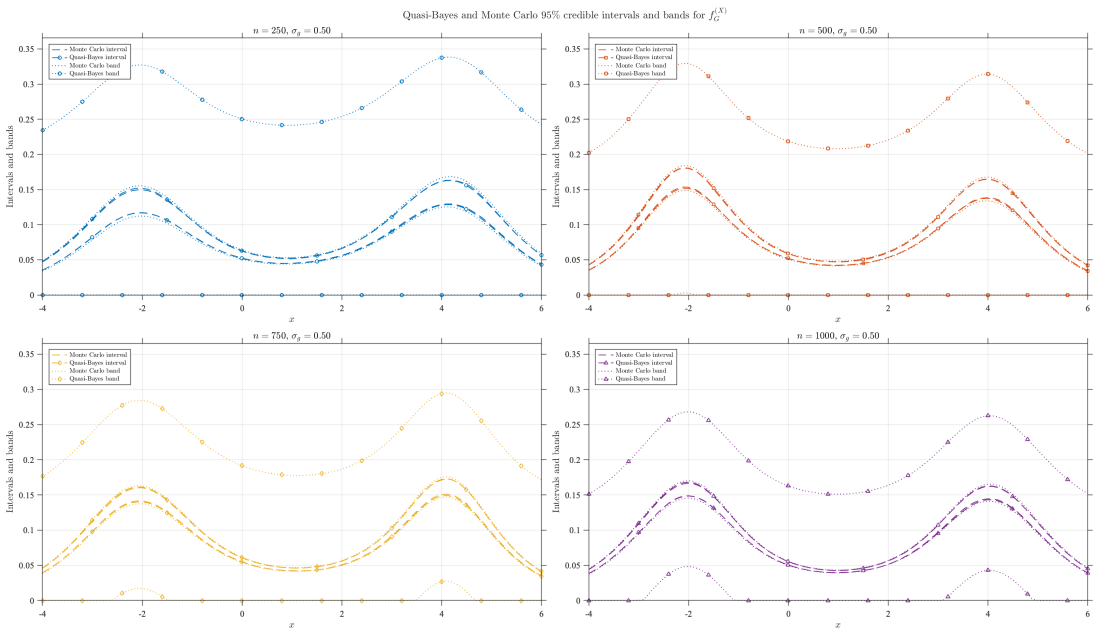


Figure 26: Bimodal example II, with Gaussian noise ($\sigma_g = 0.50$): quasi-Bayes versus Monte Carlo credible intervals and bands.

Lemma D.2. *Under the assumptions of Theorem D.1, there exists, for λ-almost every $x \in \mathbb{R}$ a constant $C(x)$ such that for every $n \geq 1$*

$$|f_{\widetilde{g}_n}^{(X)}(x) - f_{g^*}^{(X)}(x)| \leq C(x)||\widetilde{g}_n - g^*||_2,$$

where $\widetilde{g}_n$ and g^* are interpreted as vectors in $\mathbb{R}^d$ and $||\cdot||_2$ is the Euclidean norm. Furthermore, there exists a constant C such that

$$\int |f_{\widetilde{g}_n}^{(X)}(x) - f_{g^*}^{(X)}(x)|\lambda(dx) \leq C||\widetilde{g}_n - g^*||_2,$$

Proof. By the assumption A3), there exists for λ-almost every x, a constant $C(x)$ such that $\sup_\theta k(x \mid \theta) \leq C(x)$. Thus

$$\begin{aligned}
|f_{\widetilde{g}_n}^{(X)}(x) - f_{g^*}^{(X)}(x)| &\leq \int_\Theta k(x \mid \theta)|\widetilde{g}_n(\theta) - g^*(\theta)|\mu(d\theta) \\
&\leq C(x) \int_\Theta |\widetilde{g}_n(\theta) - g^*(\theta)|\mu(d\theta) \\
&\leq C(x)||\widetilde{g}_n - g^*||_2.
\end{aligned}$$

Moreover, again by assumption A3), the exists a constant C such that $\int \sup_{\theta\in\Theta} k(x \mid \theta)\lambda(\mathrm{d}x) \leq C$. Hence,

$$\begin{aligned}
\int |f_{\widetilde{g}_n}^{(X)}(x) - f_{g^*}^{(X)}(x)|\lambda(dx) &\leq \iint_\Theta k(x \mid \theta)|\widetilde{g}_n(\theta) - g^*(\theta)|\mu(d\theta)\lambda(dx) \\
&\leq \int \sup_{\theta\in\Theta} k(x \mid \theta)\lambda(\mathrm{d}x) \int_\Theta |\widetilde{g}_n(\theta) - g^*(\theta)|\mu(d\theta) \\
&\leq C||\widetilde{g}_n - g^*||_2
\end{aligned}$$

which concludes the proof. □

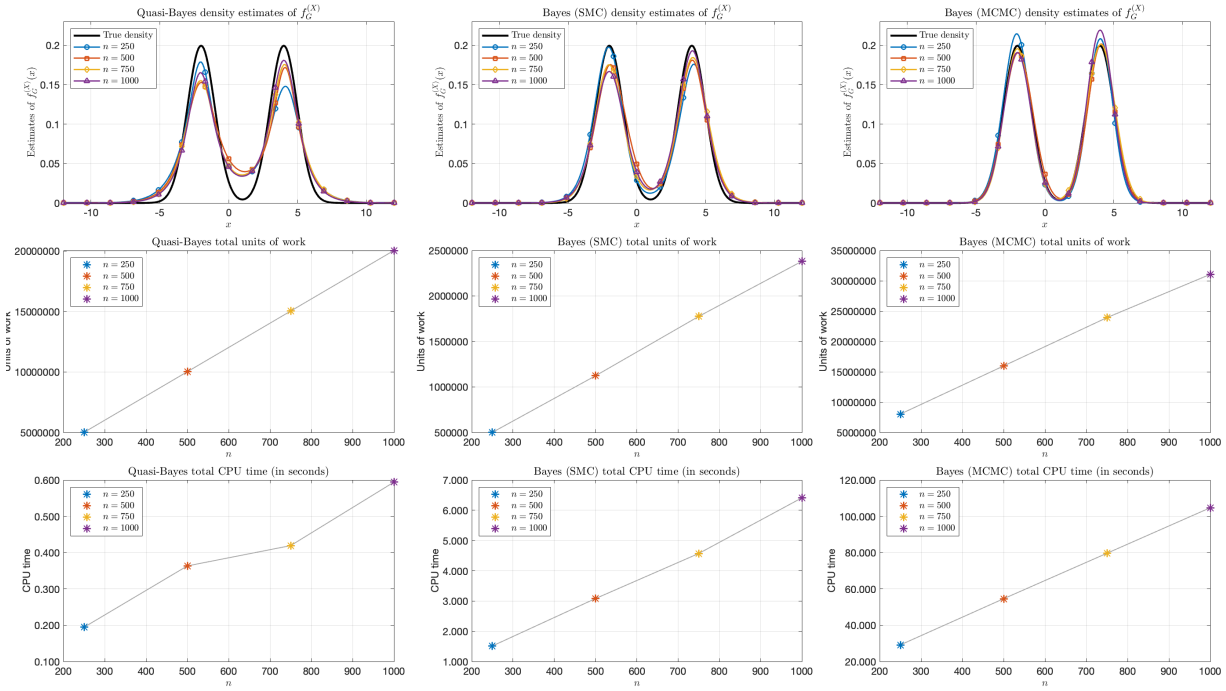


Figure 27: Bimodal example II, with Laplace noise ($\sigma_l = 0.25$): comparison between quasi-Bayesian and Bayesian approaches.

Lemma D.3. *Let g_n and g^* be the representations of g_n ($n \geq 0$) and g^* in $\mathbb{R}^d$. Under the assumptions of Theorem D.1, for every $\delta < 1 - 1/(2\gamma)$*

$$||g_n - g^*||_2 = o(n^{-\delta}) \quad P^* - a.s.$$

Proof. Since Θ is finite, then the assumptions F1)–F2) hold. Therefore we can apply Corollary 4.1 and deduce that the random vector $\widetilde{g}_n$ converges to g^*, P^*-a.s. Moreover, $(\widetilde{g}_n)$ satisfies the stochastic approximation in $\mathbb{R}^d$:

$$\begin{aligned}\widetilde{g}_{n+1} &= \widetilde{g}_n + \widetilde{\alpha}_{n+1}\widetilde{g}_n \circ \left(\frac{\widetilde{k}_{Y_{n+1}}}{f^{(Y)}_{\widetilde{g}_n}(Y_{n+1})} - \mathbf{1} \right) \\ &= \widetilde{g}_n + \widetilde{\alpha}_{n+1} h(\widetilde{g}_n) + \widetilde{\alpha}_{n+1}\epsilon_{n+1},\end{aligned}$$

with initial value $\widetilde{g}_0$, where

$$h(g) = g \circ \left(\int \frac{\widetilde{k}_y}{f^{(Y)}_g(y)} f^{(Y)}_{g^*}(y)\lambda(dy) - \mathbf{1} \right),$$

and

$$\epsilon_{n+1} = g \circ \left(\frac{\widetilde{k}_{Y_{n+1}}}{f^{(Y)}_g(Y_{n+1})} f^{(Y)}_{g^*}(Y_{n+1}) - \int \frac{\widetilde{k}_y}{f^{(Y)}_g(y)} f^{(Y)}_{g^*}(y)\lambda(dy) \right).$$

The thesis follows from Theorem 3.1.1 in Chen (2005) if we can prove that the following conditions hold:

F3.1 $\widetilde{\alpha}_n \to 0$, $\sum_{n=1}^{\infty} \widetilde{\alpha}_n = \infty$, $\widetilde{\alpha}_{n+1}^{-1} - \widetilde{\alpha}_n^{-1} \to 0$;

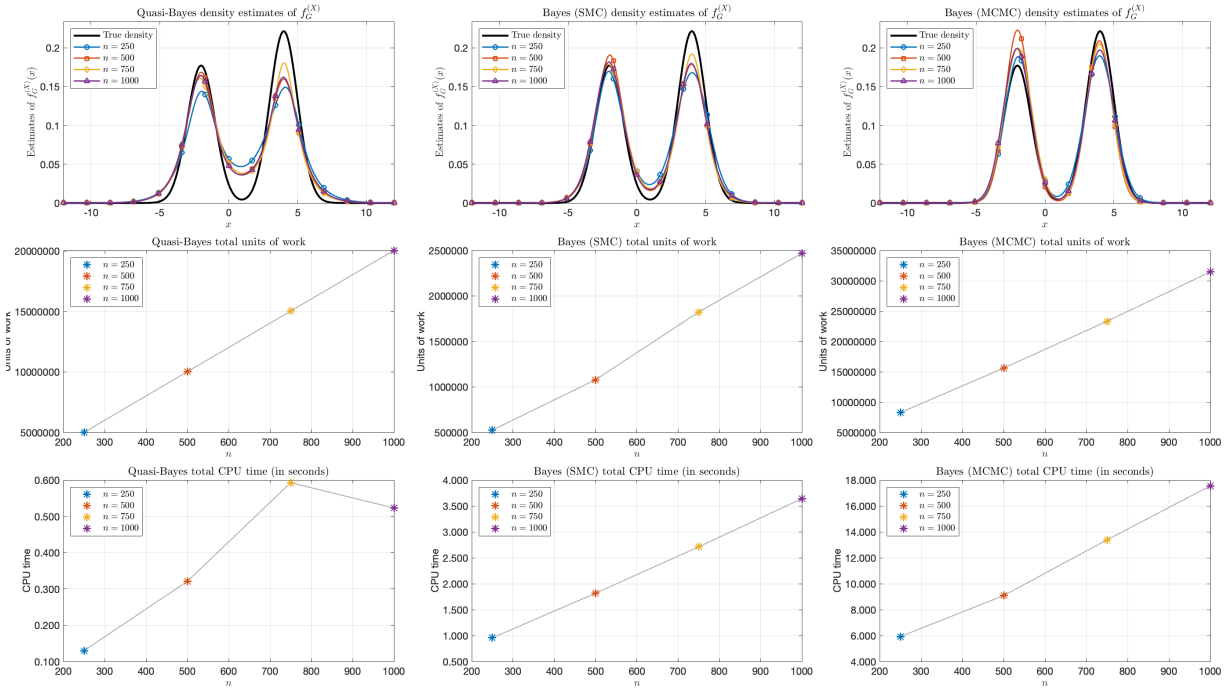


Figure 28: Bimodal example II, with Gaussian noise ($\sigma_g = 0.25$): comparison between quasi-Bayesian and Bayesian approaches.

F3.2 $\sum_{n=1}^{\infty} \widetilde{\alpha}_n^{1-\delta/2} \epsilon_n$ converges, P^*-a.s.;

F3.3 h is measurable and locally bounded, and is differentiable at g^*. Let H be a matrix such that, as $g \to g^*$,

$$h(g) = H(g - g^*) + r(g), \quad r(g^*) = 0, \quad ||r(g)||_2 = o(||g - g^*||_2).$$

All the eigenvalues of the matrix H have negative real parts.

Remark D.4. *Theorem 3.1.1 in Chen (2005) also requests the following condition:* F3.4 There exists a continuously differentiable function $\mathcal{I} : \Delta^d \to \mathbb{R}$ such that

$$\sup_{\delta_1 \leq \mathrm{d}(g,g^*) \leq \delta_2} h(g)^T \nabla_g \mathcal{I} < 0,$$

for every $\delta_2 > \delta_1 > 0$. *However, a close inspection of the proof of Theorem 3.1.1 in Chen shows that F3.4 is employed only to ensure convergence of $\widetilde{g}_n$ towards g^*. In our case, the convergence has been already proved.*

F3.1 is obvious. To prove F3.2, notice that $(\sum_{k=1}^{n} \widetilde{\alpha}_k^{1-\delta/2} \epsilon_k)_{n \geq 1}$ is a martingale. Moreover, since Θ is finite and

$$\int \int_{\Theta} g(\theta) \frac{\widetilde{k}(y \mid \theta)}{f_g^{(Y)}(y)} f_{g^*}^{(Y)}(y) \lambda(dy) \mu(d\theta) = 1,$$

then $|\epsilon_{n,i}| \leq 1$ for every i. Together with $\gamma(2 - \delta) > 1$, this implies that, for every $i = 1, \ldots, d$,

$$\sup_n E^*((\sum_{k=1}^{n} \widetilde{\alpha}_k^{1-\delta/2} \epsilon_{k,i})^2) = \sup_n \sum_{k=1}^{n} \widetilde{\alpha}_k^{2-\delta} E^*(\epsilon_{k,i})^2$$

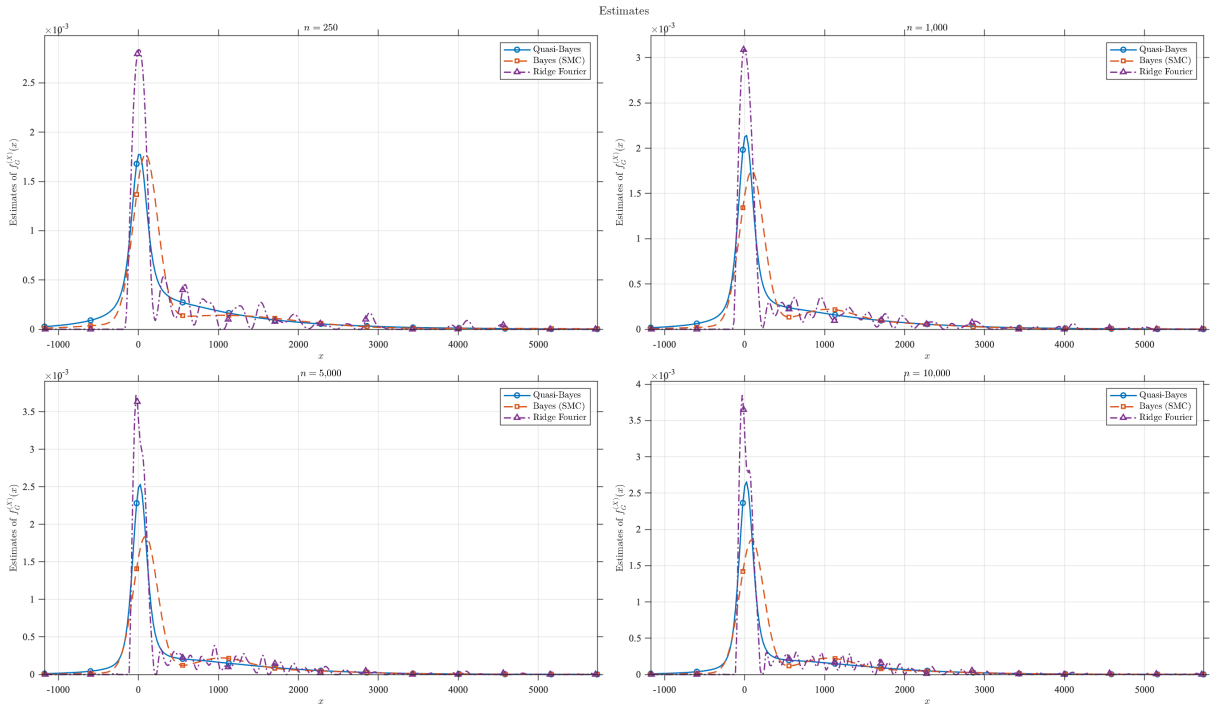


Figure 29: Flow-cytometry data, with Laplace noise: comparison between quasi-Bayesian, Bayesian and ridge-regularized Fourier deconvolution approaches.

$$\leq \sum_{k=1}^{\infty} \left(\frac{1}{\alpha + k} \right)^{\gamma(2-\delta)} < +\infty.$$

Thus, $(\sum_{k=1}^{n} \widetilde{\alpha}_k^{1-\delta/2} \epsilon_k)_{n \geq 1}$ is a uniformly integrable martingale, and therefore it converges P^*-a.s. We now prove F3.3. The function h is continuously differentiable at g^* and

$$H(\theta, \theta') = -g^*(\theta) \int \frac{\widetilde{k}_y(\theta) \widetilde{k}_y(\theta')}{f_{g^*}^{(Y)}(y)} \lambda(dy).$$

The matrix H can be seen as the product of the diagonal matrix $\mathrm{diag}(g^*)$, which is positive definite, times $-M$, with

$$M(\theta, \theta') = \int \frac{\widetilde{k}_y(\theta) \widetilde{k}_y(\theta')}{f_{g^*}^{(Y)}(y)} \lambda(dy).$$

We now prove that M is positive definite. Given a vector $u = [u(\theta)]_{\theta \in \Theta} \in \mathbb{R}^d$ with $||u|| = 1$, we can write that

$$\begin{aligned} u^T M u &= \iint_{\Theta^2} u(\theta) u(\theta') \int \frac{\widetilde{k}_y(\theta) \widetilde{k}_y(\theta')}{f_{g^*}^{(Y)}(y)} \lambda(dy) \mu(d\theta) \mu(d\theta') \\ &= \int \frac{1}{f_{g^*}^{(Y)}(y)} \left(\int_{\Theta} u(\theta) \widetilde{k}_y(\theta) \mu(d\theta) \right)^2 \lambda(dy) \geq 0, \end{aligned}$$

with equality if and only if $\int_{\Theta} u(\theta) \widetilde{k}_y(\theta) \mu(d\theta) = 0$, λ-almost everywhere. The only solution of the above equation in u is $u(\theta) = 0$ for every $\theta \in \Theta$, which is inconsistent with $||u|| = 1$. Since the

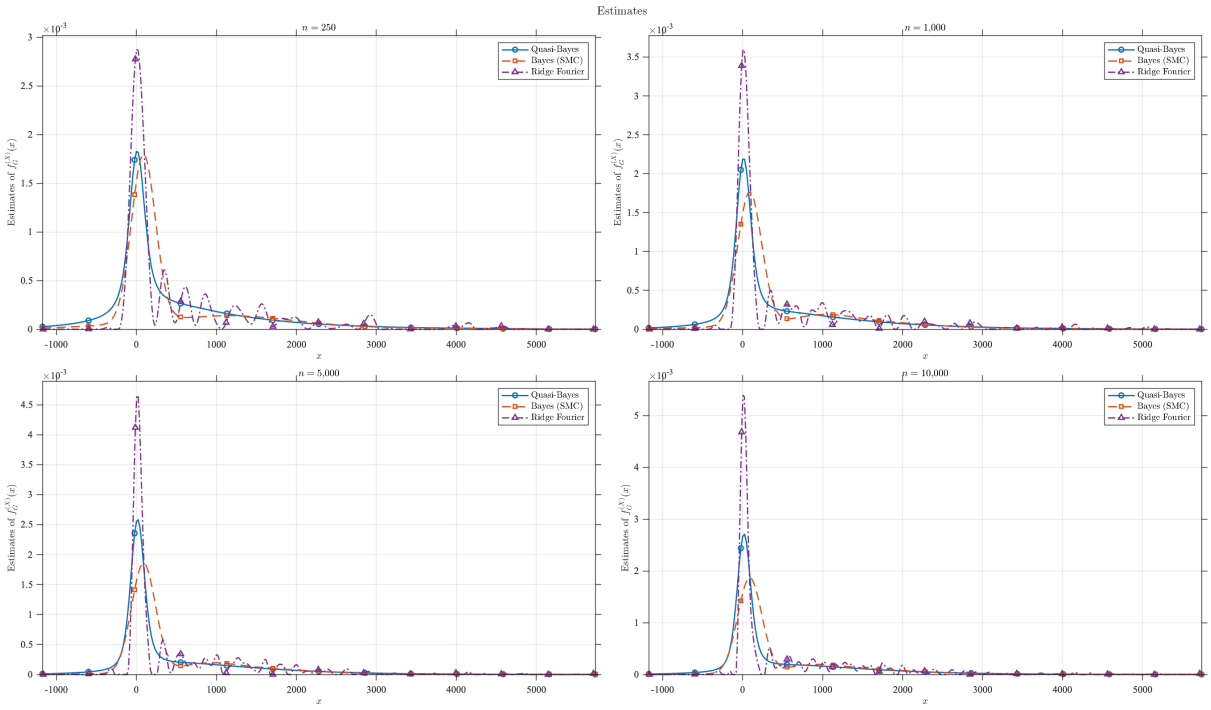


Figure 30: Flow-cytometry data, with four-component Gaussian-mixture noise: comparison between quasi-Bayesian, Bayesian and ridge-regularized Fourier deconvolution approaches.

function $u^T M u$ is continuous on the compact set $||u|| = 1$, then the minimum is attained, and by the above reasoning it is strictly positive. It follows that M is positive definite. □

Proof of Theorem D.1. The proof is an immediate consequence of Lemma D.2 and Lemma D.3. □

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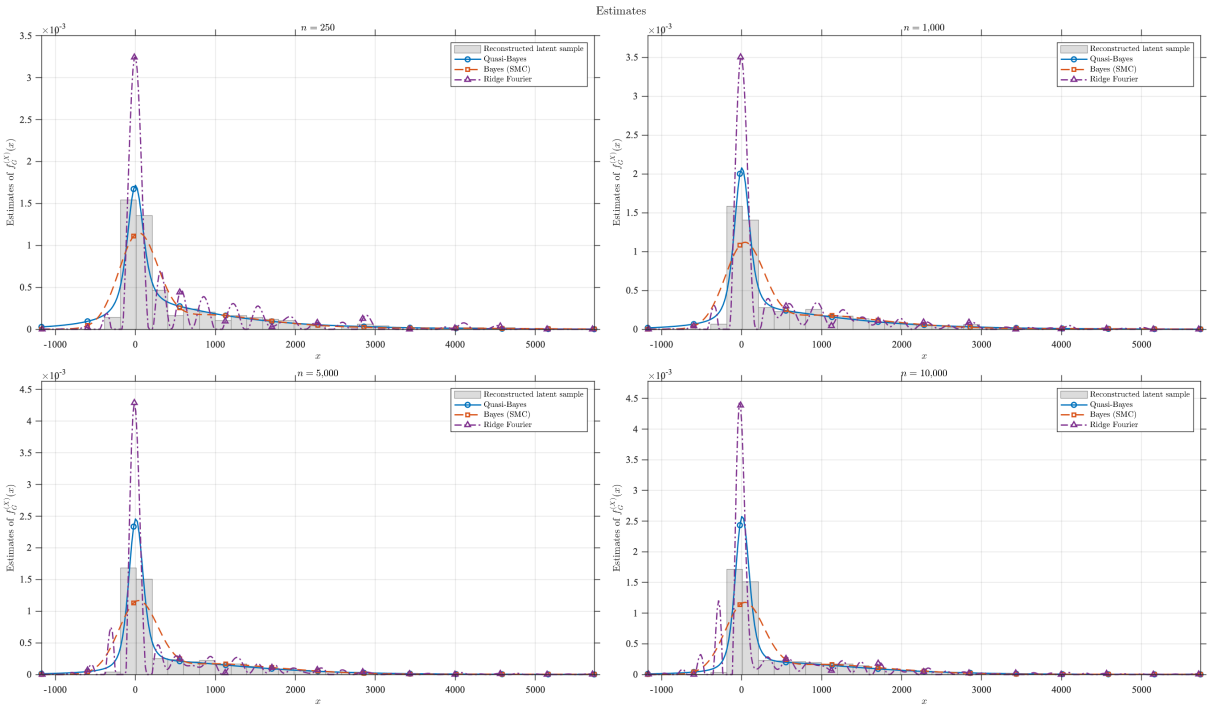


Figure 31: Flow-cytometry data, with Gaussian noise: comparison between quasi-Bayesian, Bayesian and ridge-regularized Fourier deconvolution approaches.

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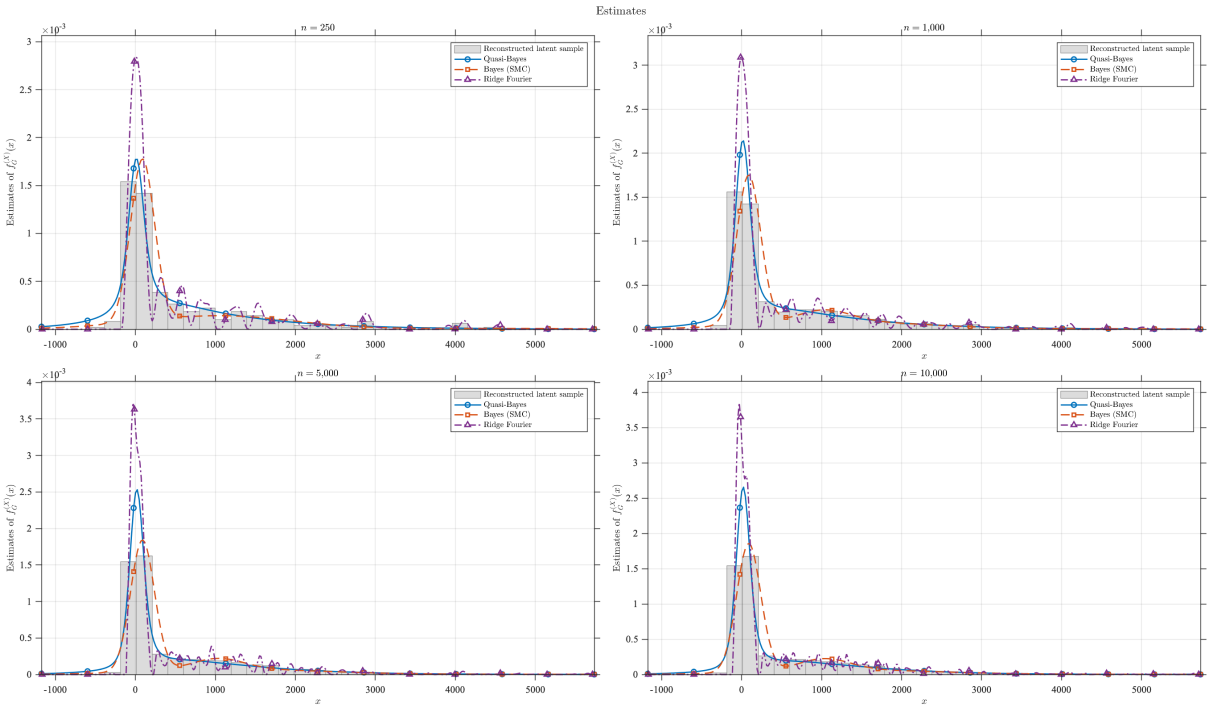


Figure 32: Flow-cytometry data, with Laplace noise: comparison between quasi-Bayesian, Bayesian and ridge-regularized Fourier deconvolution approaches.

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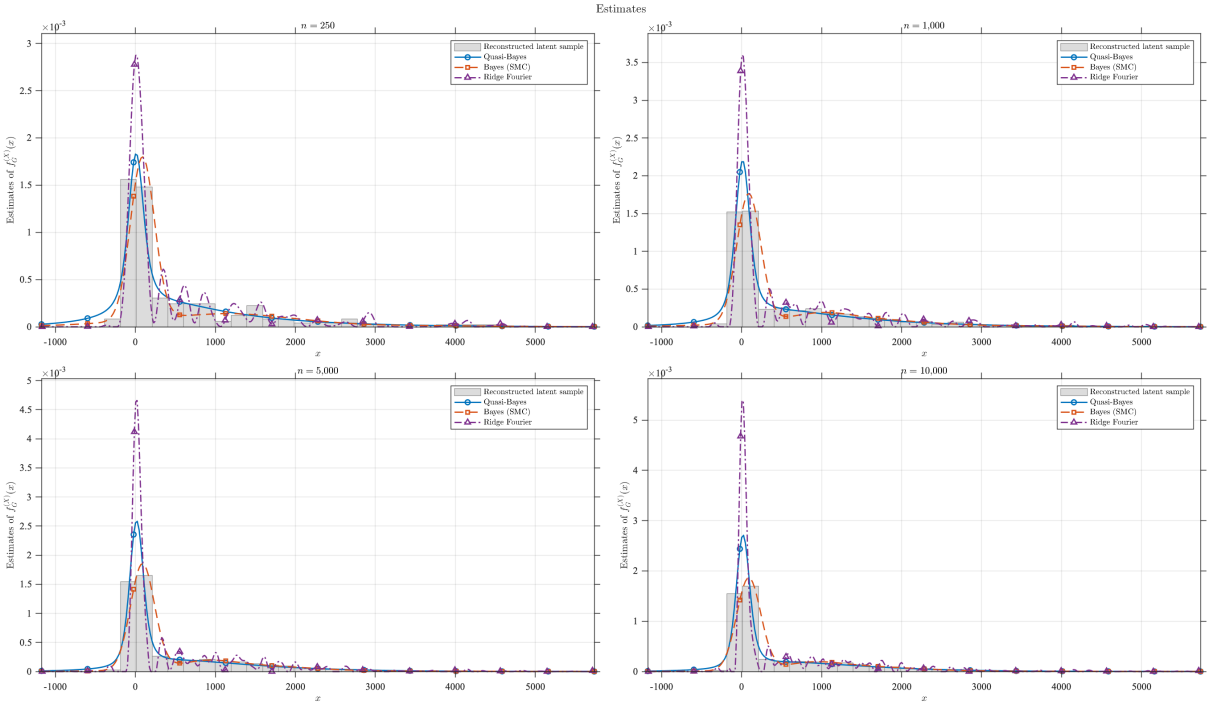


Figure 33: Flow-cytometry data, with four-component Gaussian-mixture noise: comparison between quasi-Bayesian, Bayesian and ridge-regularized Fourier deconvolution approaches.

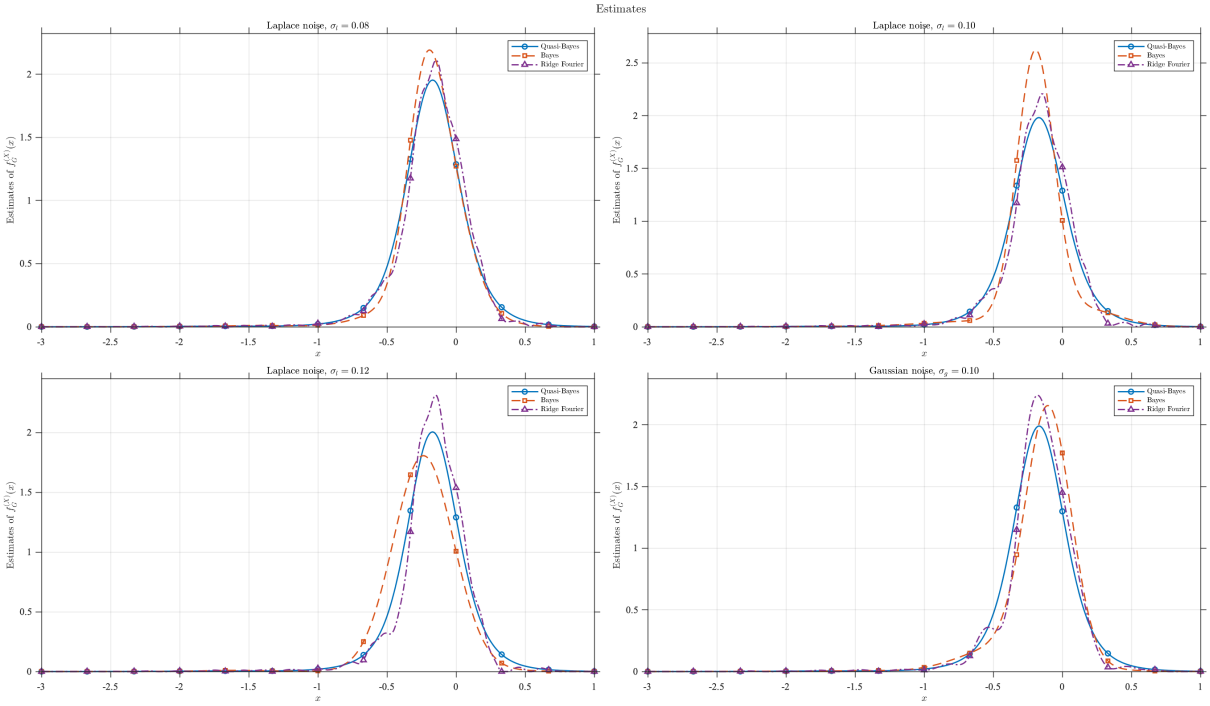


Figure 34: Stellar metallicity data, with Laplace and Gaussian noise: comparison between quasi-Bayesian, Bayesian and ridge-regularized Fourier deconvolution approaches.

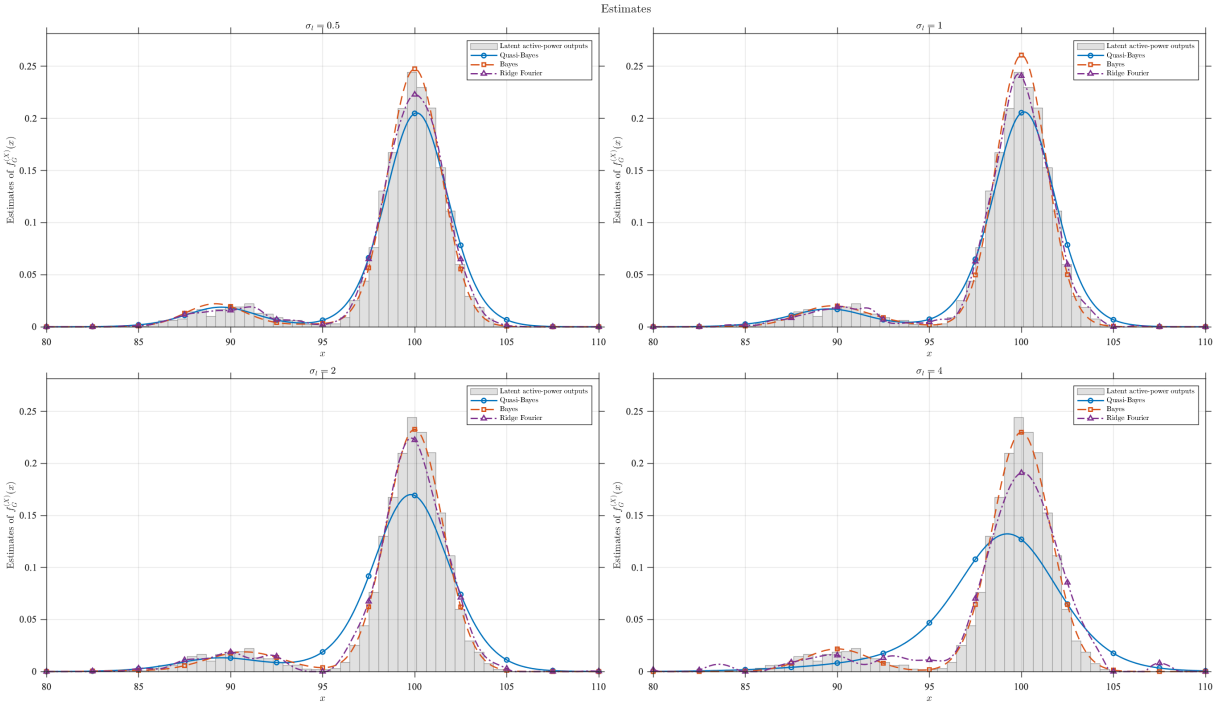


Figure 35: Active power-output data, with Laplace noise: comparison between quasi-Bayesian, Bayesian and ridge-regularized Fourier deconvolution approaches.

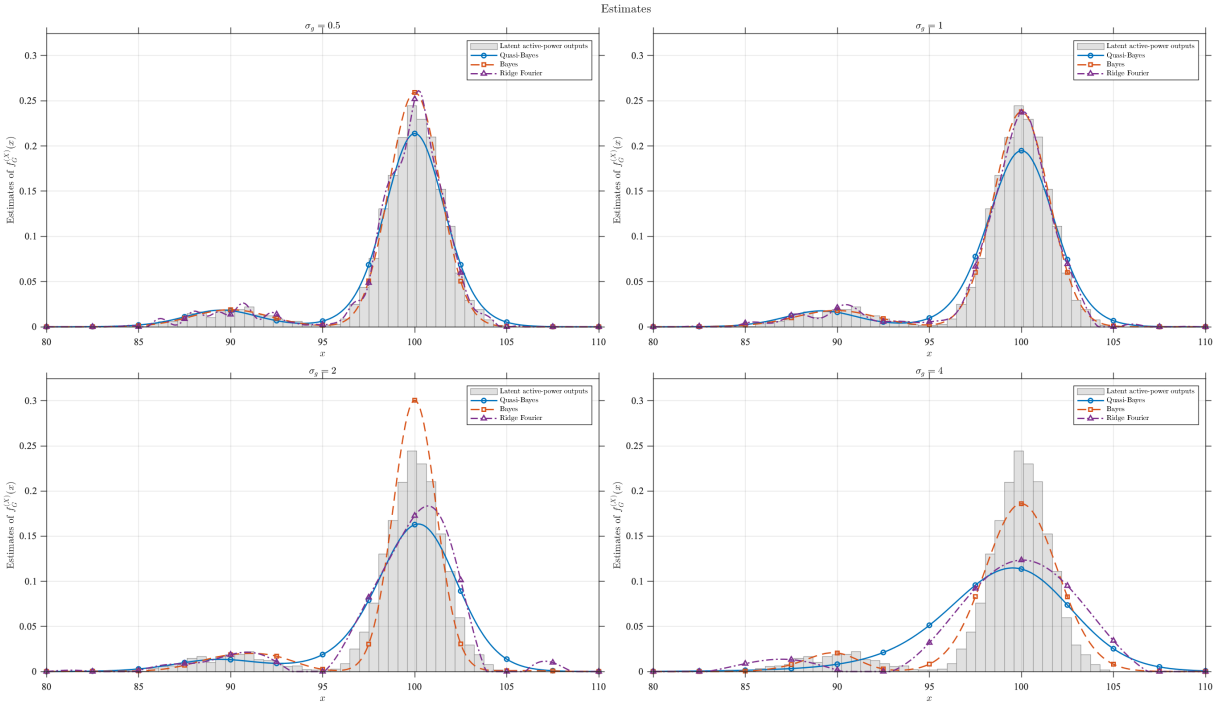


Figure 36: Active power-output data, with Gaussian noise: comparison between quasi-Bayesian, Bayesian and ridge-regularized Fourier deconvolution approaches.